Caisson 4/5 Spare

COVER SHEET – STATEMENT OF WORK (SOW)

N4523A26R1001

SOLICITATION COPY

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Statement of Work (SOW)

1.0 **General Requirements:** This document invokes the requirements applicable to the docking and maintenance work to be performed on One each Caisson 4/5 Spare. Work performed and accomplished shall be in accordance with the highest quality workmanship standards and practices commonly accepted in the commercial marine industry and the technical references listed herein. The Contractor is responsible for understanding and complying with all requirements specified herein. The requirements listed herein work in conjunction with the technical drawings and other references to describe the essential features, functions, and requirements of the work performed.

1.1 **Asset Characteristics:**

Caisson 4/5 Spare			
Length:	147 FT – 5 IN	Draft:	25 FT – 11 IN
Width:	27 FT	Full Displacement:	1565 L Tons
Height:	53 FT – 11 IN	Light Load	884 L Tons
Hull Type:	Steel		

1.2 **Scope Overview:** Coordinate with the Contracting Officer, via the SUPERVISOR to deliver the Asset to the Contractor's facility. Upon docking Asset, wash and clean Asset: Perform various base metal and structural repairs, Replace the port rubber seal and perform required inspections and maintenance of starboard seal. Conduct ultrasonic and visual inspections of the hull, weather deck, tanks, piping and operations deck. Clean and pump tanks. Maintain/replace deck hatches. Install new mimic board. Test alarm systems. Clean ventilation. Replace all valves. Replace dewatering pumps. Replace grease lines and steady bearings. Replace actuator stands. Replace actuators. Replace valve-stem extension shafts. Remove, clean and reinstall rubber fendering. Preserve hull, weather deck, weather deck handrails, tanks, piping and operations deck. Replace zinc anodes. Conduct operational testing. Undock the Asset. Arrange and coordinate the delivery of the Asset back to the Government with the Contracting Officer via the SUPERVISOR.

1.3 **Definitions:** As used throughout this Specification, the following definitions and terms have meanings as indicated:

1.3.1 Unless otherwise stated, “as shown”, “as indicated”, “as detailed”, or other words of similar import refer to the contractual documents including reference documents and publications.

1.3.2 The words “as directed”, “as approved”, “when authorized” or words of similar import refer to the direction, requirements, permission, approval, or acceptance by the Contracting Officer.

1.3.3 The term “Objective Quality Evidence” (OQE) shall mean current, accurate, and complete objective documentation and information submitted together as a complete data package sufficient that a qualified and disinterested third party could analyze and evaluate whether technical requirements were achieved.

1.3.4 Where “in accordance with Regulatory Body requirements” is stated, the industrial efforts shall be in accordance with the appropriate commercial regulatory body requirements unless otherwise authorized by the Government.

1.3.5 The term “marine” or “marine quality” means an item constructed of materials unaffected by moisture, sea spray, extremes in temperature or other hazards of the marine environment and designed and constructed to perform its intended functions under marine operating conditions.

1.3.6 The term “herein” means the applicable section(s) of this Specification.

1.3.7 The term “NAVSEA” means the Naval Sea Systems Command, one of five system commands, responsible for engineering, building, supporting, and maintaining the Navy’s fleet of vessels.

1.3.8 The term “Normal Work Hours” for the purposes of this contract means work hours is from 0720 to 1602 during the work week of Monday thru Friday. Notifications and reports would be expected to be in this timeframe otherwise it would be considered as notification conducted for the following normal working day. Weekends and Holidays are considered outside the normal work hours.

1.3.9 The term “Stakeholder” means customer, Government, any agency conducting work during the contracted availability timeline. The term “SUPERVISOR” is defined as the local Government activity responsible for the execution and contract administration of Navy maintenance and modernization work. For the purposes of this SOW/CDRLs this activity is the assigned Code 101.13 representative.

1.3.10 The term “OEM” or Original Equipment Manufacturer is a company whose goods are used as components in the products of another company. An OEM creates parts and components that are used by other companies in their finished products.

1.3.11 The term “AIT” or Alteration Installation Team is defined as a Navy team (military, civilian or civilian contractor) under the direct supervision of Navy personnel, that is trained and equipped to accomplish specific Alterations or temporary modifications on specified ships.

1.3.12 Worksite is defined as within the specific compartment or space where physical productive work is occurring. For work performed on Navy boats and craft or service craft, the “worksite” is defined as on or near the Asset.

1.3.13 Foreign Material Exclusion (FME) is used to maintain system cleanliness. Accomplishment of cleanliness restoration will be invoked within the work item when cleanliness is lost or suspected of being lost.

1.3.14 Key Event: An event that, if slippage occurs, could impact or delay the overall schedule, or prevent timely delivery of the Asset. Key Events are identified by the contract, the SUPERVISOR, or the contractor.

1.3.15 Milestone: A significant event identified by the Maintenance Team. Milestones are used as a scheduling aid and establish significant points where progress must be evaluated and confirmed. Accumulated failure to achieve Milestones on schedule may result in missed Key Event. Milestones may be identified by either the contractor or the SUPERVISOR.

1.3.16 Critical Path: That sequence of Work Activities which forms the work and test chain of the longest duration and directly affects the completion of the availability. Factors that influence when a Work Activity is on the Critical Path include: time duration required for the Work Activity, space limitations, manpower available, and the predecessor/successor relationships between Work Activities. Typically, the Critical Path is determined by automated schedule analysis and will include any sequential set of Work Activities forming the longest chain of events extending throughout the schedule and which has the least Total Float.

1.3.17 Total Float: The total number of days that a path of Work Activities can be delayed without affecting the project finish date. A path of Work Activities is established by predecessor and successor relationships.

1.3.18 The term "workday" is defined as any day that falls on Monday through Friday, excluding Federal holidays.

1.3.19 The acronym "GDO" stands for Government Docking Officer.

a. The GDO may be present at the time of docking a vessel/ asset using a floating or graving dock, or the vertical lift facility is raised and shall remain at the dock or vertical lift facility until all blocks are well covered to make sure that no blocks come adrift or are misplaced to endanger the ship when landing.

b. The GDO shall be notified and provided blocking plan, for any bump, shift of blocks, vessel/asset fleeting, or vessel/asset transition while within the Contractor's possession or facility. GDO will ensure calculations and block placement is consistent with engineering requirements, and blocks align with proper vessel/asset structure for the safety of the vessel/asset.

c. The GDO may be present at the time of undocking a vessel/ asset using a floating or graving dock, or the vertical lift facility is lowered ensure that no blocks or block caps remain affixed to the vessel/asset and the vessel/asset is seaworthy, without list, as not to endanger the ship.

1.3.20 The term "drydocking" refers to an evolution in which a vessel/asset is landed on blocks at a drydocking facility, in accordance with GDO-approved docking and blocking plans, enabling production work to be performed

1.3.21 The term "undocking" refers to an evolution in which a vessel/asset is removed from a drydocking facility for purposes returning a vessel to the water.

1.3.22 The term(s) "fleeting" including "bumping" or "shifting" refer to a vessel/asset that is re-floated and re-landed on the blocks without leaving a facility for purposes of docking or undocking other vessels or for shifting the position of the ship on the blocks for underbody coating system application or other production work.

1.4 **Arrival Conference:** The Government will arrange an arrival conference with the Contractor after contract award.

1.5 **Required Documents:** Administrative requirements of this SOW shall commence at contract award and be updated throughout the contract period.

1.5.1 Prepare and keep current schedules necessary for the purpose of establishing an orderly and systematic accomplishment of production work and material ordering. (REQUIRED REPORT CDRL A001)

1.5.1.1 Production Schedule shall be in Gantt format

1.5.2 Maintain a list of all subcontractors with the extent of work to be accomplished by each subcontractor. (REQUIRED REPORT CDRL A002)

1.5.3 Maintain a list of manufacturer's Technical Representatives to be provided per the particular specification requirement. (REQUIRED REPORT CDRL A003)

1.5.4 Maintain a Welder Qualification List, with name, type of qualification(s) and date of last qualification test for each qualification. (REQUIRED REPORT CDRL A004)

1.5.5 Maintain manufacturer's product data sheets for all materials to be used by the Contractor to accomplish the requirements of this contract. (REQUIRED REPORT CDRL A005)

1.5.6 Maintain Safety Data Sheets (SDS) for all materials to be used by the Contractor to accomplish the requirements of this contract. (REQUIRED REPORT CDRL A006)

1.5.7 Maintain a list of key Contractor personnel with names, positions, certifications, telephone numbers, e-mail addresses, and emergency contact information. (REQUIRED REPORT CDRL A007)

1.5.8 Maintain a copy of the Contractor's Inspection and Quality Control System with copies of all related forms, reports, and other supporting documentation. (REQUIRED REPORT CDRL A008)

1.5.9 Maintain a copy of the Contractor's Environmental Protection and Hazardous Waste Management Plan, including Risk Management/Mitigation plans specific to this contract. (REQUIRED REPORT CDRL A009)

1.5.10 Maintain copies of required federal, state and local environmental notifications and permits for performance of the work (or a schedule of when they will be obtained).

1.5.11 When subcontractors are delegated responsibility for tests/inspections, the subcontractor's qualified and currently certified inspectors must be included on this list.

1.5.12 The copies shall be available for Government review within 2 hours of request.

1.6 **Progress Meetings:** Weekly progress meetings shall be held at the Contractor's facility near the Asset. The day and time of the meeting shall be set by the SUPERVISOR upon consultation with the Contractor. A review of percentage completion for each work item shall be conducted including the Contractor's assessment of schedule adherence. Production schedule, corrective action reports, material lead times and other issues shall be discussed. Publish the updated production schedule, and the test and inspection plan no less than 24 hours prior to the progress meeting.

1.6.1 Production Start – first day of the Period of Performance.

1.6.1.1 Major discussion points shall include all growth work identified prior to the 20 percent of scheduled Period of Performance, Milestones, and Schedule. (**Production Start +37 calendar days**)

1.6.1.2 Review conference with all stakeholders shall be conducted at the availability 25 percent of scheduled Period of Performance. (**Production Start +46 calendar days**)

1.6.1.3 Review conference with all stakeholders shall be conducted at the availability 50 percent of scheduled Period of Performance. (**Production Start +92 calendar days**)

1.6.2 Major discussion points shall include all critical path work, testing work items, Milestones, and Schedule.

1.6.3 Review conference with all stakeholders shall be conducted at the availability 75 percent of scheduled Period of Performance**. (Production Start +138 calendar days**)

1.6.3.1 Major discussion points shall include testing items, Milestones, and Schedule.

1.7 **Availability Requirements:**

1.7.1 Arrange for and transport the Caisson from the Government facility at Puget Sound Naval Shipyard (PSNS) in Bremerton, WA to the Contractor's repair facility. Upon completion of all work coordinate delivery of the Asset to the Government facility at PSNS in Bremerton, WA. Provide all materials, labor, and insurance bonds necessary for the safe transport of the Asset to and from the Contractor's facility and for the period that the Asset is located at the Contractor's repair facility.

1.7.1.1 Asset pickup and delivery shall be scheduled with the SUPERVISOR 5 working days prior to Asset pickup and delivery.

1.7.1.2 Flood through doors shall be closed prior to delivery of Caisson back to Puget Sound Naval Shipyard.

1.7.2 Provide a separate, lockable, and secure office space for use by (2 EA) Government on-site representatives furnished with desk, chair, lights, heat, air conditioning, electrical outlets, sanitary facilities, internet service, janitorial service, fire extinguisher, and 24-hour service from 7 working days prior to start of availability to 2 working days after availability completion. Contractor will not relocate offices after they have been turned over to the Government except with the concurrence of the Contracting Officer.

1.7.2.1 The facility must have a minimum of 300 square feet of floor space.

1.7.2.2 The facility must be located within one-quarter mile of the Asset.

1.7.2.3 The office space shall have direct access to a dedicated restroom facility with a water closet and hot / cold running water.

1.7.2.4 The office space shall be lockable, secure, and safe for personnel and equipment.

1.7.2.5 Provide a parking area adjacent to the facility. The area must be lighted and accommodate 3 automobiles simultaneously. Designate parking area with sign stating "PSNS PARKING ONLY".

1.7.2.6 Provide QTY (3 EA) keys for all doors and QTY (3 EA) master keys to fit all locks to the SUPERVISOR on the first day the SUPERVISOR is on site.

1.7.2.7 Provide Internet service separate from contractor's computer servers/network with QTY (2 EA) CAT-5 cable connection or WIFI at 50 MBPS minimum download speed.

1.7.2.8 Desktop Lighting must provide 28-foot candles of illumination measured at the desktop level.

1.7.2.9 Provide QTY (One EA) 15-pound fire extinguisher.

1.7.2.10 Heating and air conditioning must be capable of maintaining the temperature between 65 and 78 degrees Fahrenheit.

1.7.2.11 Provide QTY (3 EA) double-pedestal desk, 30-inches wide by 48-inches long.

1.7.2.12 Provide QTY (3 EA) swivel chair with adjustable arm rests, and QTY (2 EA) straight-back chair.

1.7.2.13 Provide QTY (One EA) 115-volt, shock resistant, double electrical receptacle for each desk. Each receptacle must be a convenient height and located adjacent to each desk.

1.7.2.14 Provide dedicated sanitary facilities equipped with drains, hot and cold potable water.

1.7.2.14.1 Hot water must be maintained at 120 to 140 degrees Fahrenheit.

1.7.2.14.2 Provide QTY (One EA): lavatory, water closet, soap dispenser, and towel dispenser.

1.7.2.15 Provide janitorial services to include sweeping, mopping, and trash pickup on a weekly basis.

1.7.2.16 Provide QTY (2 EA) 3 drawer lockable filing cabinets.

1.7.2.17 Provide QTY (One EA) paper shredder with paper receptacle.

1.7.2.18 Provide a private conference room with tables and chairs to accommodate a minimum of five persons for the entire contract period. This requirement can be satisfied by having reasonable access to Contractor facilities.

1.7.2.19 Facility must be in accordance with local building codes, sanitary and current fire regulations. The facility must include smoke alarms and sprinkler systems.

1.7.2.20 All furnishings and equipment per paragraph 1.7 shall be maintained in fully operable condition by the Contractor.

1.7.2.21 Facility must be delivered to the Government clean, sanitary, damage free, and vermin free.

1.7.2.22 Provide QTY (One EA) trash can with each desk and in the lavatory.

1.7.2.23 Provide QTY (One EA) copier capable of printing A3, and A4 size paper. Provide A3 and A4 size printing paper throughout the availability.

1.17.2.24 Facilities must be provided 7 working days prior to contract start date through 2 working days after the contract completion date unless otherwise specified.

(V) "STRUCTURAL INSPECTION"

1.7.3 Furnish One each well-lighted gangway and landing platform while the Asset is alongside the pier, moored to another vessel, float. barge, or in dry dock. Prior to use by Government shipyard personnel shall inspect the newly erected gangway and landing platform. Acceptance criteria are as follows: (REQUIRED REPORT CDRL A010)

1.7.3.1 Gangway and landing platform shall be a minimum of three feet in width and designed and constructed to support 75 pounds per square foot.

1.7.3.2 Maintain gangways free of dirt, debris, ice, snow, and foreign matter.

1.7.3.3 Inclination of gangway shall not exceed 30 degrees.

1.7.3.4 Illuminate the entire length of the gangway and adjacent areas with weather tight lights shielded to prevent temporary blinding of personnel.

1.7.3.5 The landing platform (vessel brow) will remain unobstructed for the duration of its use.

1.7.4 Provide temporary 440 volt, 60 Hertz, 3-Phase, 400 amp electrical service for the Asset for the entire availability period. Connect electrical service via the Asset's shore power receptacle using an Appleton AR40144 Female connector or equivalent.

1.7.5 Document the result of each inspection requirement throughout this Statement of Work.

1.7.6 The Contractor will operate existing, newly installed, or repaired shipboard equipment in the performance of the Contractor's work effort only while the caisson is not waterborne and is in dry-dock. Only navy shipyard personnel will operate shipboard equipment while the caisson is waterborne in the drydock or pier side. Navy shipyard personnel will operate all equipment during all tests and trials required by this statement of work. Equipment repaired or modified during this availability shall be operated in accordance with applicable technical manuals.

1.7.7 The Contractor shall provide alternate means for discharge and disposal of the ballast water and potable water, at no additional charge to the Government, if local regulations prohibit pier side discharge.

1.7.7.1 Tanks will be pumped to low suction prior to delivery to contractor.

1.7.8 Document each existing electrical cable hookup data prior to each removal or disconnection, for use during each installation. (REQUIRED REPORT CDRL A011).

1.7.8.1 Connect each new or existing cable or wire using retained electrical hook-up data.

1.7.8.2 Install new cable identification tags at each end of each cable and on both sides of each penetration. Each tag shall indicate cable source and cable designation. Installation shall be in accordance with MIL-STD-2003-3B(SH).

1.7.8.3 Conduct Insulation resistance measurements of each new or existing cable or wire using TABLE 3.10.1 for guidance. (REQUIRED REPORT CDRL A012)

1.7.8.4 Perform electrical continuity test on each new or existing cable or wire using TABLE 3, S9086-KC-STM-010/CH-300 and Note 11 for guidance. (REQUIRED REPORT CDRL A013)

1.7.8.5 Template exact sizes, locations, and configurations from existing Asset conditions and/or each item listed in paragraph 4.0.

1.7.9 Access to Asset for safety shall be maintained at all times, unless for berth or docking shift moves.

1.7.10 Provide photographic evidence of arrival conditions, to include hull, freeboard, weather deck, machinery deck and all tanks/voids. (REQUIRED REPORT CDRL A014)

1.7.10.1 Photographs shall be clear, to provide enough visual detail for overall assessment of condition.

1.7.10.1.1 QTY (4 EA) Max photographs per 8” x 11” sheet.

1.7.10.2 Provide QTY (50 EA) photographs upon arrival at contractor discretion to meet the requirements of 1.7.12.

1.7.10.3 Provide QTY (50 EA) photographs at SUPERVISOR direction.

1.7.11 Tanks and voids opened shall be clean, dry, and pass Government inspection prior to being closed.

1.7.12 Access to vessel. The vessel will be accessible by the Supervisor, and associates of the Government, or other prime contractors with the Government and their Subcontractors, and officers, employees, and associates of offerors on other contemplated work, admission to the Contractor's facilities and access to the vessel without any further request for indemnification from any party.

1.7.12.1 No person not known to be a U.S. citizen shall be eligible for access to naval vessels, work sites and adjacent areas when said vessels are under construction, conversion, overhaul, or repair. The Contractor shall establish procedures to comply with this requirement and NAVSEAINST 5510.2D.

1.8 **Care of Asset:** The Contractor shall be responsible for the care of Asset equipment, materials, and components related to the work being performed. The Contractor shall exercise reasonable care, as agreed upon with the Supervisor, to protect the vessel from fire, and shall maintain a system of inspection over the activities of its welders, burners, riveters, painters, pipe fitters, and similar workers, and of its subcontractors. Fire hoses shall be maintained by the Contractor, ready for immediate use on the asset at all times, and while the asset is berthed alongside the Contractor's pier or in dry dock. The Contractor shall maintain a fire watch aboard the vessel in areas where the Contractor is working.

1.8.1 Asset shall be maintained in a broom clean condition on a daily basis. Asset shall be free of visual contaminants such as oxide films, oils, grease, dust, dirt, or foreign material.

1.8.1.1 The Contractor shall at all times keep the site of the work on the asset free from accumulation of waste material or rubbish caused by its employees, or the work performed by the Contractor in accordance with this contract, and at the completion of such work shall remove all rubbish from and about the site of the work on a daily basis.

(V) "CLEANLINESS INSPECTION"

1.8.1.2 Weekly cleanliness inspection shall be accomplished, to ensure jobsites are maintained at 1.8.1 level of cleanliness. (REQUIRED REPORT CDRL A015)

1.8.2 Temporary protective measures shall be taken to avoid wear and damage to the Asset and all associated components. Water shall not be allowed to accumulate in the hull or structure during work performance.

1.8.3 Open areas in structure shall be covered to prevent water entry.

1.8.4 Install rat guards on each mooring line immediately upon mooring vessel. Each rat guard shall remain installed during 100 percent of moorage time.

1.8.5 While at the Contractor's facility and when the temperature becomes as low as thirty five degrees Fahrenheit, the Contractor shall assist the Government when requested in keeping all pipelines, fixtures, traps, tanks, and other receptacles on the asset drained to avoid damage from freezing, or if this is not practicable, the asset shall be kept heated to prevent such damage.

1.9 **Workmanship:** Workmanship provided by Contractor employees shall be consistent with the highest standards of commercial marine practices and regulatory bodies. The Government reserves the right to accept or reject material and workmanship nonconformance. Noncompliance with Contractor's quality control processes will influence past performance rating provided on contract evaluation. Plating and structural member cuts shall be neatly and accurately made with edges cleaned and prepared for welding. Sharp edges shall be ground smooth to avoid injury to personnel. Equipment, machinery and associated support systems shall not leak oil, water or other types of fluids. Machinery and equipment shall be protected against weather hazards.

1.9.1 At the time of delivery, work performed shall meet the performance requirements specified. Materials and articles used for work performed shall be of commercial marine quality, conforming to the requirements of the Regulatory Bodies listed herein, unless specified to be otherwise.

1.9.2 The decision that an area is inaccessible shall be determined by inspection and agreed to by the SUPERVISOR prior to proceeding with the requirements of this SOW.

1.10 **Corrective Action Request (CAR):** The Contractor is solely responsible for implementation of the Inspection and Quality Control System of 1.5.8. Written responses to contractor generated corrective actions will be provided to SUPERVISOR when requested. (Government observer(s) will monitor production work performed at any time where work is ongoing. If, during the performance of work, the Government observer(s) witness work that: fails to meet specifications within this SOW, would lead to an unsatisfactory end product, damage to Government Property, is a safety violation; the observer will alert the SUPERVISOR. Conditions that fail to meet specifications within this SOW may result in the Contractor being issued a CAR. Contractor must respond in writing to each issued Method B/C/D CAR within 3 working days unless otherwise specified by the SUPERVISOR. Response must include immediate corrective action taken, preventive action, preventive action for recurrence of identified nonconformance, root cause analysis (TABLE ROOT CAUSE ANALYSIS), and Objective Quality Evidence (OQE) for corrective action completed to included estimated completion dates. Contractor must inform the SUPERVISOR when corrective actions are complete for each issued Method A CAR. Response must state that the non-conformance has been corrected. When a contractor CAR response is unsatisfactory, revised response is required within 3 working days unless otherwise specified by the SUPERVISOR.

1.10.1 Method “A” CAR – Any minor nonconformance that can be corrected/adjusted within 2 working days of discovery or/and the remediation cost is less than or equal to $50,000.

1.10.2 Method “B” CAR – Nonconformance where repetitive Method “A” CARs of similar nature and circumstance have failed to result in a satisfactory product or a single nonconformance where corrective action cannot be achieved within three working days or remediation cost is greater than $50,000 but less than or equal to $500,000.

1.10.3 Method “C” CAR – A systemic nonconformance where multiple Method “B” CARs were issued for similar nonconformities, when a previous Method “B” CAR failed to result in a satisfactory product or a single nonconformance where remediation cost is greater than $500,000 but less than or equal to $1,000,000.

1.10.4 Method “D” CAR - When a Method “C” CAR was ineffective in obtaining satisfactory performance or the event was of such a nature that immediate top-level management attention is required to properly address and correct the nonconformance. Factors to consider are costs greater than $1,000,000, significant impact to scheduled completion, multiple areas of failure within the Quality Management System.

TABLE ROOT CAUSE ANAYLSIS

Root Causes	Category Description

• Personnel
    ○ Work Practices
    ○ Training
    ○ Supervision
• Procedures
• Design
    ○ Technical Documentation
• Material
	• Work Practices- Craftsman knows or understands the requirements, but fails to follow them
	• Training- Training of employees and subcontractors.
	• Supervision- Lack of preparation or follow through for the original planned event.
	• Procedures- Issues with procedures provided by outside agency or another activity utilized during the unplanned event.
	• Tech Documentation- Issues with DWGs, Specifications or Design aspect.
	• Material- Failure of material to perform under designed conditions and uses.

Category	Attribute
Work Practices	• Failure to follow Procedure • Use of incorrect or outdated Procedures • Inattention to detail • Improper tools/use of tools
Training	• Nonexistent training or qualification • Content is inadequate • Inadequate training or qualification frequency
Supervision	• Assignment of unqualified personnel • Inadequate direction provided • Inadequate review of worksite or Documents
Procedures	• Procedure contains inadequate or unclear direction • Procedure contains incorrect direction
Technical Documentation	• Error in Drawing or Technical Document • Design deficiency
Material	• Material Failure

1.11 <u>**Condition Found / Inspection Discrepancy Report (CFR/IDR):**</u>

Utilize FIGURE A (or Company correspondence containing same information) in transferable media, to communicate to the Contracting Officer SUPERVISOR concerning all conditions found, discrepancies noted, and other information requiring formal documentation. Provide the Contractor's recommendation for corrective action to correct the condition found including potential schedule impact. Contractor's estimated pricing (man-hours and materials) shall be provided separately to the Contracting Officer. Provide all inspection results including detailed measurements, photographs, sketches, and other means to accurately detail inspection results using FIGURE A, with attachments as necessary. The Government will review and respond to the submitted report. Additional repairs related to the found condition will be tasked if determined necessary by the Government. In the event difficulty is encountered in meeting requirements or difficulty is anticipated in complying with the contract schedule dates, immediately notify the Contracting Officer via verbal communication with the SUPERVISOR Government receipt of notification is not to be construed as a waiver of the requirements, delivery schedule, or waiver of rights or remedies provided by law or under this contract, or any other requirements in the SOW relating to jeopardy of contract schedule dates. Reports and notifications do not authorize the Contractor to perform additional work not covered by the contract or authorized by the Contracting Officer. (REQUIRED REPORT CDRL A016)

FIGURE A - CONDITION FOUND/INSPECTION DISCREPANCY REPORT

CFR Number________ **Date: ______________**

From: _________________________________ (Contractor Authorized Representative)
To: PSNS & IMF, NWRMC IND MAN, C400, Contracting Officer
Via: PSNS & IMF, NWRMC IND MAN, C101.13, SUPERVISOR

Asset **: Cassion 4/5 Spare**

Section of Contract Affected: ____

Description:

Impact:

Recommended Action:

C101 Response:

____ ___________________

Signature of C101.13 Representative **Date**

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1.12 **<u>Interferences:</u>** Contractor pricing offered shall include removal and re-installation of all interferences associated with performance of the required work. The fact that interference is not shown on a plan or specifically identified in the specification item is not justification for a contract change. A physical check of each work site prior to bid submittal is strongly encouraged.

1.13 **<u>Safety Requirements:</u>** Comply with the requirements of 29 CFR 1915, 29 CFR 1910, and 29 CFR 1926, state and local laws, rules, and ordinances during contract performance.

1.13.1 <u>General:</u> Contractor shall be responsible for the compliance of the above requirements by all subcontractor personnel performing work related to this contract. In cases of conflict between the above regulations, the stricter of the requirements shall take precedence.

1.13.2 <u>Accidents and Injuries:</u> Immediately report injuries to Contractor and subcontractor employees to the SUPERVISOR. Investigation shall be completed within 7 working days of each accident/injury. Document a report that includes narratives of the circumstances leading up to and including the injury and any actions (immediate and contemplated) taken to mitigate future reoccurrences of injuries. (REQUIRED REPORT CDRL A017)

1.13.3 <u>Scaffolding:</u> Scaffolds and ladders shall meet the requirements of 29 CFR 1915.71, and 29 CFR 1915.72.

1.13.3.1 Install a swinging gate guardrail at each scaffold ladder and through-platform access opening.

1.13.3.2 Install standard railing around the non-access sides and a swinging gate guardrail on the entrance side of through-platform access openings.

1.13.3.3 Install standard toe-boards around all scaffold floor openings, except at the entrance to the swinging gate guardrail, on all exposed sides of through-platform access openings.

1.13.3.4 Develop detailed engineered drawings and specifications for scaffolds when OSHA, ANSI, or manufacturer's designed guidelines do not exist, such as for "hanging scaffolding". Drawings and specifications shall be designed by a registered professional engineer and meet the following:

1.13.3.4.1 Maintain on the jobsite a copy of the detailed engineering drawings and specifications showing the sizes and spacing of members and the calculations developed for the engineered design(s).

1.13.3.4.2 Construct, load, and maintain scaffolding in accordance with the registered professional engineer's design.

1.13.3.5 Install a commercially available scaffold tag system on all scaffolds. The tagging system, as a minimum, shall be prominently attached to the scaffold structure, consist of green colored "Okay to Use" tag(s), red colored "Danger" tag(s), and accomplish the following:

1.13.3.5.1 Apply all tags on the scaffold structure as close as practicable to each ladder access and/or entry point.

1.13.3.5.2 Apply a red tag to indicate to users that the scaffold is being dismantled, is not yet completely erected, or for some reason is unsafe for use.

1.13.3.5.3 Apply a green tag to indicate to users that the scaffold is safe for use and is compliant with all OSHA regulations and other applicable requirements.

1.13.3.5.4 Ensure green tags contain the following information as a minimum: the location of the scaffolding structure; a unique reference number to identify each separate scaffolding structure, the date the scaffolding structure was first erected; the name(s) of the scaffold builder(s); the load rating of the scaffold structure; and the Competent Person's printed name and signature.

1.13.3.5.5 Green tags shall be checked and initialed for daily by a scaffolding competent Person.

1.13.3.6 Scaffolding shall be free standing and self-supporting, free from any attachment to any lifeline, handrail, antenna, safety equipment, electronic fixture, or Asset equipment.

1.13.3.7 Structures and Openings shall meet the requirements of 29 CFR 1915.73.

1.13.4 The following shall be accomplished no later than period of performance start +3 working days. Install temporary access scaffolding in the main ballast tank at the forward and aft ladders that comply with the requirements of 1.13.3. Closable access openings shall be placed at girder levels 2 and 3. A landing shall be constructed at girder level One that provides ladder access to the concrete level at the bottom of the caisson. Each closable access opening shall be capable of supporting 1000 pounds. Temporary access scaffolding shall remain in place for the duration of production. Temporary access scaffolding will not be removed until after dock trials are complete.

1.13.4.1 Install temporary closeable access openings in the forward and aft trim tanks that comply with the requirements of 1.13.3. Closable access openings shall be placed at each existing access opening at each girder level and at each breast hook level. Each closable access opening shall be capable of supporting 1000 pounds. Each closable access opening shall be secured to the surrounding caisson structure and shall remain in place for the duration of production. Temporary closeable access openings shall remain in place for the duration of production. Temporary closeable access openings will not be removed until after dock trials are complete.

1.13.4.2 Place temporary removable covers in each opening greater than 11 inches in diameter in the deck of each forward and aft trim tank breast hook and girder level. Each temporary removable cover shall be capable of supporting 1000 pounds. Temporary removable covers shall remain in place for the duration of production. Temporary covers will not be removed until after dock trials are complete.

1.14 **<u>Hot Work/Welding:</u>** Prior to performance of any hot work on the Asset, certify a safe atmosphere exists in and around the work area before starting any work that may produce the heat of ignition, sparks or flame. Accomplish steps to make each area safe for hot work. A copy of the gas free certification shall be posted in accordance with applicable OSHA requirements.

1.14.1 All welding equipment and materials shall comply with the applicable requirements of American National Standards Institute/American Welding Society (ANSI/AWS) requirements. Welding shall be sequenced to keep the thermal expansion of structures to a minimum.

1.14.2 Remove existing preservation / coatings prior to each expected welding, cutting, and hot work operation.

1.14.2.1 Preservation / coatings shall be removed 4 inches in each direction and each opposite side.

1.14.3 Perform visual inspection of new welds, to determine conformance, in accordance with the requirements of American Society for Testing and Materials (ASTM) International. All new welds shall be 100 percent visually inspected. All defects shall be repaired in accordance with approved procedures. All welders shall be certified in accordance with AWS requirements. When invoked by the requirements of section 3.0 of this SOW Document the following visual inspections.

(I) "FIT UP INSPECTION"

1.14.3.1 Accomplish fit up inspection in accordance with ANSI/AWS D1.1. (REQUIRED REPORT CDRL A018)

(I) "VISUAL INSPECTION"

1.14.3.2 Accomplish nondestructive testing visual inspection (VT) in accordance with ANSI/AWS D1.1. (REQUIRED REPORT CDRL A019)

1.14.4 Perform non-destructive testing of new welds, when invoked by the requirements of section 3.0 of this SOW, in accordance with the requirements of American Society for Testing and Materials (ASTM) International.

(I) "MAGNETIC PARTICLE TEST (MT)"

1.14.4.1 Accomplish nondestructive testing magnetic particle inspection (MT) in accordance with ASTM E-709-01. (REQUIRED REPORT CDRL A020)

1.14.5 Maintain documentation of each proposed access cut and a list of each proposed bolted/riveted access removal. Documentation is not required where access cuts are authorized per a NAVSEA approved drawing.

1.14.6 Develop a PCP with a Weld Procedure Specification (WPS) for each weld procedure and/or material to be utilized. Each WPS shall be in accordance with ANSI/AWS D1.1.

1.14.7 Chip and grind surfaces flush and smooth in way of removals.

1.15 **Threaded Fasteners:** Except as otherwise specified herein, or on the drawings listed, threaded fasteners shall be in accordance with applicable ASTM standards. Coarse thread series shall be used unless the component's design indicates a necessity to use fine thread series. Hole diameters for mounting bolts shall not exceed the diameters produced using standard size drill bits. Black oxide coated brass and cadmium plated fasteners are prohibited. Where drawings specify the use of brass fasteners on a fuel or fuel oil tank, CRES 316 fasteners shall be substituted in lieu of brass. Consult the requirements of MIL-STD-777 and Naval Ships' Technical Manual (NSTM) Chapter 075 for guidance in instances where there is a question as to the material to be utilized.

1.15.1 Threads on externally threaded fasteners, after being installed and tightened, shall protrude a minimum of 2 threads to a maximum of 5 threads beyond the face of the nut with the exception of minimum thread protrusion for self-locking (plastic insert) nut installations. For those installations, bolt or stud end minimum thread protrusion shall be flush with the face of the nut. For self-locking nuts, the bolt or stud end shall protrude a maximum of 5 threads beyond the face of the nut after the nut is tightened.

1.15.2 Clean, chase, and tap exposed threaded areas of each fastener and threaded socket.

1.15.3 Tighten threaded fasteners in the sequence, patterns, and the torque values, when not otherwise specified, using the requirements of Section 4, paragraph 075-4.1 through 075-4.6.2 of NSTM Chapter 075 for guidance.

1.16 **PRESERVATION**:

1.16.1 Measure, record, and maintain Objective Quality Evidence (OQE) information on Appendices One through 8 throughout the entire performance of preservation operations. OQE information for the in-process being performed must be made immediately available for onsite review upon request by the SUPERVISOR. (REQUIRED REPORT CDRL A021)

1.16.1.1 Appendices One through 8 shall be in accordance with Reference QA_Appendix_Forms_C411_2023_Contractors.pdf.

1.16.1.1.1 All spaces that are not applicable shall be marked N/A.

1.16.1.1.2 Unused sections shall be crossed out and marked N/A.

1.16.1.1.3 Any errors shall be one lined through; initialed and dated by the person making the correction.

1.16.1.1.4 Any markings that are illegible or unclear shall be considered invalid.

1.16.2 Comply with the coating manufacturer's requirements for the establishment and maintenance of environmental conditions and for coating thickness applications throughout the entire removal and preservation operation(s).

1.16.2.1 Calibrated Electronic Data Logger record and maintain environmental conditions OQE during the entire preservation operations. Data shall be made immediately available for onsite review upon request by the SUPERVISOR. (REQUIRED REPORT CDRL A022)

1.16.3 Apply contrasting colors between the primer, stripe, intermediate, and topcoat using coating products that are from the same manufacturer and compatible with each other. Other system preservation coatings shall be different in color from both the preservation coating over which it is being applied and the next coating in the system; no two coatings can be similar in color and shall be applied in accordance with the manufacturer's product data sheet.

1.16.4 Apply stripe coat to all edges, weld seams, welds of attachments and appendages, cutouts, corners, butts, foot/handholds, including back side of piping, underside of beams and other mounting hardware (non-flat surface) after each previous full coat has dried. The stripe coat shall encompass and extend a minimum of One-inch beyond the intersection of two flat surface planes, each radii edge, and all perpendicular shapes, edges, or angles.

1.16.5 Mask and/or install temporary expandable foreign material exclusion (FME) plugs, blanks, or other hard and durable protective covers, painted blaze orange, in hull openings and surfaces to prevent system contamination or equipment damage from abrasive blasting, surface preparation, and preservation operations in accordance with 1.28.

1.16.6 Solvent clean surface prior to the start of each blasting, and preservation operation. Accomplish the requirements of Surface Preparation

Specification SSPC-SP-1 of Systems and Specifications, SSPC Painting Manual, Volume 2.

1.16.7 Accomplish touch-up preservation for new and disturbed surfaces shall match surrounding coating system(s) in accordance with the manufacturer's product data sheets.

1.16.8 Feather edges of well adhered paint remaining after cleaning for all surface preparation methods. Visible areas of defective existing paint shall be removed until an area of completely intact and adhering paint is attained around the defective area by feathering (tapering) the edges of tightly adhering existing paint at an approximate 30-degree slope into the newly prepared bare metal surface thus preventing application of new paint over loose or cracked paint.

1.16.9 Paints shall not be thinned.

1.16.10 Hull and freeboard preservation coatings shall be fully cured prior to undocking. All other preservation coatings shall be fully cured prior to final delivery to the Government.

1.16.10.1 Preservation coatings are considered fully cured when they meet the requirements of the applicable manufacturer's product data sheets minimum cure to service time schedule and are dry throughout their thicknesses.

1.16.11 Conduct Surface Chloride or Surface Conductivity measurements.

1.16.11.1 Surface Chloride; shall not exceed 3 ug/cm^2 (30 mg/m^2) for immersed or 5 ug/cm^2 (50 mg/m^2) non-immersed applications. Surface Conductivity shall not exceed 30 micro-siemens/cm immersed or 70 micro-siemens/cm non-immersed applications.

1.16.11.2 Readings shall be documented on Appendices 4 of 1.16.1.1.

1.17 **<u>Environmental and Hazardous Waste Management</u>:** Comply with all federal, state, and local laws, codes, regulations, acts, ordinances, and rules for Environmental Protection and Hazardous Waste Management / Disposal, including obtaining required licenses, generator identification number, and permits. Designate one employee responsible for Environmental and Hazardous Waste Management. All contract, manifests, invoices, and other documents related to the removal, storage, handling, transportation or disposal of hazardous waste shall bear a generator identification number issued to the Contractor pursuant to applicable law. Take actions to mitigate creation of hazardous waste by recycling and other available methods. All hazardous waste generated by work under this contract is anticipated to be Contractor generated and shall be the responsibility of the Contractor. Estimated hazardous waste associated with work required in these Specifications is listed below:

Product		**Amount**
Cleaning Solvents		150 gallons
Paint (enamel, latex, epoxy, thinners, oil based, rubber, non-skid, lacquer, varnishes, removers)		650 gallons
Rags (contaminated with a known source)		1,000 pounds
Blast Media and paint chips contaminated with heavy metals and other contaminants		163 tons
Oil (Synthetic)		20 gallons
Sludge (Contaminated with a known hazardous waste)		300 gallons

Hazardous waste generated during the actual performance of the work may vary in type or amount from the waste listed above. The above is an estimate of the amount of waste anticipated to be generated; not an estimate of the amount of work in generating that waste. (REQUIRED REPORT CDRL A023)

1.17.1 The report shall include analysis or other method used to identify the waste and state whether each listed waste was hazardous (with generator assignment), non-hazardous, or did not exist.

1.17.2 The Contractor shall ensure that their cost proposal for the work proposed herein includes reasonable compensation for all anticipated cost of managing and disposing of the wastes listed above, and/or created or generated in the execution of the delivery order issued, in accordance with the requirements of 42 USC 6901-6992 Rev 1996, 49 USC 5103, 10 USC 7311 Rev 1/22/11. The Contractor shall include a 5-way break down of the anticipated costs associated with this effort in their offer.

1.17.3 Consider coatings to contain heavy metals (e.g. lead, chromium, cadmium), hexavalent chromium, crystalline silica, and/or other toxic or hazardous substances until laboratory analysis establishes otherwise. Comply with all procedures and precautions required for employee protection, sampling, removal, and disposal.

1.17.4 Consider insulation, lagging, and mastics to be asbestos-containing material (ACM) until it can be established by laboratory analysis, or other reliable method(s), that the material does not contain asbestos. Ensure compliance with the requirements of 29 CFR 1915.1001.

1.17.5 Consider materials, debris, and/or residue associated with ventilation systems to contain toxic/hazardous substances including but not limited to cadmium, chromium, and PCB's until laboratory analysis establishes otherwise.

1.18 **Materials:** All Contractor furnished materials (CFM) installed shall be new. Materials shall be as indicated on the listed drawings, publications or as specified in this Specification. The use of asbestos, polychlorinated biphenyl (PCB), magnesium, cadmium, or mercury in any materials is prohibited.

1.18.1 Procure/Order 100 percent of materials within 5 working days of contract award, to support the production schedule, milestones of 1.29, and delivery of the Asset within the contracted period of performance.

1.18.2 Provide initial CFM status report at Award + 5 working days and update weekly. Document 100 percent of CFM that is required to accomplish the requirements of this SOW. (REQUIRED REPORT CDRL A024)

1.19 **Weight Control:** Institute a weight control program in which actual weights, for removed, relocated and installed equipment/structure, are measured by either weighing or from manufacturer certified data. The location of the center of the load must be recorded along with each weight entry in the weight control report. This location shall show the distance vertically from the baseline of the Asset, longitudinally from amidships and tangentially from the centerline of the Asset. Items that are permanently removed and replaced with the identical item, in the same location, are not required to be included in the weight control program. Items that are replaced with like items of differing material or quantity shall be included in the weight control program. Items that are new installations shall be included in the weight program. (REQUIRED REPORT CDRL A025)

1.20 **Label Plates:** Label plates installed in locations exposed to the weather shall be CRES and shall be prepared by the "Metal-Photo" process and sealed to exclude moisture between the plate and the mounting surface. Label plates shall be installed to assure maximum visibility. Label plates shall be attached with corrosion-resistant steel (CRES) screws or adhesive, using drawing S2803-980209 Rev L for guidance. Where adhesive is used, surface preparation shall be in strict accordance with the adhesive manufacturer's instructions. Test label plates shall be attached on or adjacent to all equipment or components tested without interference to equipment or component operation. The label shall identify the equipment or component, the test load, the dynamic and static load (if applicable), the rated load, the date of testing, and the activity performing the test, using the layout format of drawing S2803-980209 Rev L for guidance.

1.21 **Integrated Logistic Support:** Maintain legible original copies in approved transferable media, of the following original equipment manufacturer's (OEM) information: operating instructions, manuals, drawings, catalog cut sheets, warranty information, and all other supporting documents, for all new equipment, parts, and supplies installed. (REQUIRED REPORT CDRL A026)

1.22 **Gas Free:** Provide certified Marine Chemist, Certified Industrial Hygienist services, and Shipyard Competent Person services as required by 29 CFR 1915 for Contractor and Government Personnel. (REQUIRED REPORT CDRL A027)

1.22.1 Consolidate and maintain the Contractor's current confined space entry/gas free certification processes into a single and comprehensive document as part of a confined space/gas free certification program.

1.22.2 If rescue plan utilizes a third party, provide documentation of written agreement from third party. (REQUIRED REPORT CDRL A028)

1.22.3 The list shall of 1.22 be available for Government review within one hour of request.

1.22.4 Maintain a list of Contractor's National Fire Protection Association (NFPA) Marine Chemist certifying gas-free conditions for this project and a list identifying Contractor employees designated as "Competent Persons" in accordance with 29 CFR 1915.7. (REQUIRED REPORT CDRL A029)

1.22.4.1 Post, update, and maintain gas free certificates in accordance with 29 CFR 1915 during the entire contract period of performance.

1.23 **General Requirements for Testing:** Test all Contractor affected equipment, systems and subsystems for compliance with this SOW and Commercial Marine standards and practices. Demonstrate workmanship, alignment, strength, rigidity, water tightness, functionality, and suitability for the purpose intended.

1.23.1 Develop a Test and Inspection Plan for each Test and Inspection required by zero and first tier references, each symbol (I) (V) (Q) (G) test/inspection and by this SOW, containing paragraph number or listed applicable reference requiring Tests and/or Inspection; identification of each component/system; description of each Test and/or Inspection; accept/reject criteria; date scheduled; Government notifications; date completed; and each Test and/or Inspection result Tests and inspections the contractor deems necessary to substantiate product conformance shall be included. (REQUIRED REPORT CDRL A030)

1.23.1.1 Fully integrate the Test and Inspection Plan with the Contractor's production schedule.

1.23.1.2 Provide the number of tests for a given system, component, or installation required to demonstrate required satisfactory performance. This section specifies tests to be performed in support of the work performed.

1.23.1.3 Be maintained at a contractor location.

1.23.2 Provide a detailed description of preoperational checks to be conducted, operations to be performed, and the parameters to be satisfied. Develop data sheets which provide spaces for recording the quantitative values measured during the test, for each test procedure. Specify the minimum and maximum allowable tolerance limits for each measured value. Develop signature sheets for fill-in by the person conducting the test, the Government witness present, and the date of the test. Block diagrams, simplified schematics or diagrams may be used to clarify the procedure or test method. Include a comment sheet to record significant events and observations by either the Government witness or Contractor.

1.23.3 Provide materials, supplies, utilities, equipment, calibrated equipment/ instrumentation, services and personnel to accomplish each test and/or inspection. Incorporate safety precautions into each test and/or inspection procedure. Prepare all test and/or inspection documentation during the development and implementation of the acceptance test.

1.23.3.1 Whenever Government support is required for the operation of Asset equipment and/or systems, a minimum of 3 workday's prior notification of the operational/test support requirement shall be provided to the SUPERVISOR.

1.23.3.2 Tests/inspections and equipment/systems operation requiring Government and/or Ship's Force operational support shall be scheduled for accomplishment during normal day shift working hours, as much as practicable.

1.23.3.3 Provide a minimum of 3 working days prior notification to the SUPERVISOR for tests/inspections and equipment/system operations requiring Government operational support for work being performed on weekends, after normal dayshift working hours, and immediately before and/or after Federal holidays.

1.23.3.4 Notify the SUPERVISOR a minimum of 3 working days prior to each (I), (V), (Q), or (G)-test/inspection. Equipment shall not be adjusted during acceptance testing. If SUPERVISOR is present during test/inspection, Supervisor will annotate "Concur" or "Do Not Concur" on the applicable test record, and countersign. The SUPERVISOR's signature validates the inspection was performed correctly and the results were documented as specified by the contract.

1.23.3.5 Gage range shall be such that test pressure is in the middle third of each scale.

1.23.4 Perform tests/inspections using the applicable test procedure for systems and subsystems. Demonstrate compliance with the Specification requirements by satisfactory completion of the test. Complete all work prior to initial testing of that particular system or subsystem and in a timeframe sufficient to allow for correction of deficiencies prior to dock trials, sea trials, or other applicable milestones established in these specifications. Perform adjustments and testing prior to each scheduled acceptance test.

1.23.4.1 Stop the test/inspections if the equipment, system, or subsystem tested fails to satisfy the acceptance criteria of the established test procedure. Identify the cause and correct.

1.23.5 Document the result of each test/inspection requirement throughout this SOW.

1.23.6 Accomplish (G)-Point (government notification) as follows:

1.23.6.1 (G) is a symbol inserted in a SOW paragraph to establish a point in the sequence of accomplishment of work at which time the SUPERVISOR must be notified by the prime contractor in all cases to permit observation of a specific test or inspection (I) (V) by the government. When the symbol (G) precedes tests or inspections in a SOW paragraph which are applicable to more than one action, the symbol (G) must identify the action required, e.g., (G) "HYDROSTATIC TEST". When more than one unit is involved, the (G) notification requirement applies to each unit.

1.23.6.2 For tests and inspections involving (I), (V), (Q), (G)-points, records must be documented upon acceptance or rejection and electronic copy provided to the SUPERVISOR within 24 hours of conclusion of each (G)-Point. (REQUIRED REPORT CDRL A031)

1.23.6.3 Government will consider the work in accordance with SOW paragraph incomplete if the contractor's documentation and records are not complete.

1.23.6.4 Proceed with the test or inspection if the Government Representative is not present, provided the required advance notice has been furnished to the SUPERVISOR and the contractor has completed and documented the preceding tests and inspections.

1.23.6.5 A partial test or inspection requiring (G) notification may be accomplished in the event that all work cannot be completed and work progress would be delayed in waiting for total completion of work. Comply with the requirements of 1.23.6 when the incomplete work is completed and ready for the remainder of the test or inspection. Note partial inspections on the test or inspection form.

1.23.6.6 A qualified contractor representative must be present to accomplish, accept or reject and document tests or inspections associated with the symbol (G).

1.23.7 Accomplish (I) (V) and (Q) checkpoints as follows:

1.23.7.1 Accomplish (I), (V) and (Q) tests/inspections that do not have associated (G)-points, with qualified and/or currently certified personnel where required by the technical documents (e.g., NBPI, NACE, nondestructive testing, electrical cableway inspection, Oxygen Cleanliness Inspector, etc.)

1.23.7.2 (I) inspections require verification and documentation by a separate individual, other than the person who has accomplished the work, who is qualified as an inspector.

1.23.7.3 (V) inspections require verification and documentation by the qualified tradesperson, trade supervisor, or inspector.

1.23.7.4 (Q) inspections require verification and documentation by a qualified Technical Representative in accordance with 1.24 and associated PCP requirements.

1.23.7.5 The authority to accomplish, document, accept and reject (I) and (V) inspections may be delegated to qualified subcontractor personnel, without regards to geographical location, subject to SUPERVISOR approval.

1.23.7.6 The authority to witness or perform, document and accept/reject (I)(G), (Q)(G), and (V)(G) tests and inspections is a prime contractor's responsibility but, subject to SUPERVISOR approval within a 50-mile radius of the contractor’s plant nearest to the place of performance of the contract, may be delegated to subcontractors who are MSRA or ABR agreement holders, SSPC QP1 certified, NDT certified, or have a current QMS accepted by the SUPERVISOR.

1.23.7.7 The contractor may delegate responsibility to subcontractors to perform, document and accept/reject (I)(G) and (V)(G) tests and inspections performed at plants located outside a 50-mile radius of the contractor's plant nearest to the place of performance of the contract subject to SUPERVISOR prior approval.

1.23.7.7.1 For work being performed outside a 50-mile radius of the place of contract performance, the prime contractor must submit via approved transferrable media, of purchase orders to the SUPERVISOR within 2 working days or otherwise as directed by the Government, prior to issue of purchase order and shipment of equipment. For contractors who do not utilize purchase orders as a vehicle for accomplishing work within their company, a report identifying the delineation of the specific SOW requirements, in lieu of the purchase order must be submitted to the SUPERVISOR.

1.24 **<u>Process Control Procedure (PCP) Requirements</u>**: Develop a PCP ***when directed by the work requirements of 3.0***. Work within the submitted PCP shall not commence until a Government review of the PCP has been completed. Format in accordance with the requirements of paragraphs 1.24.1 through 1.24.4. (REQUIRED REPORT CDRL A032)

1.24.1 Section 1 - Identification:

1.24.1.1 Include the process title and procedure number, revision, and date developed.

1.24.1.2 List the work item title and paragraph that the PCP fulfills.

1.24.1.3 Include Contractor/Subcontractor names and addresses.

1.24.1.4 Include space for the Approval Signature and title of the Contractor's representative, date developed, date of submission, and scheduled start date of the PCP.

1.24.2 <u>Section 2 - Personnel Qualifications:</u>

1.24.2.1 List the qualification requirements for the personnel performing the work.

1.24.2.2 Include a statement that a briefing will be conducted prior to beginning work to ensure personnel have direct knowledge of the requirements of the procedure and the safety requirements of the job.

1.24.3 <u>Section 3 - Process Description:</u>

1.24.3.1 List any specialized or critical equipment needed to perform the work.

1.24.3.2 List any specialized or critical personal protective safety equipment required to be worn by personnel.

1.24.3.3 Identify Government notification points for the start and during the Contractor identified inspection points during the accomplishment of the PCP.

1.24.3.4 Describe the PCP as related to the sequence of work including critical factors which have a direct bearing on the process quality and safety.

1.24.3.5 List the acceptance and rejection criteria used for determining satisfactory PCP completion.

1.24.3.6 Inspection and data document forms which will be utilized as OQE to record and ensure satisfactory completion of the PCP.

1.24.3.7 The method to ensure an up-to-date copy of the PCP, are at the worksite during the performance of the work.

1.24.3.8 The method utilized to control the PCP.

1.24.4 <u>Section 4 - Hazardous Material:</u>

1.24.4.1 State if no hazardous material/waste will be used or generated.

1.24.4.2 Identification of hazardous materials which will be used in the PCP and hazardous waste that will be generated by the accomplishment of the PCP. Include a SDS for each hazardous material that will be used aboard the Asset.

1.24.4.3 Include the methodology which will be utilized to minimize the quantity of hazardous materials which will require control and disposal.

1.24.4.4 Description of the methods and safeguards to be utilized for the capture, containment, control, and disposal of hazardous material or hazardous.

1.25 **Compartment Arrival / Closeout Inspections**:

1.25.1 Accomplish an inspection of 100 percent of each Asset compartment, identifying damage, deterioration, missing components, discrepancies, and areas of concern prior to the start of work on the Asset.

1.25.1.1 Notify the SUPERVISOR no later than 3 working days prior to conducting the inspection.

1.25.2 Develop a compartment closeout schedule no later than the 75 percent completion point of the Asset availability schedule. (REQUIRED REPORT CDRL A033)

1.25.2.1 Closeout schedule shall list each compartment on the Asset by compartment name and number.

1.25.2.2 Closeout schedule shall identify work completed, work in progress, work remaining to be accomplished, and expected completion dates.

(V) FINAL CLOSEOUT INSPECTION

1.25.3 Accomplish a closeout inspection upon completion of production work and prior to Asset delivery; with the SUPERVISOR in each Asset compartment to include paragraph 3.2 underwater hull, 3.4 tanks, 3.5 weather deck and 3.6 operations deck. (REQUIRED REPORT CDRL A034)

1.25.3.1 Notify the SUPERVISOR no later than 3 working days prior to conducting the inspection.

1.26 **Technical Representative:** Provide the services of a qualified on-site Technical Representative to provide assistance in the process or processes, repair and testing of the equipment specified in the invoking SOW paragraphs. Technical Representative will only be contracted by the prime contractor and the same company cannot provide both the Technical Representative and perform the actual production work.

1.26.1 Notify the SUPERVISOR of the prime contractor's identifying the OEM authorized service provider. Provide certification from the OEM to the SUPERVISOR that the vendor is an OEM-authorized service provider. (REQUIRED REPORT CDRL A035)

1.26.2 The Technical Representative must meet the following minimum qualification requirements:

1.26.2.1 Have technical knowledge of the specified equipment or process and have a documented history of successful performance or repairs on similar equipment or processes.

1.26.2.2 Have demonstrated competency in analyzing repair requirements and process performance and making recommendations based on process or disassembly inspection results.

1.26.2.3 Have current/active documented and verified access to Original Equipment Manufacturer (OEM) proprietary plans, specifications, procedures, material, parts, special tools, and equipment.

1.26.2.4 Provide certification via approved transferable media, from the OEM to the SUPERVISOR that the individual is an authorized service provider.

1.26.3 The Technical Representative must review, sign, and date all required reports, for technical adequacy prior to submittal to the SUPERVISOR. Associated Process Control Procedures must be reviewed, signed, and dated by the Technical Representative prior to the start of work.

1.26.4 Inspect new and repaired areas and component parts of the equipment prior to assembly to ensure compliance with technical manual requirements and drawings.

1.26.4.1 Any deviations or departure from the specifications and/or the requirements of 1.26.4 require an approval from the SUPERVISOR prior to equipment assembly.

1.26.5 Inspect and provide technical guidance and assistance during process performance, equipment assembly and adjustment. Verify assembly procedures, sizes, and clearances comply with manufacturer's requirements, technical manual requirements, and drawings when specified. (REQUIRED REPORT CDRL A036)

1.26.5.1 Verify and document mechanical and electrical alignments, final closing sizes, and clearances.

1.27 **Heavy Weather Plan:** Maintain a written plan that must be implemented during gales, storms, hurricanes, and destructive weather, including mooring calculations in accordance with 845-6686999 US Navy Vessel Water Depth, Mooring and Hull/Appendage/Clearance Requirements for Transit and Berthing and DDS 582-1, Design Data Sheet, Calculations for Mooring Systems, using UFC 4-159-03, Mooring Design for guidance. The documented Heavy Weather Plan must be submitted for a document review and acceptance. The Contractor shall have an acceptable documented Heavy Weather Plan in place no later than 15 working days prior to availability start date. The Heavy Weather Plan must be available for periodic conformity audits throughout the contract. (REQUIRED REPORT CDRL A037)

1.27.1 Ensure the plan designates responsibility and implements procedures for prevention of damage to the vessel. This includes periods when the vessel is physically located in private contractors' plants; during times when work on the vessel requires openings to hulls or decks; and when contractor owned/furnished floating equipment is tied alongside the vessel. The plan must contain specific responsibilities and detailed actions to be taken during the weather conditions listed below.

1.27.1.1 Conditions where there is substantial advance warning for approaching adverse weather are addressed by the following 4 categories:

1.27.1.1.1 Gale/Storm/Hurricane Condition IV: Trend indicates a possible threat of destructive winds of force indicated within 72 hours.

1.27.1.1.2 Gale/Storm/Hurricane Condition III: Destructive winds of force indicated are possible within 48 hours.

1.27.1.1.3 Gale/Storm/Hurricane Condition II: Destructive winds of force indicated are anticipated within 24 hours.

1.27.1.1.4 Gale/Storm/Hurricane Condition I: Destructive winds of force indicated are anticipated within 12 hours or less.

1.27.1.2 Conditions where there is little or no advance warning for approaching adverse weather are addressed by the following 2 categories:

1.27.1.2.1 Thunderstorm/Tornado Condition II: Destructive winds accompanying the phenomenon indicated are reported or expected in the general area within 6 hours. Lightning and thunder are also anticipated.

1.27.1.2.2 Thunderstorm/Tornado Condition I: Destructive winds accompanying the phenomenon are imminent. Lightning and thunder are also anticipated.

1.27.1.3 Ensure that the plan contains, as a minimum, the following information as dictated by conditions listed in 1.27.1:

1.27.1.3.1 Steps to be taken to remove or secure staging items or equipment on decks of the vessel, pier or dry dock, including cranes that could become windborne.

1.27.1.3.2 Protection of the vessel from damage from other floating equipment.

1.27.1.3.3 Provisions for protection of government equipment and material in custody of the Contractor from damage by pier side flooding.

1.27.1.3.4 Provisions for removal of temporary hoses, welding lines, air lines, oxygen/acetylene lines, etc., extending through watertight closures.

1.27.1.3.5 Provisions for security, emergency fire and flooding protection, emergency shipboard dewatering and fire main capability, emergency shipboard electrical generation, and emergency shipboard communications.

1.27.1.3.6 One portable dewatering pump and associated equipment for each 100 feet of ship’s length in addition to the dewatering equipment provided by 845-6686999 US Navy Vessel Water Depth, Mooring and Hull/Appendage/Clearance Requirements for Transit and Berthing must be available on the damage control or main deck adjacent to damage control lockers or temporary damage control equipment boxes. Pumps must be capable of providing a minimum of 200 gal/min at a discharge head of 50 feet of dewatering capacity and can be used at the scene of a casualty within 3 minutes of receiving an alarm. Additional dewatering capacity to provide 1,000 gal/min at a discharge head of 50 feet at the scene must be available within 15 minutes. During the waterborne overhaul period, no damage control system associated with flooding prevention and control, or any portion thereof must be removed or made inoperable without prior notification of the COR and to the casualty-control station and until a back-up system has been established.

1.27.1.4 Ensure provisions for access to the ship for personnel and emergency equipment during and immediately following the storm consistent with prudent safety precautions. Assurance that all hull/deck openings are made watertight. Steps to be taken to secure floating piers during high winds/high tides.

1.27.1.5 The name and telephone number (business and residential) of the Contractor's single point of contact. This person must have the authority to commit the Contractor to take necessary actions as requested by the COR.

1.27.2 Ensure the plan contains the following mooring related information:

1.27.2.1 Specify steps to be taken to secure the vessel to contractor's pier, dry dock, graving dock, marine railway, or Contractor's other facility. Information must define specific precautions to be taken and supporting calculations, to include limits of docking blocks and dock stability for both normal and heavy weather conditions. Calculations for heavy weather configurations must include wind and tidal considerations.

1.27.2.2 Provide the heavy weather state at which the vessel must be undocked.

1.27.2.3 Submit mooring calculations for the worst anticipated loading condition during the availability. For vessels with a self-compensating fuel system, the loading condition must show the self-compensation fuel system full of water, fuel, or some combination of fuel and water, projecting the worse possible condition as shown in calculations for maintaining ship’s stability. Calculations may require re-submittal if significant changes occur from the original estimate on which the calculations were based.

1.27.2.4 For ships in dry dock, provide limits and supporting calculations for listed conditions. Analyze both the "normal" dock configuration and the "heavy weather" configuration.

1.27.2.4.1 Maximum safe wind speed and surge for side block strength and stability. Include maximum loading of the side blocks on the vessel.

1.27.2.4.2 Maximum safe wind speed and surge for dry dock strength and stability.

1.27.2.4.3 Surge required to float the vessel.

1.27.2.4.4 Table or graph showing safe combinations of wind speed and surge.

1.27.2.5 For vessels pier side, provide limits and supporting calculations for vessel loading conditions specified in 1.27.2.3. Analyze the "heavy weather" mooring configuration that would be used during the conditions specified in 1.27.1. Analyze worst-case wind directions including frontal, broadside, and quartering.

1.27.2.5.1 Maximum safe wind speed for mooring strength. Include strength of pier, pier fittings, mooring lines, and shipboard fittings. Maximum applied load on any mooring line must be the breaking strength of the mooring line divided by 2.5 (factor of safety of 2.5).

1.27.2.5.2 Maximum safe surge for mooring.

1.27.2.5.3 Maximum safe elongation of mooring lines. Include the following information: Size and type of mooring line; Percent elongation of mooring line at failure; Tattletale-free length and length between attachments.

1.27.2.5.4 Sketch, showing size, type, and location (vertical and horizontal angles) of all securing devices including fenders, bumpers, and camels.

1.27.2.6 Include the following statement, providing the necessary data: Vessel __________ can be safely moored to withstand a maximum of ___ mph winds with a __ knot current and a __ foot storm surge.

1.28 **<u>Foreign Material Exclusion (FME):</u>** Install and maintain temporary expandable FME plugs, blanks, or other hard and durable protective covers, that are painted blaze orange to maintain system cleanliness from environmental dangers.

1.28.1 Install and maintain blanks/plugs, studs, nuts and bolts, painted blaze orange for use foreign material exclusion (FME) immediately upon openings in equipment, valves, and piping systems not subject to pressure to prevent entry of foreign material and protect flanges and threaded areas.

1.28.1.1 Existing system fasteners used for blanking that will be used for installation are excluded from the requirement for blaze orange color.

1.28.1.2 FME may be used for systems normally under pressure but are tagged out for maintenance.

1.28.1.3 Wood plugs, wood, or wood products are prohibited for use in tanks/voids, and as blanks on pressurized systems.

1.28.1.4 Wood products, including plugs, are permitted for use as FME external to the vessel for hull penetrations, and non-pressurized systems to include gravity drain piping.

1.28.1.5 The use of cloth, polyvinyl sheet, paper, tape and rubber sheeting, as FME prevention devices is prohibited.

1.28.2 FME shall be install in hull openings and surfaces to prevent system contamination or equipment damage from abrasive blasting, surface preparation, and preservation operations.

1.28.3 Establish an FME log, record, and identify the location of each protective covering installed and removed, including each corresponding and specific measurement(s), location(s), and/or area(s) in relationship to structural members or from suitable hull reference points. (REQUIRED REPORT CDRL A038)

1.28.3.1 Remove each temporary FME plug, blank, and protective covering, record removal date in FME log, and restore each system to its original configuration upon final completion of testing, blasting, and preservation operations.

1.29 **Milestone Requirements:** The Contractor shall meet the following Contract Production Milestones:

1.29.1 Production Start – means the first day of the period of performance (POP).

1.29.2 All inspections reports shall be submitted no later than the first 20 percent of the scheduled POP. (**Production Start +37 calendar days**). Only the inspection reports of 1.29.4, and 1.29.6, exempted from this requirement.

1.29.3 **No later than Production Start + 25 calendar days:** All dewatering pumps, valves, motor valve operators and valve operating shafts removed from caisson.

1.29.4 **No later than Production Start + 60 calendar days:** Exterior hull, port rubber seal mating surface and weather deck SSPC-SP-6 and grit removal complete. Submission of all CDRLs related to steel visual inspections, ultrasonic inspections, and steel repair completion inspections for the exterior hull, port rubber seal mating surface and the weather deck submitted to the SUPERVISOR.

1.29.5 **No later than Production Start + 70 calendar days:** Government answers inspection CDRLs for exterior hull, port rubber seal mating surface and weather deck.

1.29.6 **No later than Production Start + 80 calendar days:** All tanks and piping SSPC-SP-6 and grit removal complete. Submission of all CDRLs related to steel visual inspections, ultrasonic inspections, and steel repair completion inspections for all tanks and piping submitted to the SUPERVISOR.

1.29.7 **No later than Production Start + 90 calendar days:** Government answers inspection CDRLs for all tanks, and piping.

1.29.8 **No later than Production Start +100 calendar days:** Exterior hull, port rubber seal mating surface and weather deck weld repairs completed.

1.29.9 **No later than Production Start +120 calendar days:** Tank and piping weld repairs complete.

<u>1.30 Work Required to be Accomplished in Drydock:</u> The following work items in Table A shall be accomplished while in dry-dock:

TABLE A WORK REQUIRED TO BE ACCOMPLISHED IN DRYDOCK	
Paragraph	**Description**
3.1	Docking and Undocking the Vessel
3.2	Hull Repairs
3.3	Structural Repairs

3.4	Tank/Voids with the exception of 3.4.1 (Open, Pump, Clean, Ventilate, and Gas Free Tanks), 3.4.2 (Identify Color Schemes and Markings) and 3.4.19 (Tank Close Out Inspection)
3.6.5.2	Structural Repair Completion: Operations Deck
3.11.7	Pump and Motor Removals WITH AN EXEPTION IN ACCORDANCE WITH 1.30.1
3.11.7.6	Install Dewatering Motor and Pump Assembly WITH AN EXEPTION IN ACCORDANCE WITH 1.30.1
3.11.7.11	Dewatering Pump Operational Test
3.11.1.2	Piping Visual Inspection
3.11.1.3	Piping Inspection
3.11.2	Remove existing valves, fasteners and gaskets and replace with new valves WITH AN EXEPTION IN ACCORDANCE WITH 1.30.1
3.11.2.5.1	Install new stainless-steel CRES 316 fasteners WITH AN EXEPTION IN ACCORDANCE WITH 1.30.1
3.11.2.5.5	Modify each existing, pipe, piping flange, spool piece flange, valve operating shaft, valve operating shaft coupling, valve stem yoke. WITH AN EXEPTION IN ACCORDANCE WITH 1.30.1
3.11.2.5.6	Align and install each valve without strain WITH AN EXEPTION IN ACCORDANCE WITH 1.30.1
3.11.2.5.7	Install new gaskets between each disturbed flange WITH AN EXEPTION IN ACCORDANCE WITH 1.30.1
3.11.2.5.8	Torque Valve Fasteners WITH AN EXEPTION IN ACCORDANCE WITH 1.30.1
3.11.2.6	Valve Operational Dry Test
3.11.2.7	Valve Seat Tightness Test
3.11.2.8	Valve Operational Wet Test
3.11.5.5	Reach Rod Shaft Operational Testing
3.11.9.7 3.11.10.4	Set Actuator Limits
3.13.1.2	In Dry Dock Baseline Testing
3.13.1.3	In Dry Dock Final Trial

1.30.1 Exceptions listed in TABLE A may be worked pier side and outside of the drydock in accordance with the following.

1.30.1.1 Valve and dewatering pump over boards shall be blanked on the seaside flange of each valve or dewatering pump overboard. Valve and dewatering pumps shall only be removed one at a time. Slight leak by of the seaside blank is acceptable so long as streams, jets or a heavy flow or leak by of water is not observed. Water dripping by is acceptable.

1.30.1.2 After the associated valve or dewatering pump has been removed, the tank side flange shall be blanked so that each valve or dewatering pump intake or overboard is double protected to the sea. Slight leak by of the tank side blank is acceptable so long as streams, jets or a heavy flow or leak by of water is not observed.

1.30.1.3 A 1/8" thick minimum thickness gasket shall be installed between each disturbed flange and blank meeting the performance characteristics of Garlock 3200.

1.30.1.4 Blanks shall consist of ½", thick ASTM 36 plate.

1.30.1.5 Prior to installing or removing Caisson dewatering pumps or valves outside of the drydock:

1.30.1.5.1 Locate 1 each temporary dewatering pump to be placed in each location in the Trim Tank NO. 1 dewatering pump sump, Trim Tank NO. 2 dewatering pump sump, Main Tank NO.1 dewatering pump sump and Main Tank NO.2 dewatering pump sump.

1.30.1.5.2 Each pump shall be operational for the duration of time that any Caisson dewatering pump or valve is not in place and while the Caisson is not in dry dock. Each pump shall and have capacity to pump 250 gallons of water per minute.

1.30.1.5.3 Each pump shall be located with the pump intake at the bottom of each sump. Pump discharge lines shall be pre-located in a manner to evacuate all discharge of water off the Caisson in accordance with all Federal, State and Local environmental regulations.

1.30.1.5.4 While the Caisson is waterborne inspection of asset draft and each tank and sump shall be performed every 4 hours to assure that there is no excessive leak by. Record asset draft and the level of leakage in each sump in the previous four hours. Note any leaking over boards. The result of the inspection shall be recorded and posted in a conspicuous location on the operations deck. Report any changes in the rate of leakage to the COR immediately.

1.30.1.5.5 Pump each sump to low suction each day or when the sump is full, whichever is sooner.

2.0 **REFERENCES:** All references shall be of the latest issue and/or revision or as indicated in the Consolidated List of References. It is incumbent on the Contractor to maintain current commercial specifications and be thoroughly familiar with Federal, State and Local laws, regulations and ordinances. Contractor is prohibited from using Government provided technical information for any purpose other than that contemplated by the specifications. Contractor is prohibited from further use, release, or disclosure of this information. References provided on beta.SAM.gov, safe.apps.mil and/or CD-ROM, including any information contained thereon, shall be destroyed in accordance with the instructions as stated below, at completion of the contract. After award, for those Contractors in receipt of reference material from beta.SAM.gov, safe.apps.mil and/or CD-ROM and not awarded a delivery order shall destroy all information received in accordance with the instructions stated below. This applies to all physical and electronic copies of reference material associated with this contract award.

DESTRUCTION NOTICE: Destroy by any method that will prevent disclosure of the contents or reconstruction of the documents.

2.1 **Order of Precedence:** Unless otherwise noted within this specification, hierarchy, priority, or order of precedence of requirements shall be as follows:

2.1.1 Specification Definite Item
2.1.2 Specification General Requirement
2.1.3 Military/Navy Drawings
2.1.4 Military/Navy Technical Manuals
2.1.5 Commercial Drawings
2.1.6 Commercial Technical Manuals
2.1.7 Commercial or Industrial Standards
2.1.8 Commercial Practices

2.2 **Discrepancies:** Drawing, technical reference, or other discrepancies associated with this specification shall be reported using the Request for Specification Clarification form, during proposal, or the “Condition Found / Inspection Discrepancy Report” form (FIGURE A) during delivery order period of performance.

CONSOLIDATED LIST OF REFERENCES		
The following publication and drawing requirements are applicable to this SOW:		
	Publication Number	**Title**
1)	29 CFR 1910	Occupational Safety and Health Standards
2)	10 USC 7311	Repair or Maintenance of Naval Vessels: Handling of Hazardous Waste
3)	29 CFR 1915	Occupational Safety and Health for Shipyard Employment
4)	29 CFR 1926	Safety and Health Regulations for Construction
5)	42 USC 6901 to 6992	Resource Conservation and Recovery Act of 1976
6)	49 USC 5103	Hazardous Materials Transportation Authorization Act of 1994
7)	ANSI/AWS D1.1	Structural Welding Code, Steel
8)	MIL-STD-777	Schedule of Piping, Valves, Fittings and Associated Piping Components for Naval Surface Ships
9)	NSTM Chapter 074 Volume One	Welding and Allied Processes
10)	NSTM Chapter 075 Fasteners	Threaded Fasteners
11)	SSPC Painting Manual, Volume 2	Systems and Specifications
12)	QA_Appendix_Forms_C411_2023_Contractors.pdf	QA Appendices
13)	MIL-STD-2003-3B(SH)	Electric Plant Installation
14)	302-5302690 Standard Cable installation Requirements	Req. for NDT

CONSOLIDATED LIST OF REFERENCES		
The following publication and drawing requirements are applicable to this SOW:		
	Publication Number	**Title**
15)	MIL-STD-1310 REV H	Shipboard Bonding, Grounding, and Other Techniques for Electromagnetic Compatibility
17)	MILSTD 1689 Rev A	Fabrication, Welding, and Inspection of Ships Structure
18)	T9074-AS-GIB-010/271	Requirements For Non Destructive Testing Methods
19)	S9086-RK-STM-010	Pumps
20)	S9086-KE-STM-010/CH-302	Electric Motors and Controllers

	Drawing Number	Title
1)	S2803-980209 Rev L	Bureau of Ships, Label Plates
2)	S2804-921819 Rev C	Distinguishing Figures
3)	803-921865R	Cathodic Protection Hull Zinc Anodes
4)	2 Inch Reach Rod Packing Gland	2 Inch Reach Rod Packing Gland
5)	111-7283069	Rubber Seal
6)	401-6896568	Spare Caisson TLI
7)	505-6814399	Mod Grease line
8)	804-5184155	Lifeline Life Rail Awnings
9)	980CAS-CNTRPNL-21	Control Panel Install
10)	980CAS-RRMOD-22	Reach Rod Modification
11)	980CAS-TLIRLC-21	TLI Panel Relocation
12)	997-6901533	PWD60761
13)	PW 47978	Redline DK Lighting and FDN Mods
14)	70178	Kick Pipe
15)	70187	Caisson 3 fire and Alarm System
16)	73742 73743 73744	Spare Caisson Valve Replacement
17)	139752	Caisson Sect & Breast Hooks
18)	16002080	DD3 Caisson Replacement Struct Notes
19)	16002112	DD3 Caisson Replacement Misc Metal Details
20)	PW C-1120	Valves
21)	CAS Hinged Cover Plate Hardware Replacement Detail	CAS Hinged Cover Plate Hardware Replacement Detail
22)	One Line Diagram Caiss 5	One Line Diagram Caisson 5
23)	PW 15957	Caisson Shell Plating & Details
24)	PW 15958	Caisson Section & Breast Hooks
25)	PW 15959	Caisson Girders G4, G5 & G6
26)	PW 15960	Caisson Girders G1, G2 & G3
27)	PW 15961	Caisson Sections
28)	PW 15962	Blkhd Sect & Dets
29)	PW 15969	Pumping Equipment
30)	PW 15970	Pumping Equipment
31)	PW 15971	Electrical Work
32)	PW 48053	CA5 Power Mods
33)	PW 48054	Weather Deck Lighting
34)	PW 50746	Flood Tubes for Spare Caisson
35)	PW 51109	SHT2 Elect Actuator
36)	980 CAPSARE Stand 21	Spare Caisson Stand Modification

37)	B-3739	Caisson 4,5 and Spare Lightweight Ramp
38)	SK 10749	Spare Caisson Boarding Ladder
39)	Spare Caisson 4&5 UT Grid	Caisson Spare UT Inspection Sites
40)	Caisson Door Sill Sketch	Caisson Door Sill Sketch
41)	WDL-SKTCH-980	Weather Deck Lighting
42)	980CAS_RR-UHMW_MOD 21	Valve Reach Rod Steady Bearings
43)	Tank Stanchions Sketch Rev A	Tank Stanchions Sketch Rev A
44)	PW 70176	Lighting and Bill of Material
45)	Alarm Light Bracket	Alarm Light Bracket
46)	845-6686999	US Navy Vessel Water Depth, Mooring and Hull/Appendage/Clearance Requirements for Transit and Berthing
47)	DDS 582-1	Design Data Sheet, Calculations for Mooring Systems
48)	UFC 4-159-03	Mooring Design
49)	Caisson 4-5 Spare Hull and Tank Anodes	Caisson 4-5 Spare Hull and Tank Anodes
50)	48055	Puget Sound Naval Shipyard, Bremerton, Wash. Flood Alarm System Circuit "FD"
51)	Spare Caisson 4&5 Piping UT Grid	Spare Caisson 4&5 Piping UT Grid
52)	47983	Caisson 5 & Spare Deck Lighting
53)	PW 70147	DD 3 Caisson Replacement Misc. Metal Details
54)	M-1646 TRF Caisson 2 Fall Protection Anchorage	M-1646 TRF Caisson 2 Fall Protection Anchorage

3.0 WORK REQUIREMENTS:

3.1 DOCKING AND UNDOCKING THE ASSET:

3.1.1 Personnel and Equipment: Furnish qualified and experienced personnel, facilities, and the necessary equipment to safely dock and undock the Asset, including:

3.1.1.1 A Dock Master currently qualified and certified by the Contractor and accepted by the Government Docking Observer (GDO).

3.1.1.2 Supporting employees to accomplish the docking and undocking evolutions.

3.1.1.3 Tugs, pilots, and divers required for the complete transfer, docking, and undocking of each Asset. At the docking conference, the GDO shall review and approve the number and size of the tugs to be used.

3.1.1.4 The Caisson shall not be dry docked or undocked on a long weekend associated with a Federal Holiday. If the contractor elects to dry dock or undock the Caisson on any other weekend, a written request shall be submitted to the SUPERVISOR at least 3 working days in advance of the proposed event.

3.1.1.5 Provide alternate/emergency backup electrical power distribution for safely dry docking and undocking the caisson, in the event of shore power failure.

3.1.1.6 Notify the SUPERVISOR, a minimum of 3 working days prior to an evolution involving the actual dry docking, shifting, and/or undocking of the caisson.

3.1.1.7 Connect and ground the caisson's hull, when in dry dock, at the bow and stern of the craft with minimum of 500,000 circular mil cables, and to the ground connection in dry dock for protection against lightning and other static charges.

3.1.1.7.1 Connect the static ground leads to the caisson being docked before water is pumped to one foot above keel level.

3.1.2 Docking Procedure: Develop a docking procedure prior to dry docking, which shall include the following: (REQUIRED REPORT CDRL A039)

3.1.2.1 Operating practices.

3.1.2.2 Safety and security procedures for the protection of personnel and equipment prior to, during, and while the caisson is in dock.

3.1.2.3 Special precautions or actions required by the characteristics of the dock, marine railway, or other docking methods.

3.1.2.4 Pumping/flooding schedule.

3.1.2.5 Specific list, trim and stability requirements.

(I) "BLOCKING ALIGNMENT"

3.1.3 <u>Blocking Plan:</u> Develop a blocking plan to set and align the docking blocks in accordance with the requirements of drawing 645-7281587 Rev A, using drawing PW 15969, PW 15970, PW 15957, PW 15958, PW 15961, PW 15962, 997-6901533, PW C-1120, W 50746, PW 10749, Caisson 4/5 Spare Hull and Tank Anodes for guidance. (REQUIRED REPORT CDRL A040)

3.1.3.1 Develop a docking plan in the form of a scaled drawing including: block size, block spacing, and location of Asset with respect to the blocks at landing. Position blocking to ensure that equipment on the surface of or protruding from the hull shall not be damaged.

3.1.3.2 Determine the docking clearances utilizing references for guidance and the following requirements: The minimum clearance between the side of the Asset (including fixed projections) and the permanent dock structure (such as wing walls, crane rails, piers, etc.) shall be 12 inches during Asset movement into or out of dock and allow for safe work practices accomplished inside the dock. Minimum vertical clearance between the keel of the Asset and the dock floor shall allow for safe accomplishment of work and provide sufficient access for all inspections. Hull openings shall not be obstructed.

3.1.3.2.1 Minimum vertical clearance between the keel of the Asset and the dock floor shall be 40 inches.

3.1.3.3 The blocking plan shall be accompanied by a written request for approval to the GDO a minimum of 5 working days prior to the scheduled docking date. The use of haul, fixed, or fitted blocks shall be clearly stated.

3.1.3.4 The use of universal docking blocks is prohibited.

3.1.3.5 Maintain supporting data substantiating that:

3.1.3.5.1 The block loading pressures are limited to a maximum of 20 long tons per square foot.

3.1.3.5.2 The number, distance from centerline, and square footage of the side blocks per side is sufficient to resist overturning moments.

3.1.3.5.3 At least 90 percent of blocks land on longitudinal strength members or main transverse bulkheads.

3.1.3.6 Blocking shall be constructed of hardwood or composite hardwood/concrete and soft caps, with at least 2 inches of soft cap thickness.

3.1.3.6.1 The term “hardwood”, for the purposes of this contract, includes White Oak, California Laurel, Oregon Myrtle, Iron Wood, Blue Gum, American Rock Elm, or preserved Red Oak. The normal lifespan for hardwood blocking is approximately 10 years.

3.1.3.6.2 Woods acceptable for use as soft caps are Douglas Fir, Tamarack, Long Leaf Pine, or Hemlock.

3.1.3.6.3 Composite blocking showing evidence of spalling and cracks or chipped and damaged concrete is not acceptable.

3.1.3.7 Blocking showing evidence of excessive crushing, warping, cracking, checking, rotting, or damage from dogging, loss of contact at edges caused by cracking or unequal shrinkage or deterioration to an extent no longer being capable of supporting a prescribed load over full bearing areas is not acceptable and shall be replaced.

3.1.4 Crib and Brace Blocking: Ensure that crib and brace blocking are in accordance with the requirements noted below, using the drawing applicable to the docking facility.

3.1.4.1 Blocks in excess of 6 feet in height but below 8 feet 6 inches in height shall be considered high blocks. Blocks which are 8 feet 6 inches or greater in height shall be considered extra-high blocks.

3.1.4.2 High keel blocks shall be cribbed or butted in the aft one-third and forward one-third of the keel block line. All extra-high keel blocks shall be cribbed or butted.

3.1.4.3 Side blocks above 6 feet in height (measured from the bottom of the block to the highest corner of the soft cap) shall be tied together longitudinally in pairs by means of steel bracing, or joined by cribbing, or be constructed as individual pyramids. If the side blocks are haul blocks, they must be hauled in pairs.

3.1.5 Setting and Alignment of Docking Blocks: Furnish, set, record, and align the blocks in accordance with the Government approved Contractor blocking plan. Prepare dry dock for inspection a minimum of a work day prior to the proposed docking. Notify and accomplish an inspection of the blocking with the GDO in a timeframe sufficient to make blocking changes. Develop documentation of the inspection showing blocking, sighting marks on dock coping relative to block locations, spacing, offsets, heights, and shaping to the GDO at least 4 hours prior to flooding the dry dock. Immediately prior to flooding the dry dock, perform a final inspection with the GDO to ensure that no loose trash, wood, equipment, and other extraneous matter could float up during flooding and interfere with/delay the docking operation.

3.1.6 Stability Calculations: Calculate stability. This shall include draft, list trim, strength, and stability calculations.

3.1.7 Docking Conference: Perform a docking conference attended by the GDO and SUPERVISOR, the Contractor's Dock Master, the pilot who will be present during the docking evolution and required dive personnel within 4 working days prior to the scheduled docking of the Asset. The Dock Master shall discuss the time of docking, tide situation, weather conditions, tug and pilot arrangements, line handling, electrical shore power, grounding, safety issues, safety concerns, other services and procedures.

3.1.8 Docking the Asset: Dry dock the Asset without strain. Immediately after docking, electrically connect the Asset hull at the bow and stern of Asset for protection against lightning and other static charges in accordance with the requirements of NSTM Chapter 074 Volume One.

3.1.8.1 Notify the GDO a minimum of 3 working days prior to actual docking and undocking of the Asset.

3.1.8.1.1 Contact the SUPERVISOR for current GDO contact information.

3.1.8.2 Dry dock facilities shall have an alternate / back up electrical power generating and distributing source to furnish all electrical requirements for docking and undocking of Asset, in the event of a loss of main power source.

3.1.9 Post Docking: Prepare and inspect Asset underwater hull surfaces.

3.1.9.1 Immediately after docking, perform a visual inspection of the underwater portions of the Asset with the GDO for damage and deterioration. Inspect the fit on the blocks and furnish necessary shimming between the blocking and the hull to prevent hull movement and to ensure 100 percent block contact with the hull structure.

3.1.9.2 Within 8 hours of docking, wash the underwater hull and fittings with fresh water at a minimum of 3,000 PSIG nozzle pressure to remove dirt, slime, marine growth, fouling and other foreign substances.

3.1.9.2.1 Accomplish 100 percent removal and disposal of marine growth from underwater hull, exterior surface of each flood through piping, QTY (8 EA) Flood Through Covers, QTY (2 EA) Sea Valve Piping Screens, QTY (2 EA) Dewatering Valve Piping Screens, and QTY (2 EA) Sinking Valve Piping Screens no later than 2 working days after dry docking.

3.1.9.3 Comply with federal, state, and local laws and ordinances for disposal of wash water and debris removed from underwater portions of the Asset.

3.1.10 <u>Undocking the Asset:</u> After completion of work, undock the Asset at a time and date agreed upon by the GDO, SUPERVISOR and the Contractor. Perform an undocking conference similar to the Docking Conference of 3.1.7 within 3 working days of undocking. Immediately prior to flooding the dry dock, perform an inspection with the GDO to ensure that no loose trash, wood, equipment, and other extraneous matter could float up during flooding and interfere with/delay the undocking operation.

3.1.10.1 Immediately after the hull penetrations are submerged and before the Asset lifts off the blocks, stop flooding the dry dock. Accomplish a watertight integrity check of the hull. Continue flooding of the dock when directed by the GDO.

3.1.10.2 After the Asset has cleared the blocks, perform a visual inspection of the underwater hull surfaces to ensure that no soft caps are adhered to the hull surfaces. Perform a visual inspection of the docking blocks to account for all soft caps and to ensure that no blocks are adhered to the underwater hull structure of the Asset.

3.1.11 <u>Docking Observer:</u> The Docking Observer is the Government official certified by the Government as qualified to execute the duties of a Docking Observer. The presence of the Docking Observer is for protecting the

Navy's interests, not to give direction to the Contractor. However, the Docking Observer shall have the authority to stop the docking, undocking or launching evolution when the Asset safety or other Navy interests are jeopardized. This authority does not imply limitations on or responsibility for the actions of the Contractor's Dock Master.

3.1.12 Final Docking Report: Develop a final docking report (NAVSEA 9997/1 through 9997/5, as necessary) upon completion of undocking. Electronic versions of the final docking report, in portable document format (PDF) and MSWord format are available from the GDO upon request. (REQUIRED REPORT CDRL A041)

3.2 **HULL REPAIRS:**

3.2.1 Identify Distinguishing Figures and Markings: Identify and record existing distinguishing underwater body, hull, and freeboard color schemes, markings, and figures in sketch form listing colors, locations, sizes, and distinguishing marking(s), and figure(s) information, using drawings PW 15957, PW 15959, PW 15960, SK 10749, 997-6901533 and S2804-921819 Rev C for guidance. (REQUIRED REPORT CDRL A042)

3.2.2 Rubber Fenders: Remove existing rubber fender materials and rubber fender fasteners from the Asset and sweep blast clean to remove sea growth, paint, and other contaminants from the fender surfaces, using drawings SK 10749 or guidance.

3.2.2.1 Temporarily store the cleaned rubber fender materials.

3.2.2.2 Inspect mounting studs, for signs of corrosion, deterioration, galling, damage, and distortion. (REQUIRED REPORT CDRL A043)

3.2.2.3 Install the cleaned rubber fender materials and rubber fender fasteners to their original locations on the Asset after full cure time has elapsed after final application of the hull and freeboard surface preservation.

Long Lead: The following materials are considered Long Lead Time Material and shall be ordered by the Contractor within 5 working days of contract award.

3.2.2.3.1 Remove existing and install new QTY (76 EA) Nut, One-inch, Monel 400, hex head in the locations as directed by the SUPERVISOR and based on the inspection report of 3.2.2.2.

3.2.2.3.2 Remove existing and install new QTY (76 EA) Washer, One-inch, Monel 400 in the locations as directed by the SUPERVISOR and based on the inspection report of 3.2.2.2.

3.2.2.3.3 Apply anti-seize compound that is similar or equal to the performance characteristics of CID-A-A-59004 for each fastener.

3.2.3 Zinc Anodes: Remove existing and install new zinc anodes and associated CRES fasteners, using drawing 803-921865R, Caisson 4-5 Spare Hull and Tank Anodes 2011, PW 15959 and PW 15960 for guidance.

3.2.3.1 Remove existing and install new QTY (66 EA) hull zinc anodes conforming to the requirements of Type ZHC-42 on the underwater body of the Asset.

3.2.3.2 Remove existing and install new QTY (132 EA) Nut self-locking, new QTY (132 EA) Washer tooth lock; new fasteners shall conform to drawing 803-921865R.

3.2.3.3 Identify and record exact location of each zinc anode. (REQUIRED REPORT CDRL A044)

3.2.3.3.1 List and identify corresponding specific measurements, locations, and/or areas in relationship to fixed hull structural members and docking block locations, or with measurements from suitable hull reference points.

3.2.3.3.2 A sketch of the underwater body and freeboard plating surfaces with appropriate data and measurements is an acceptable alternative.

3.2.3.4 Accomplish an electrical resistance test for each new zinc anode in accordance with General Note 4 of drawing 803-921865R. (REQUIRED REPORT CDRL A045)

3.2.3.5 Ensure that new zincs remain free of preservation coatings.

3.2.4 <u>Hull and Freeboard Ultrasonic Test (UT) Inspection:</u> Accomplish UT measurement inspection of the underwater body hull. Accomplish in accordance with Spare Caisson 4&5 UT Grid, using drawings PW 15957, PW 15959, PW 15960, SK 10749, T9074-AS-GIB-010/271, 997-6901533 and TABLE ONE for guidance. Record and submit readings on the drawing Spare Caisson 4&5 UT Grid. (REQUIRED REPORT CDRL A046)

TABLE ONE			
Original Plating Thickness (inch)			Minimum Plate Thickness
Fractional	Decimal	Pound/SF	25% Wastage
1/16	0.063	2.55	0.047
1/8	0.125	5.10	0.094
3/16	0.188	7.65	0.141
1/4	0.250	10.20	0.188
5/16	0.313	12.75	0.235
3/8	0.375	15.30	0.281
7/16	0.438	17.85	0.329
1/2	0.500	20.40	0.375
9/16	0.563	22.95	0.422
5/8	0.625	25.53	0.469
3/4	0.750	30.63	0.563
7/8	0.875	35.74	0.656
1	1.000	40.84	0.750

3.2.4.1 List in one column the design thickness.

3.2.4.2 Record in a separate column if reading is taken over an existing doubler or insert plate.

3.2.4.3 List in another column, next to the design thickness, the actual readings taken.

3.2.4.4 List and identify corresponding specific measurements, locations, and/or areas in relationship to structural members or with measurements from suitable hull reference points.

3.2.4.5 A sketch of the underwater body and freeboard plating surfaces with appropriate data is an acceptable alternative.

3.2.4.6 Calculate, identify, and highlight measurements indicating a difference (wastage) greater than or equal to 25 percent of the designed plate thickness, using TABLE ONE for guidance.

3.3.4.7 Perform UT measurements for the entire length of the port rubber seal steel mating surface at 2-foot intervals, in three equally spaced locations per interval. Measure and record in columnar form data in accordance with 3.2.4.1 through 3.2.4.6.

3.2.5 Hull SP-6 Blasting: Commercial blast existing preservation system and inspect 100 percent of the entire underwater body, and hull surfaces of the caisson from port to starboard and from the keel up to the upper edge of the main deck coaming, including the entire keyway, from bow to stern. Including flood through cover and each exterior surface of the flood through piping, and each: hull opening, overboard, attachment, sea screen grate, and each appendage. Including the port rubber seal mating surface. Use drawings PW 15957, PW 15959, PW 15960, SK 10749, 997-6901533 and S2804-921819 Rev C for guidance.

(I) "VISUAL INSPECTION (VT)"

3.2.6 Hull Visual Inspection: Accomplish a Visual Inspection (VT) after completion of SP-6 blast, grit removal and cleanup of the surfaces of 3.2.5 in the locations of 3.2.5. Use drawings PW 15957, PW 15959, PW 15960, SK 10749, 997-6901533 and S2804-921819 Rev C for guidance. (REQUIRED REPORT CDRL A047)

3.2.6.1 Search for, mark, identify, and record defective (actual and suspect) areas that exhibit signs of severe pitting, excessive erosion, deterioration, damage, or which may be detrimental to the structural and watertight integrity of the caisson.

3.2.6.1.1 Previously unidentified damaged or deformed structure shall be reported within 48 hours of discovery to the SUPERVISOR.

3.2.6.1.2 Identify abandoned unused hull appendages to be removed and ground flush on the entire hull, from the keel to the top of each weather deck kick plate.

3.2.6.1.3 Create a two dimensional, or Autocad 2016, sketch or drawing showing hull/structural configurations, arrangements, relationships, and a listing of the inspection location's data is an acceptable alternative.

3.2.6.2 The locations and dimensions of structural repairs identified by the Ultrasonic Inspection and Visual Inspection Reports of 3.2.4 and 3.2.6. will be listed on the structural metal repair plan of 3.3.1.1.

3.2.6.2.2 Unidentified damaged hull surfaces shall be reported within 48 hours of discovery to the SUPERVISOR.

(I)"STRUCTURAL REPAIR COMPLETION: HULL"

3.2.6.3 Accomplish repair of 100 percent of underwater hull structure in paragraph 3.3.1 including paragraph 3.3.6.7, prior to accomplishing the requirements of 3.2.7. Verify all repairs are accomplished by the Contractor in the locations and dimensions as approved by the SUPERVISOR in the structural metal repair plan as identified on the declining log prior to final SP-10 Blast. (REQUIRED REPORT CDRL A048)

3.2.7 Underwater Hull SP-10 Surface Preparation: Accomplish SP-10 for the surfaces of 3.2.5.

3.2.7.1 Near white blast clean 100 percent of surfaces listed in 3.2.5 and 3.11.6. Accomplish the requirements of Surface Preparation Specification SSPC-SP-10 of Systems and Specifications, SSPC Painting Manual, Volume 2. Achieve a surface profile in accordance with the manufacturer's product data sheet.

3.2.7.2 Power tool clean inaccessible areas where abrasive blasting equipment cannot reach. Accomplish the requirements of Surface Preparation Specification SSPC-SP-11 of Systems and Specifications, SSPC Painting Manual, Volume 2. Achieve a surface profile in accordance with the Manufacturers Product Data Sheet.

Long Lead: The following materials are considered Long Lead Time Material and shall be ordered by the Contractor within 5 working days of contract award.

3.2.8 Hull Preservation: Apply a preservation coating system, anticorrosive and antifouling coating system: in accordance with the requirements of MIL-PRF-24647, Type II; 2 coats anticorrosive (AC); 3 coats antifouling (AF); installed in accordance with the manufacturer's product datasheets to the entire surfaces prepared in 3.2.7. Final topcoat color configuration shall match the previously existing final color configuration using the information recorded in 3.2.1 for guidance. Use drawings PW 15957, PW 15959, PW 15960, SK 10749, 997-6901533 and S2804-921819 Rev C for guidance.

3.2.8.1 All AC and AF coats in the product system shall be from the same manufacturer.

3.2.9 Apply Asset Markings: Apply Polysiloxane type coating to markings and distinguishing figures identified in 3.2.1. Coatings shall be in accordance with the requirements of MIL-PRF-24635 Type V or VI Grade B; and manufacturer's product data sheets using drawing S2804-921819 Rev C for guidance.

(V) "UNDERWATER HULL INSPECTION PRIOR TO STAGING REMOVALS"

3.2.10 Underwater Hull Pre-Staging Removal Inspection: Accomplish a pre-staging removal inspection of the underwater body, and hull surfaces including the keyways and rubber seals of the caisson from the keel up to the upper edge of the main deck, from bow to stern after completion of preservation, and prior to staging removals.

3.2.10.1 Inspect cleanliness, cured paint, and completion of work prior to staging removals.

3.2.10.2 Each deficiency shall be recorded on the checkpoint form and corrected prior to staging removals.

3.2.10.3 Each corrected deficiency shall be verified by the SUPERVISOR as satisfactorily corrected prior to staging removals.

(V) "POST STAGING REMOVAL INSPECTION"

3.2.11 Underwater Hull Post Staging Removal Inspection: Accomplish a Post Staging Removal Inspection of the underwater body, and hull surfaces of 3.2.7 no later than 24 hours after staging removals.

3.2.11.1 Demonstrate that each location listed in 3.2.11 is in a broom clean condition. Locations shall be free of visual contaminants such as oxide films, oils, grease, dust, dirt, blast grit, rust, rust stains or foreign material.

3.2.11.2 Each deficiency shall be recorded on the inspection form and corrected within 24 hours.

3.2.11.3 All corrected deficiencies shall be verified by the SUPERVISOR as satisfactorily corrected prior to the resumption of any other production.

3.3 **STRUCTURAL REPAIRS:**

3.3.1 Structural Metal Repair: Accomplish structural metal repairs of 3.3.1.1.1 through 3.3.1.1.5, in the SUPERVISOR approved locations as listed in the structural metal repair plan of 3.3.1.1. Distribution of work to be based on the structural metal repair plan locations submitted by the Contractor and approved by the SUPERVISOR.

3.3.1.1 Based on the results of the inspections of 3.2.6.2 (hull), 3.3.4 (port rubber seal - sealing surface), 3.4.6.2 (tanks and voids), 3.5.5.2 (weather deck), 3.6.5 (operations deck) and 3.11.1.3 (piping) prepare a structural metal repair plan with the following data. (REQUIRED REPORT CDRL A049)

3.3.1.1.1 Record the square inches of each proposed weld build up repair. Record the location of each proposed repair.

3.3.1.1.2 Record the linear inches of each proposed "V" out excavate and weld fill repair. Record the location of each repair.

3.3.1.1.3 Record the pounds and design thickness of each proposed structural steel crop out, replace with new and weld. Record the location of each proposed repair.

3.3.1.1.4 Record the pounds and design thickness of each proposed plate steel crop out, replace with new and weld install. Record the location of each proposed repair.

3.3.1.1.5 Record the linear inches of each proposed remove, chip and grind flush and smooth unused/abandoned hull appendages. Record the location of each proposed repair.

3.3.1.2 Upon approval of the Contracting Officer via the SUPERVISOR accomplish the following base metal repairs for each approved repair location in the structural metal repair plan of 3.3.1.1:

3.3.1.2.1 Accomplish quantity not to exceed 8,000 square inches weld buildup repair. Any amount less than 8,000 square inches that is not required will be descoped in accordance with 3.3.1.

3.3.1.2.2 Accomplish quantity not to exceed 600 linear inches "V Out" excavate and weld fill repair. Any amount less than 600 linear inches that is not required will be descoped in accordance with 3.3.1.

3.3.1.2.3 Accomplish quantity not to exceed 1,000 pounds structural steel crop out, replace with new and weld install. Any amount less than 1,000 pounds that is not required will be descoped in accordance with 3.3.1.

3.3.1.2.4 Accomplish quantity not to exceed quantity not to exceed 1,300 pounds plate steel crop out, replace with new and weld install. Any amount less than 1,300 pounds that is not required will be descoped in accordance with 3.3.1.

3.3.1.2.5 Accomplish quantity not to exceed 20 linear feet remove, chip and grind flush and smooth unused/abandoned hull appendages. Any amount less than 20 linear feet that is not required will be descoped in accordance with 3.3.1.

3.3.1.3 Utilize the structural references listed in sections 3.2, 3.3, 3.4, 3.5, 3.6 and 3.11 as guidance in way of structural weld repairs.

3.3.1.4 Template exact sizes, locations, and configurations from existing shipboard conditions prior to removal and re-installation.

3.3.1.5 Keep a declining log listing at a minimum the paragraph number of each repair that is approved by the SUPERVISOR. List the CFR number that each repair was identified on. List the location, design thickness of plate or design thickness of structural material and dimensions of the repair. Contractor shall provide weekly declining log updates to SUPERVISOR. (REQUIRED REPORT CDRL A050)

3.3.1.6 Record the following in the declining log:

3.3.1.6.1 Record the square inches of each weld build up repair of 3.3.1.2.1 approved by SUPERVISOR.

3.3.1.6.2 Record the linear inches of each weld repair of 3.3.1.2.2 approved by the SUPERVISOR.

3.3.1.6.3 Record the pounds of each structural steel crop out, replace with new and weld of 3.3.1.2.3 approved by the SUPERVISOR.

3.3.1.6.4 Record the pounds of each plate steel crop out, replace with new and weld install of 3.3.1.2.4 approved by the SUPERVISOR.

3.3.1.6.5 Record the linear inches of each remove, chip and grind flush and smooth unused/abandoned hull appendages of 3.3.1.2.5.

3.3.1.7 Accomplish nondestructive testing for each repair of 3.3.1.2.1 through 3.3.1.2.5 in accordance with the following:

(I) "VISUAL INSPECTION (VT)"

3.3.1.7.1 Accomplish nondestructive testing visual inspection.

(I) "FIT UP INSPECTION"

3.3.1.8 Accomplish fit up inspection for each repair of 3.3.1.2.3 and 3.3.1.2.4.

(I) "MAGNETIC PARTICLE INSPECTION (MT)"

3.3.1.8.1 Accomplish nondestructive testing magnetic particle inspection.

(I) "AIR HOSE TEST"

3.3.1.9 Accomplish a local air hose test for each new weld of 3.3.1.2.3 and 3.3.1.2.4 as follows:

3.3.1.9.1 Air hose nozzle must be placed at less than 3 inches, but greater than zero inches, from the area to be tested and pressure directed at the structure under test in a manner most likely to disclose leaks.

3.3.1.9.2 The minimum nozzle diameter must be 3/8 inch, and the nozzle pressure must be 90 plus or minus 5 PSIG as monitored at the nozzle.

3.3.1.9.3 Apply a soapy solution to the opposite side of the structure and inspect for leaks.

3.3.1.9.4 Inspect joint or fitting for leakage by observing for formation of bubbles. Allowable leakage: None. (REQUIRED REPORT CDRL A051)

(I) "RUBBER SEAL MATING SURFACE ULTRASONIC TEST (UT)"

3.3.2 Port Rubber Seal UTs: Remove existing port rubber seal and associated fasteners prior to SP-6 blasting of 3.2.5. port rubber seal. Accomplish UT measurements in accordance with drawing Spare Caisson 4&5 UT Grid. Use drawings PW 15957, PW 15959, PW 15960, SK 10749, T9074-AS-GIB-010/271, 997-6901533 and TABLE ONE for guidance. Record and submit readings on the drawing Spare Caisson 4&5 UT Grid. (REQUIRED REPORT CDRL A052)

3.3.2.1 Accomplish UT measurements of the port rubber seal steel mating surface.

3.3.2.1.1 Inspect the entire length of the port rubber seal steel mating surface at 2 foot intervals, in three equally spaced locations per interval. Measure and record in columnar form the following:

3.3.2.1.2 List in one column the design thickness.

3.3.2.1.3 Record in a separate column if reading is taken over an existing doubler or insert plate.

3.3.2.1.4 List in another column, next to the design thickness, the actual readings taken.

3.3.2.1.5 List and identify corresponding specific measurements, locations, and/or areas in relationship to structural members or with measurements from suitable hull reference points.

3.3.2.1.6 A sketch of the surfaces with appropriate data is an acceptable alternative.

3.3.2.1.7 Calculate, identify, and highlight measurements indicating a difference (wastage) greater than or equal to 25 percent of the designed plate thickness, using TABLE ONE for guidance.

(I) "VISUAL INSPECTION (VT)"

3.3.3 Port Rubber Seal Mating Surface Visual Inspection: After completion of Commercial Blast Clean NACE 3/SSPC-SP-6, of the surface listed in 3.2.5 accomplish a Visual Inspection (VT). Inspect the entire length of the exposed port rubber seal steel mating surface, bars, filler bars, seal clamps, and appendages, using drawings 997-6901533 and 111-7283069 for guidance. (REQUIRED REPORT CDRL A053)

3.3.3.1 Search for, mark, identify, and record defective (actual and suspect) areas that exhibit signs of severe pitting, excessive erosion, deterioration, damage, or which may be detrimental to the structural and watertight integrity of the caisson.

3.3.3.1.1 Unidentified damaged or deformed structure shall be reported within 48 hours of discovery to the SUPERVISOR.

3.3.3.1.2 A two dimensional, or AutoCAD 2016, sketch or drawing showing hull/structural configurations, arrangements, relationships, and a listing of the inspection location's data is an acceptable alternative.

3.3.4 The locations and dimensions of structural repairs of the port rubber seal mating surface as identified by the Ultrasonic Inspection and Visual Inspection Reports of 3.3.2 and 3.3.3 will be listed on the structural metal repair plan of 3.3.1.1.

3.3.4.1 Unidentified damaged surfaces shall be reported within 48 hours of discovery to the SUPERVISOR.

(I) "RUBBER SEAL STUD INSPECTION"

3.3.5 Rubber Seal Stud Inspection: Prior to removal of portside rubber accomplish rubber seal stud inspection on port and starboard sides for thread damage, corrosion, bent studs, and broken studs. Search for existing nuts not meeting the thread protrusion requirements of 1.15.1. Search for washers that have compressed the rubber seal greater than 1/8 inch in accordance with General Note 2.04 of 111-7283069. Submit findings no later than the first 20 percent of the scheduled period of performance.

3.3.5.1 Identify each stud on the port and starboard side that requires replacement.

3.3.5.2 **Do not remove the starboard rubber seal unless directed to do so by the Contracting Officer.**

3.3.6 Rubber Seal Stud Replace: Based on the report of 3.3.5, remove existing and weld install not to exceed 80 each new rubber seal studs. Only replace studs approved by the SUPERVISOR. Use drawing 997-6901533 and 111-7283069 for guidance. Any amount less than 80 studs that is not required will be descoped.

3.3.6.1 When fitting new studs assure that when the rubber seal is in place and fit up that newly installed nuts will meet the requirements of thread engagement in 1.15.1 and that washers will not compress the rubber seal greater than 1/8 inch.

3.3.6.2 Template exact sizes, locations, and configurations from existing shipboard conditions prior to removal and re-installation.

3.3.6.3 Accomplish nondestructive testing for each stud repair in accordance with the following:

(I) "FIT UP INSPECTION"

3.3.6.3.1 Accomplish fit up inspection.

(I) "VISUAL INSPECTION (VT)

3.3.6.4 Accomplish nondestructive testing visual inspection.

(I) "TORQUE TEST"

3.3.6.5 Torque test each newly installed stud to 807 inch-pounds (67.25 foot-pounds) for each replaced stud in accordance with paragraph 6.9.2 of MILSTD 1689 Rev A.

(I) "STRUCTURAL REPAIR COMPLETION: PORT RUBBER SEAL MATING SURFACE AND STUD REPAIR"

3.3.6.6 Accomplish repair of 100 percent of port rubber seal mating surface in paragraph 3.3.1 and port stud repairs in 3.3.6, prior to the installation of the rubber seal of 3.3.7. Verify all repairs are accomplished in the locations and dimensions as approved by the SUPERVISOR in the structural metal repair plan as identified on the declining log prior to final SP-10 Blast.

Long Lead: The following materials are considered Long Lead Time Material and shall be ordered by the Contractor within 5 working days of contract award.

3.3.7 Install New Port Rubber Seal: Remove existing port rubber seal prior to SP-6 blasting of 3.2.5, in accordance with drawing 997-6901533 and using drawing 111-7283069 for guidance.

3.3.7.1 Remove existing and install new on Port Side Rubber seal as follows:

3.3.7.2 QTY (274 EA) Nut, Monel 400, Item No. 2 of 997-6901533.

3.3.7.3 QTY (274 EA) Washer, Monel 400, Item No. 3 of 997-6901533.

3.3.7.4 QTY (As Required) Rubber Filler One Part, Item No. 4 of 997-6901533.

3.3.7.5 QTY (23 EA) Rubber Seal Type One, Item No. 5 of 997-6901533.

3.3.7.6 QTY (One EA) Rubber Seal Type 2, Item No. 6 of 997-6901533.

3.3.7.7 QTY (2 EA) Rubber Seal Type 3, Item No. 7 of 997-6901533.

3.3.7.8 QTY (One EA) Rubber Seal (corner) Type 4, Item No. 8 of 997-6901533.

3.3.7.9 Tighten fasteners in accordance with General Note 2.04 of 111-7283069.

(I) "FASTENER TORQUE"

3.3.7.10 After installation of the port rubber seal and fasteners inspect fastener torque in accordance with General Note 2.04 of 111-7283069.

3.3.7.11 Replace rubber filler; fair in rubber segments smooth.

3.3.7.11.1 All surfaces to be smooth with no more than one 1/8” defect per 4 square inches.

3.3.7.11.2 No surfaces cracks more than 2 inches long in any directions.

(V) "RUBBER SEAL VISUAL INSPECTION (VT)"

3.3.8 <u>Starboard Rubber Seal Inspect:</u> Accomplish a visual inspection of the Starboard rubber seal; using drawings 997-6901533 and 111-7283069 for guidance. Submit findings no later than the first 20 percent of the scheduled period of performance.

3.3.8.1 Search for, mark, identify, and record any areas that exhibit signs of separation, pitting, cracking, tears, excessive wear, damaged or missing rubber sealant, loose or damaged nuts, bent studs or broken studs using drawings 997-6901533 and 111-7283069 for guidance.

3.3.8.1.1 Inspect tightness of each fastener to ensure fastener has not compressed the rubber seal greater than 1/8 inch and that each fastener is no less than flush with the fastener thread and no greater than five fastener threads are exposed.

3.3.8.1.2 All surfaces to be smooth with no more than one 1/8” defect per 4 square inches.

3.3.8.1.3 No surfaces cracks more than 2 inches long in any directions.

3.3.8.1.4 Identify location of each defect with specific measurements and location in relationship to structural members or with measurements from a suitable hull reference point.

3.3.8.2 Remove existing and replace rubber filler of 3.3.8; using new Sikaflex 291 or equal. Fair in rubber segments smooth using drawings 997-6901533 and 111-7283069 for guidance.

3.3.8.3. **Do not remove starboard rubber seal unless directed to do so by the Contracting Officer via the SUPERVISOR.**

3.3.9 Porthole Assembly Repair: Accomplish clean and inspect of QTY (6 EA) Port porthole assemblies plus One blank and QTY (6 EA) Starboard porthole assemblies plus One blank.

3.3.9.1 Clean, search for, mark, identify, and record defective (actual and suspect) areas that exhibit signs of severe pitting, excessive erosion, deterioration, damage, or which may be detrimental to the structural and watertight integrity of the Asset, the equipment listed in 3.3.9. Create a repair plan. (REQUIRED REPORT CDRL A054)

3.3.9.2 Template exact sizes, locations, and configurations from existing shipboard conditions.

(I) "WATER HOSE TEST"

3.3.9.3 Accomplish a water hose test of each porthole in accordance of repair plan for 3.3.9. (REQUIRED REPORT CDRL A055)

3.3.9.3.1 Use a one and one-half inch hose with a minimum nozzle diameter of one-half inch. Pressure at the nozzle must be no less than 50 PSIG at a maximum distance of 10 feet from the surface being tested.

3.3.9.3.2 The stream of water must be directed against the structure in a manner most likely to disclose leaks. The opposite side of the structure must be inspected to detect and locate leaks. Allowable leakage: None.

3.4 **TANKS/VOIDS:**

3.4.1 Open, Pump, Clean, Ventilate, and Gas Free Tanks/Voids: Open, pump, clean, venti
late, gas free, maintain dry, and Safe for Workers/Safe for Hot Work, as applicable, each location listed in TABLE 2, using drawing/manual PW 15960 and MIL-STD-777 for guidance.

3.4.1.1 Meter, measure, and record actual amounts of residual liquids (in gallons) and solids (in pounds) removed from each location listed in TABLE 2. It is anticipated that the Asset will be delivered to the Contractor with tanks/voids at low suction and that an estimated cumulative total of 10,000 gallon of residual liquids and an estimated cumulative total of 1000 pound of residual solids will be removed from the Asset after Asset's delivery to the Contractor. (REQUIRED REPORT CDRL A056)

TABLE 2
Main Ballast Tank
Forward Trim tank
AFT Trim Tank

3.4.2 Identify Distinguishing Figures and Markings: Identify and record existing color schemes, markings, and figures in sketch form listing colors, locations, sizes, and distinguishing marking(s), and figure(s) information, using drawings PW 15958, PW 15959, PW 15960, PW 15961, PW 15962, and T9074-AS-GIB-010/27 for guidance. (REQUIRED REPORT CDRL A057)

3.4.3 Tank Anodes Replace: Remove existing type ZHS zinc anodes and install new zinc anodes and associated CRES fasteners in the Main Ballast Tank, Forward Trim Tank and AFT Trim tank, using drawing 803-921865R, PW 15957, PW 15958, PW 15959, PW 15960, PW 15961 and PW 15962 for guidance.

3.4.3.1 Remove existing QTY (154 EA) type ZHS anodes in the locations listed in Table 2.

3.4.3.2 Install new QTY (308 EA) ½ inch x 2-inch Grade 316 studs, QTY (308 EA) ½ inch Grade 316 tooth lock washers, and QTY (308 EA) ½ inch Grade 316 self-locking Nylock nuts.

3.4.3.3 Template installations to existing shipboard conditions.

3.4.3.4 Install new QTY (154 EA) type ZHC-42 anodes in place of equipment removed.

3.4.3.5 Identify and record exact location of each zinc anode. (REQUIRED REPORT CDRL A058)

3.4.3.5.1 Measure and record in sketch form the layout dimensions, configurations, and locations of each newly installed zinc anode in relationship to structural members or measurements from a suitable hull reference point in the locations.

(I) "RESISTANCE TEST"

3.4.3.6 Accomplish an electrical resistance test for each new zinc anode in accordance with General Note 4 of drawing 803-921865R. (REQUIRED REPORT CDRL A059)

3.4.3.6.1 Ensure that new zinc anodes remain free of preservation coatings.

3.4.3.6.2 A two dimensional, or Autocad, sketch or drawing showing hull/structural configurations, arrangements, relationships, and a listing of the inspection location's data is an acceptable alternative.

(I) "ULTRASONIC TEST (UT)"

3.4.4 Tank Ultrasonic (UT) Inspection: Accomplish UT measurement of interior structural surfaces of the equipment listed in 3.4.1 in each location listed in TABLE 2. Accomplish in accordance with drawing Spare Caisson 4&5 UT Grid, using drawings PW 15958, PW 15959, PW 15960, PW 15961, PW 15962, T9074-AS-GIB-010/271 and TABLE ONE for guidance. Record and submit readings on the drawing Spare Caisson 4&5 UT Grid. (REQUIRED REPORT CDRL A060)

3.4.4.1 List in one column the design thickness.

3.4.4.2 Record in a separate column if reading is taken over an existing doubler or insert plate.

3.4.4.3 List in another column, next to the design thickness, the actual readings taken.

3.4.4.4 List and identify corresponding specific measurements, locations, and/or areas in relationship to structural members or with measurements from suitable hull reference points.

3.4.4.5 A sketch of the underwater body plating and structure surfaces with appropriate data is an acceptable alternative.

3.4.4.6 Calculate, identify, and highlight measurements indicating a difference (wastage) greater than or equal to 25 percent of the designed plate thickness, using TABLE ONE for guidance.

3.4.5 Tank SP-6 Blasting: Remove existing and install new preservation systems to the 100 percent of surfaces in TABLE 2, including each attachment, appendage, combing, foundation, sounding tube, tank vent, overflow, piping, pipe, interior side of each new and existing access cover, support structural members, and new ladder using drawings PW 15957, PW 15958, PW 15959, PW 15960, PW 15961, PW 15962, PW 50746, PW 15969, PW 15970, C-1120 and 401-6896568 for guidance.

3.4.5.1 Accomplish 100 percent Commercial Blast Clean NACE 3/SSPC-SP-6, of the surfaces listed in 3.4.5.

3.4.5.2 Prior to SP-6 blast remove each interference and associated fastener to include each; existing sump grate fastener, pump and pump column fastener, grease line, grease line fastener, sounding tube or TLI pipe fastener (listed as DIP Tube on drawings), and tank vent fastener and retain each fastener.

3.4.5.2.1 Inspect each retained fastener for corrosion, excessive pitting, cracking, deformation, thread deterioration and wear.

(I) "VISUAL INSPECTION (VT)"

3.4.6 Tank Visual Inspection: Accomplish a Visual Inspection (VT) after completion of SP-6 blast, grit removal and cleanup of the entire surfaces listed in 3.4.5. Use drawings PW 15957, PW 15958, PW 15959, PW 15960, PW 15961, PW 15962, PW 50746, PW 115969, PW 15970, C-1120 and 401-6896568 for guidance. (REQUIRED REPORT CDRL A061)

3.4.6.1 Search for, mark, identify, and record defective (actual and suspect) areas that exhibit signs of severe pitting, excessive erosion, deterioration, damage, or which may be detrimental to the structural and watertight integrity of the caisson.

3.4.6.1.1 Previously unidentified damaged or deformed structure shall be reported within 48 hours of discovery to the SUPERVISOR.

3.4.6.1.2 Identify abandoned and unused appendages to be removed and ground flush.

3.4.6.1.3 A two dimensional, or AutoCAD 2016, sketch or drawing showing hull/structural configurations, arrangements, relationships, and a listing of the inspection location's data is an acceptable alternative.

3.4.6.2 The locations and dimensions of structural repairs identified by the Ultrasonic Inspection and Visual Inspection Reports of 3.4.4 and 3.4.6 will be listed on the structural metal repair plan of 3.3.1.1.

3.4.6.2.1 Unidentified damaged tank surfaces shall be reported within 48 hours of discovery to the SUPERVISOR.

(I) "STRUCTURAL REPAIR COMPLETION: TANKS AND VOIDS"

3.4.6.3 Structural inspections and repairs of 3.4, and 3.11 shall be completed prior to accomplishing the requirements of 3.4.7. Verify all repairs are accomplished in the locations and dimensions as approved by the SUPERVISOR in the structural metal repair plan as identified on the declining log prior to final SP-10 Blast.

3.4.7 <u>Tank SP-10 Surface Preparation:</u> Accomplish SP-10 for entire surfaces prepared in 3.4.5, using drawings PW 15957, PW 15958, PW 15959, PW 15960, PW 15961, PW 15962, PW 50746, PW 115969, PW 15970, C-1120 and 401-6896568 for guidance.

3.4.7.1 Near white blast clean 100 percent of surfaces listed in 3.4.5. Accomplish the requirements of Surface Preparation Specification SSPC-SP-10 of Systems and Specifications, SSPC Painting Manual, Volume 2. Achieve a surface profile in accordance with the Manufacturers Product Data Sheet.

3.4.7.1.1 Power tool clean inaccessible areas where abrasive blasting equipment cannot reach. Accomplish the requirements of Surface Preparation Specification SSPC-SP-11 of Systems and Specifications, SSPC Painting Manual, Volume 2. Achieve a surface profile in accordance with the Manufacturers Product Data Sheet.

Long Lead: The following materials are considered Long Lead Time Material and shall be ordered by the Contractor within 5 working days of contract award.

3.4.8 <u>Preserve Tank Surfaces:</u> Apply a 12 year service life, marine epoxy coating system: Single coat in accordance with the requirements of MIL-PRF-23236, Type VII, Class 7/18; installed in accordance with the manufacturer’s product data sheets to the entire surfaces listed in each location listed in 3.4.7. Final top coat color configuration shall be white. Use 3.4.2 for guidance.

3.4.8.1 Freshly preserved tank surfaces subject to damage and contamination shall be covered and protected from damage and contamination at all times. Surfaces shall be free of visual contaminants such as oxide films, oils, grease, dust, dirt, or foreign material.

3.4.9 Install Distinguishing Figures: Apply Polysiloxane type coating to markings and distinguishing figures identified in 3.4.2. Coatings shall be in accordance meeting the requirements of MIL-PRF-24635 Type V or VI Grade B; and manufacturer’s product data sheets using drawing and information recorded in 3.4.2 guidance.

(V) "SENSOR PIPE CLEANLINESS"

3.4.10 Sensor Pipe Cleaning: Accomplish TLI sensor pipe cleaning from the operations deck to the concrete level at each location listed in the main ballast tank, forward trim tank and aft trim tank, listed in TABLE 2, using drawing 401-6896568 for guidance.

3.4.10.1 Each pipe shall be 100 percent free of blast grit, rust, and debris.

3.4.11 Trim Tank Hatches Modify: QTY (1 EA) Modify existing Forward and QTY (1 EA) AFT Trim tank hatches; located at (G-5) Vertical Bulkhead; using drawings PW 15958 and PW 15959 as guidance.

3.4.11.1 Remove existing QTY (1 EA) Forward and QTY (1 EA) AFT trim tanks 15-inches by 23-inch hatches, including associate hardware and combing; located on the G-5 level.

3.4.11.1.1 Template exact size, configuration, and location from existing Asset conditions.

3.4.11.2 Install new QTY (1 EA) Forward and QTY (1 EA Inserts) AFT. One EA A36 steel insert in the location of each removed hatch listed in 3.4.11.1.

(I) "FIT UP INSPECTION"

3.4.11.3 Accomplish fit up inspection.

(I) "VISUAL INSPECTION"

3.4.11.4 Accomplish nondestructive testing visual inspection (VT).

(I) "MAGNETIC PARTICLE TEST (MT)"

3.4.11.5 Accomplish nondestructive testing magnetic particle inspection (MT).

(I) "AIR HOSE TEST"

3.4.11.6 Accomplish air hose test of newly installed blanks.

3.4.11.6.1 Air hose nozzle must be placed at less than 3 inches, but greater than zero inches, from the area to be tested and pressure directed at the structure under test in a manner most likely to disclose leaks.

3.4.11.6.2 The minimum nozzle diameter must be 3/8 inch and the nozzle pressure must be 90 plus or minus 5 PSIG as monitored at the nozzle.

3.4.11.6.3 Apply a soapy solution to the opposite side of the structure and inspect for leaks.

3.4.11.6.4 Inspect joint or fitting for leakage by observing for formation of bubbles. Allowable leakage: None.

3.4.11.7 Install new QTY (1 EA) Forward and QTY (1EA) AFT trim tank hatches equal in performance to Nabrico DF-686-5, Aluminum cover, stainless steel tumble bolts and dog handles, A-A-59588A Class 3B Grade 30 gaskets.

3.4.11.7.1 Install new QTY (One EA) Forward trim tank hatch; Frame 28, 6 feet starboard of centerline.

3.4.11.7.2 Hatch shall be 12-inches above (G-4) deck, 30-inches side installed vertical, opens to starboard.

3.4.11.7.3 Install new QTY (One EA) AFT trim tank hatch; Frame 28, 6 feet port of centerline.

3.4.11.7.4 Hatch shall be 12-inches above (G-4) deck 30-inches side installed vertically, opens to port.

3.4.11.7.5 Each hatch shall be installed between vertical structural members.

3.4.11.7.6 Each hatch hinge/cover shall operate clear of any interferences.

3.4.11.7.7 Lubricate and adjust each dogging mechanism for proper closure operation. Install anti-seize sealant compound meeting the requirements of CID-A-A-59004-A to each exposed fastener thread prior to installation.

3.4.11.7.8 New gaskets shall remain free from preservation and coatings.

(I) "AIR HOSE TEST"

3.4.11.8 Accomplish local air test of newly installed hatches in 3.4.11.7. (REQUIRED REPORT CDRL A062)

3.4.11.8.1 Air hose nozzle must be placed at less than 3 inches, but greater than zero inches, from the area to be tested and pressure directed at the structure under test in a manner most likely to disclose leaks.

3.4.11.8.2 The minimum nozzle diameter must be 3/8 inch and the nozzle pressure must be 90 plus or minus 5 PSIG as monitored at the nozzle.

3.4.11.8.3 Apply a soapy solution to the opposite side of the structure and inspect for leaks.

3.4.11.8.4 Inspect joint or fitting for leakage by observing for formation of bubbles. Allowable leakage: None.

(I) "CHALK TEST"

3.4.11.9 Accomplish chalk test of each newly installed equipment listed in 3.4.11.7 as follows: (REQUIRED REPORT CDRL A063)

3.4.11.9.1 Apply chalk to the bearing surface of the knife edge and close the door, hatch or scuttle by normal procedure.

3.4.11.9.2 When the door, hatch or scuttle is opened, the chalk from the knife must have been transferred to the gasket.

3.4.11.9.3 The chalk imprint must be in the center 3/5 of the width of the gasket with 100 percent continuous contact of knife edge to gasket.

(I) "OPERATIONAL TEST"

3.4.11.10 Accomplish an operational test of each new installed equipment listed 3.4.11.7.

3.4.11.10.1 Accomplish three complete open, close, dogging, and un-dogging cycles. Allowable sticking and binding: None.

3.4.12 Main Ballast Tank Hatches Modify: Remove existing and install new QTY (One EA) Forward and QTY (One EA) AFT Main Ballast Tank Hatches; using PW 15959 and PW 15960 as guidance.

3.4.12.1 Template exact size, configuration, and location from existing Asset conditions.

3.4.12.2 Remove existing QTY (One EA) Forward and QTY (One EA) Aft main ballast hatches; 30-inches by 42-inches, 10 dog, 4-inch combing.

3.4.12.3 Install new QTY (One EA) Forward and QTY (One EA) Aft main ballast hatches equal in performance to Nabrico DF-475-5, Aluminum cover, stainless steel tumble bolts and dog handles, A-A-59588A Class 3B Grade 30 gasket.

3.4.12.3.1 100 percent penetration weld install new flush A36 steel deck inserts in way of each removal excess opening, using new material of the same thickness as adjacent structures.

3.4.12.3.2 Installed equipment shall not interfere with adjacent ladders, steps, and stanchions.

3.4.12.3.3 Each hatch hinge shall be oriented outboard.

3.4.12.3.4 Each hatch shall lay flat in fully open position.

3.4.12.3.5 Lubricate and adjust each dogging mechanism for proper closure operation. Install anti-seize sealant compound meeting the requirements of CID-A-A-59004-A to each exposed fastener thread prior to installation.

3.4.12.3.6 New gaskets shall remain free from preservation and coatings.

(I) "FIT UP INSPECTION "

3.4.12.4 Accomplish fit up inspection.

(I) "VISUAL INSPECTION"

3.4.12.5 Accomplish nondestructive testing visual inspection (VT).

(I) "MAGNETIC PARTICLE TEST (MT)"

3.4.12.6 Accomplish nondestructive testing magnetic particle inspection (MT).

(I) "OPERATIONAL TEST"

3.4.12.7 Accomplish operational test each newly installed equipment listed 3.4.12.3.

3.4.12.7.1 Accomplish three complete open, close, dogging, and un-dogging cycles. Allowable sticking and binding: None

(I) "CHALK TEST"

3.4.12.8 Accomplish chalk test of each newly installed equipment listed in 3.4.12.3 as follows:

3.4.12.8.1 Apply chalk to the bearing surface of the knife edge and close the door, hatch or scuttle by normal procedure.

3.4.12.8.2 When the door, hatch or scuttle is opened, the chalk from the knife must have been transferred to the gasket.

3.4.12.8.3 The chalk imprint must be in the center 3/5 of the width of the gasket with 100 percent continuous contact of knife edge to gasket.

3.4.12.9 Fabricate and install 4 each new Egress Stanchions on the operations deck, at each location at (G-4) Forward End, frame 23, centerline; (G-4) AFT End, frame 23, centerline; in accordance with drawing Tank Stanchions Sketch Rev A and using drawing PW 15960 as guidance. Quantities shown in the Material List of the Tank Stanchion Sketch Rev A are for one each Fall Protection Tank Stanchion. Weld attach each Pipe Socket (piece #3) to the Operations Deck in accordance with the Tank Stanchions Sketch Rev A. Template to existing shipboard conditions.

3.4.12.9.1 Install QTY (2 EA) Forward hatch egress stanchion.

3.4.12.9.2 Install QTY (2 EA) AFT hatch egress stanchion.

(I) "FIT UP INSPECTION"

3.4.12.9.3 Accomplish fit up inspection.

(I) "VISUAL INSPECTION (VT)"

3.4.12.9.4 Accomplish nondestructive testing visual inspection.

(I) "MAGNETIC PARTICLE TEST (MT)"

3.4.12.9.5 Accomplish nondestructive testing magnetic particle inspection (MT) of each welded stanchion.

(I) "PULL TEST"

3.4.12.9.6 Accomplish a pull test of each welded stanchion in accordance with note 22 of drawing 804-5184155. (REQUIRED REPORT CDRL A064)

3.4.12.9.7 Install each nut (piece #4) to its corresponding bolt (piece #5) with 1 to 3 threads protruding from the nylon insert of the nut.

3.4.12.9.8 SP-10 blast and preserve each stanchion prior to installation on Asset.

3.4.12.9.9 Apply two coats MIL-PRF-23236, TYPE VII, CLASS 15B preservation coating. Install in accordance with the manufacturer's product data sheets to the entire surfaces listed in each location listed in Table 9. Final topcoat color shall be Federal Standard 595 color number 13591 OSHA Safety yellow.

3.4.13 Trim Tank Horizontal Hatches Modify: Modify existing QTY (One EA) at the forward trim tank horizontal tank hatch and (One EA) at the aft trim tank horizontal tank hatch; located at (G-4), Frame 27, Centerline forward and aft; using drawings PW 15958 and PW 15959 as guidance.

3.4.13.1 Remove existing and install TOTAL QTY (2 EA) One EA Forward and One EA AFT trim horizontal tank hatches, including bolting ring and combing.

3.4.13.2 Install new QTY (One EA) Forward and QTY (One EA) AFT trim tank hatches equal in performance to Nabrico DF-174-5, 15-inches by 23-inches, Aluminum cover, stainless steel tumble bolts and dog handles, A-A-59588A Class 3B Grade 30 gaskets.

3.4.13.2.1 Lubricate and adjust each dogging mechanism for proper closure operation. Install anti-seize sealant compound meeting the requirements of CID-A-A-59004-A to each exposed fastener thread prior to installation.

3.4.13.2.2 New gaskets shall remain free from preservation and coatings.

3.4.13.3 Crop existing deck and fit up, and weld install TOTAL QTY (2 EA) new A36 steel flush insert, One EA forward and One EA aft in way of fit up for new hatch installation at the locations listed in 3.4.13.1.

(I) "FIT UP INSPECTION"

3.4.13.4 Accomplish fit up inspection.

(I) "VISUAL INSPECTION"

3.4.13.5 Accomplish nondestructive testing visual inspection (VT).

(I) "MAGNETIC PARTICLE TEST (MT)"

3.4.13.6 Accomplish nondestructive testing magnetic particle inspection (MT).

(I) "AIR HOSE TEST"

3.4.13.7 Accomplish air hose test of newly installed hatches in 3.4.11. (REQUIRED REPORT CDRL A065)

3.4.13.7.1 Air hose nozzle must be placed at less than 3 inches, but greater than zero inches, from the area to be tested and pressure directed at the structure under test in a manner most likely to disclose leaks.

3.4.13.7.2 The minimum nozzle diameter must be 3/8 inch and the nozzle pressure must be 90 plus or minus 5 PSIG as monitored at the nozzle.

3.4.13.7.3 Apply a soapy solution to the opposite side of the structure and inspect for leaks.

3.4.13.7.4 Inspect joint or fitting for leakage by observing for formation of bubbles. Allowable leakage: None.

(I) "CHALK TEST"

3.4.13.8 Accomplish chalk test of for each newly installed equipment listed in 3.4.13.2 as follows:

3.4.13.8.1 Apply chalk to the bearing surface of the knife edge and close the door, hatch or scuttle by normal procedure.

3.4.13.8.2 When the door, hatch or scuttle is opened, the chalk from the knife must have been transferred to the gasket.

3.4.13.8.3 The chalk imprint must be in the center 3/5 of the width of the gasket with 100 percent continuous contact of knife edge to gasket.

(I) "OPERATIONAL TEST"

3.4.13.9 Accomplish an operational test of each newly installed equipment listed 3.4.11.

3.4.13.9.1 Accomplish three complete open, close, dogging, and un-dogging cycles. Allowable sticking and binding: None

3.4.14 Main Ballast Tank Vent Repair: Accomplish cleaning and inspect of QTY (2 EA) Main Ballast Tank Vents; using drawings PW 15969 and PW 15970 as guidance.

3.4.14.1 Accomplish cleaning of QTY (2 EA) Main Ballast Tank Vents, to be free of debris and foreign matter.

3.4.14.1.1 QTY (One EA), 8 Inch Vent, Steel Pipe Including Screen, Located 10-Inches Below Weather Deck (G-6) to 18 Inches Below Operating Deck (G-4) at Forward Frame 4.5 on Centerline.

3.4.14.1.2 QTY (One EA), 8 Inch Vent, Steel Pipe Including Screen, Located 10-Inches Below Weather Deck (G-6) to 18 Inches Below Operating Deck (G-4) at Aft Frame 4.5 on Centerline.

(V) "BORESCOPE INSPECTION"

3.4.14.2 Accomplish a borescope visual inspection of each interior surface and a visual inspection of each exterior surface of the equipment listed in 3.4.14. Acceptable cleanliness: No air flow obstructing debris or foreign matter.

3.4.15 Tank Ladders Inspect: Inspect existing tank ladders, structural mounting bracket and fasteners; located in Forward Trim Tank, Main Ballast Tank and AFT Trim Tank; using drawings PW 15958, PW 15959, PW 15960, PW 15961 and PW 15962 as guidance. (REQUIRED REPORT CDRL A066)

3.4.15.1 Search for, mark, identify, and record defective (actual and suspect) areas that exhibit signs of severe pitting, excessive erosion, deterioration, damage, or which may be detrimental to the structural integrity of the caisson tank ladders.

3.4.16 Tank Level Indicator (TLI) Mounting Brackets Repairs: Remove existing and install new mounting brackets located in Forward Trim Tank, Main Ballast Tank and AFT Trim tank, using drawings PW 15958, PW 15959, PW 15960, PW 15961 and PW 15962 as guidance.

3.4.16.1 Template exact sizes, locations, and configurations from existing shipboard conditions.

3.4.16.2 Remove existing and install new QTY (10 EA) structural mounting bracket, hanger, clamp, appendage, rubber liner and fastener for each location listed in 3.4.16.

3.4.16.2.1 New mounting brackets shall be equal to the performance characteristics of the Vertical Mount galvanized iron, beam mount bracket McMaster-Carr #8868T85.

3.4.16.2.2 Install new CRES 316 new nylock nuts, bolts, and washers for each item listed in 3.4.16.

(V) "TANK INSPECTION PRIOR TO STAGING REMOVALS"

3.4.17 Tank Pre-Staging Removal Inspection: Accomplish a pre-tank closeout inspection in each location listed in TABLE 2 after completion of preservation, and prior to staging removals.

3.4.17.1 Inspect cleanliness, cured paint, and completion of work prior to staging removals.

3.4.17.2 Each deficiency shall be recorded on the checkpoint form and corrected prior to staging removals.

3.4.17.3 Each corrected deficiency shall be verified by the SUPERVISOR as satisfactorily corrected prior to staging removals in the locations listed in TABLE 2.

(V) "POST STAGING REMOVAL INSPECTION"

3.4.18 Tank Post Staging Removal Inspection: Accomplish a Post Staging Removal Inspection no later than 24 hours after staging removals.

3.4.18.1 Demonstrate that each location listed in TABLE 2 is in a broom clean condition. Locations shall be free of visual contaminants such as oxide films, oils, grease, dust, dirt, blast grit, rust, rust stains or foreign material.

3.4.18.2 Each deficiency shall be recorded and corrected within 24 hours.

3.4.18.3 All corrected deficiencies shall be verified by the SUPERVISOR as satisfactorily corrected prior to the resumption of any other production in the locations listed in TABLE 2.

(V) "TANK CLOSEOUT INSPECTION"

3.4.19 Tank Closeout Inspection: Schedule and accomplish a tank/void closeout inspection of each location listed in TABLE 2 prior to final close of each tank/void. (REQUIRED REPORT CDRL A067)

3.4.19.1 Ensure each location is clean, dry, and free of industrial debris, industrial equipment, refuse, dirt, scrap materials, and other extraneous materials.

3.4.19.2 Notify the SUPERVISOR via email, 2 working days prior to accomplishing the tank/void closeout inspection(s).

3.5 WEATHER DECK REPAIRS:

3.5.1 Identify Distinguishing Figures and Markings: Identify, measure, and record layout of the existing Weather Deck non-skid areas including locations and topcoat colors of each: existing deck border, curb, distinguishing figure, lifting pad-eye deck fitting, bitt, cleat, chock, fitting, line handling area, stanchion, receptacle housing, foundation and stand, shore power receptacle housing, and raised lettering; in sketch form or with photos using drawings PW 15959 and PW 15959 for guidance.

3.5.1.1 Record all name plate locations, material and data on each name plate.

3.5.1.2 Protect each name plate from blast and preservation damage. Alternatively, nameplates can be removed prior to blast and re-installed after weather deck preservation is complete.

3.5.2 Fresh Water Wash Down: Accomplish a freshwater wash down using a minimum of 3000 PSIG nozzle pressure prior to abrasive blasting.

3.5.2.1 Install and maintain protective covering for protection of existing equipment during freshwater wash down operations.

(I) "ULTRASONIC TEST (UT)"

3.5.3 Weather Deck Ultrasonic Test (UT) Inspection: Accomplish a UT measurement inspection of the weather deck surfaces, from Port to Starboard and from forward to aft. Accomplish in accordance with drawing Spare Caisson 4&5 UT Grid. Using drawings PW 15957, PW 15958, PW 15959, PW 15960, PW 15961, PW 15962, T9074-AS-GIB-010/271 and TABLE ONE for guidance. Record and submit readings on the drawing Spare Caisson 4&5 UT Grid. (REQUIRED REPORT CDRL A068)

3.5.3.1 List in one column the design thickness.

3.5.3.2 Record in a separate column if reading is taken over an existing doubler or insert plate.

3.5.3.3 List in another column, next to the design thickness, the actual readings taken.

3.5.2.4 List and identify corresponding specific measurements, locations, and/or areas in relationship to structural members or with measurements from suitable hull reference points.

3.5.3.5 A sketch of the weather deck plating surfaces with appropriate data is an acceptable alternative.

3.5.3.6 Calculate, identify, and highlight measurements indicating a difference (wastage) greater than or equal to 25 percent of the designed plate thickness, using TABLE ONE for guidance.

3.5.4 Weather Deck SP-6 Blasting : Remove existing preservation system of the weather deck surfaces, from port to starboard and from forward to aft and all vertical surfaces including each; appendage, foundation, topside fitting, interior and exterior of each overboard, weather deck bolted access hatch, horizontal and vertical attachment point, main deck mooring bitt and foundation, towing bitt and foundation, cleat and foundation, chock and foundation, deck penetration, lifting pad eye, apron plate, apron plate foundation, each boarding ramp, exterior of each stanchion and interior and exterior each handrail deck socket, weather deck handrail stanchion, weather deck handrail, and main deck coaming. Remove preservation from each receptacle foundation and stand, receptacle housing, alarm stand and foundation, security light stand, and from each shore power receptacle housing. Use drawings PW 15959 and PW 15959 for guidance.

3.5.4.1 Accomplish 100 percent Commercial Blast Clean NACE 3/SSPC-SP-6, of the surfaces listed in 3.5.4 prior to the accomplishment of 3.5.5.

(I) "VISUAL INSPECTION (VT)"

3.5.5 Weather Deck Visual Inspection: Accomplish a Visual Inspection (VT) after completion of SP-6 blast, grit removal and cleanup of the surfaces of 3.5.4. Use drawings PW 15957 and PW 15959 for guidance.

3.5.5.1 Search for, mark, identify, and record defective (actual and suspect) areas that exhibit signs of severe pitting, excessive erosion, deterioration, damage, or which may be detrimental to the structural and watertight integrity of the caisson.

3.5.5.1.1 Unidentified damaged or deformed structure shall be reported within 48 hours of discovery to the SUPERVISOR.

3.5.5.1.2 Identify abandoned appendages to be removed and ground flush on the weather deck, from Port to Starboard and from forward to aft.

3.5.5.1.3 A two dimensional, or AutoCAD 2016, sketch or drawing showing hull/structural configurations, arrangements, relationships, and a listing of the inspection location's data is an acceptable alternative.

3.5.5.2 The locations and dimensions of structural repairs identified by the Ultrasonic Inspection and Visual Inspection Reports of 3.5.3 and 3.5.5. will be listed on the Structural Metal Repair Plan of 3.3.1.1.

3.5.5.2.1 Unidentified damaged weather deck surfaces shall be reported within 48 hours of discovery to the SUPERVISOR.

(I) "STRUCTURAL REPAIR COMPLETION: WEATHER DECK"

3.5.5.3 Accomplish repair of 100 percent of weather deck structure in paragraph 3.3.1, prior to accomplishing the requirements of 3.5.6. Verify all repairs are accomplished in the locations and dimensions as approved by the SUPERVISOR in the structural metal repair plan as identified on the declining log prior to final SP-10 Blast. (REQUIRED REPORT CDRL A069)

3.5.6 Weather Deck SP-10 Surface Preparation: Accomplish SP-10 for entire surfaces prepared in 3.5.4, and each newly installed hatch and each enclosure of 3.5.8, each boarding ramp of 3.5.9 using drawings PW 15957 and PW 15959 for guidance.

3.5.6.1 Near white blast clean 100 percent of surfaces listed in 3.5.6. Accomplish the requirements of Surface Preparation Specification SSPC-SP-10 of Systems and Specifications, SSPC Painting Manual, Volume 2. Achieve a surface profile in accordance with the Manufacturers Product Data Sheet.

3.5.6.1.1 Power tool clean inaccessible areas where abrasive blasting equipment cannot reach. Accomplish the requirements of Surface Preparation Specification SSPC-SP-11 of Systems and Specifications, SSPC Painting Manual, Volume 2 in accordance with the manufacturer’s product data sheet.

Long Lead: The following materials are considered Long Lead Time Material and shall be ordered by the Contractor within 5 working days of contract award.

3.5.7 Weather Deck Preservation: Apply a preservation system with primer, stripe coat, non-skid and top coat in accordance with the requirements of MIL-PRF-24667, Type I, V, VI Comp G; top coat in accordance with the requirements of MIL-PRF-24667 Type V, Comp G. Final top coat color configuration shall match the previously existing final color configuration as identified in 3.5.1. Accomplish in accordance with the Manufacturers Product Data Sheet.

3.5.7.1 Apply the primer and non-skid to the surfaces prepared in 3.5.6 forward and aft and from side to side with the non-skid peaks and valleys coarsely and reasonably uniform, in the same direction, reasonably uniform in color, presenting a roughly uniform course appearance over the entire surface receiving non-skid, and using the layout information recorded in 3.5.1 for guidance.

3.5.7.2 Install new painted safety mark stripe to the entire inboard side of existing port and starboard toe combing.

3.5.7.2.1 Final topcoat color to be Federal Standard 595 Color Number 14120 OSHA Safety Green for STBD, and Color Number 11140 OSHA Safety Red for PORT.

3.5.7.3 Apply a topcoat coating to prepared surfaces of 3.5.6 not receiving non-skid, and all vertical surfaces, in accordance with the requirements of MIL-PRF-24635, Type V or VI Grade B and manufacturer's product data sheets using the layout information recorded in 3.5.1 for guidance.

3.5.7.3.1 Final topcoat color shall match the previously existing final color configuration using 3.5.1 for guidance.

3.5.7.3.2 Final top coat color configuration of stanchions and locations listed as yellow in 3.5.1 shall be Federal Standard 595 color number 13591 OSHA Safety yellow.

3.5.7.3.3 Faying surfaces shall not be painted.

3.5.8 Forward and AFT Access Hatches and Enclosures Install: Remove existing hatches and install new QTY (One EA) Forward and QTY (One EA) AFT Access Enclosures; using drawings PW 70147, 139752, PW 15958, and Spare Caisson Door Sill Sketch as guidance, and in accordance with 16002080, and 16002112.

3.5.8.1 Template exact size, configuration, materials, and locations from existing Asset conditions.

3.5.8.2 Remove existing equipment QTY (One EA), Watertight Deck Hatch Cover and fastening components, Forward, Starboard, between frames 20 and 24 and QTY (One EA) Watertight Deck Hatch Cover and fastening components, AFT, Port, between frames 20 and 24.

3.5.8.2.1 Grind surfaces flush and smooth in way of removals.

3.5.8.3 Remove existing equipment QTY (One EA), Watertight Deck Hatch Combing, Forward, Starboard, between frames 20 and 24 and QTY (One EA), and Watertight Deck Hatch Combing, AFT, Port, between frames 20 and 24.

3.5.8.3.1 Lower the existing combing on the 3 non-access sides to 1.5 inches above the deck.

3.5.8.3.2 Grind flush and smooth the access side of the combing to the deck.

3.5.8.3.3 Terminate the non-access sides of the combing at a 45-degree angle where it meets the access sides of the combing that is being ground flush.

3.5.8.4 Remove existing equipment QTY (One EA) counterweight and associated foundations forward, starboard, between frames 20 and 24 and QTY (One EA), counterweight and associated foundations AFT, port, between frames 20 and 24.

3.5.8.4.1 Grind surfaces flush and smooth in way of removals.

3.5.8.5 Weld install new QTY (One EA) new access enclosure at each location listed in 3.5.8.

(I) "FIT UP INSPECTION"

3.5.8.5.1 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.5.8.5.2 Accomplish nondestructive testing visual inspection (VT).

(I) "MAGNETIC PARTICLE TEST (MT)"

3.5.8.5.3 Accomplish nondestructive testing magnetic particle inspection (MT).

3.5.8.6 Preserve the interior and exterior surfaces of the newly installed forward and aft enclosures of 3.5.8 and each handrail extension of

3.5.8.7. each watertight door of 3.5.8.8 and each weather deck boarding ramp of 3.5.9 in accordance with paragraphs 3.5.6 and 3.5.7.

3.5.8.7 Extend top portion of QTY (2 EA), Access Ladder Handrails, 1-Inch Pipe Rail, Forward, Starboard, between frames 20 and 24; QTY (2 EA), Access Ladder Handrails, 1-Inch Pipe Rail, AFT, Port, between frames 20 and 24., using drawing PW 70147 139752, PW 15958 as guidance.

(I) "FIT UP INSPECTION"

3.5.8.7.1 Accomplish fit up inspection.

(I) "VISUAL INSPECTION"

3.5.8.7.2 Accomplish nondestructive testing visual inspection (VT).

(I) "MAGNETIC PARTICLE TEST (MT)"

3.5.8.7.3 Accomplish nondestructive testing magnetic particle inspection (MT).

3.5.8.8 Weld install new QTY (One EA), Hatch, Equal to the following: Aluminum, Type B, Manly Marine Model D-020, Quick Acting Watertight door, A-A- 59588A Class 3B Grade 30 gasket, in the forward enclosure; and QTY (One EA), Hatch, Equal to the following: Aluminum, Type B, Manly Marine Model D-020, Quick Acting Watertight door, A-A- 59588A Class 3B Grade 30 gasket, in the aft enclosure, on each new access enclosure door opening, using drawing 16002112 as guidance.

(I) "FIT UP INSPECTION"

3.5.8.8.1 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.5.8.8.2 Accomplish nondestructive testing visual inspection (VT).

(I) "MAGNETIC PARTICLE TEST (MT)"

3.5.8.8.3 Accomplish nondestructive testing magnetic particle inspection (MT).

3.5.8.8.4 Lubricate and adjust each new hatch dogging mechanism for proper closure operation.

3.5.8.8.5 Lubricate 100 percent of each threaded surface with anti-seize compound meeting the requirements of CID-A-A-59004-A.

3.5.8.8.6 New gaskets shall remain free from preservation and coatings.

3.5.8.8.7 New Hatch shall have new locking hasp installed.

(I) "OPERATIONAL TEST"

3.5.8.8.8 Accomplish three complete open, close, dogging, and un-dogging cycles. Allowable sticking and binding: None

(I) "CHALK TEST"

3.5.8.9 Accomplish chalk test of for each newly installed equipment listed in 3.5.3.7.

3.5.8.9.1 Apply chalk to the bearing surface of the knife edge and close the door, hatch or scuttle by normal procedure.

3.5.8.9.2 When the door, hatch or scuttle is opened, the chalk from the knife must have been transferred to the gasket.

3.5.8.9.3 The chalk imprint must be in the center 3/5 of the width of the gasket with 100 percent continuous contact of knife edge to gasket.

(I) "WATER HOSE TEST"

3.5.8.10 Accomplish a water hose test of for each newly installed equipment listed in 3.5.3.7.

3.5.8.10.1 Use a one and one-half inch hose with a minimum nozzle diameter of one-half inch. Pressure at the nozzle must be no less than 50 PSIG at a maximum distance of 10 feet from the surface being tested.

3.5.8.10.2 The stream of water must be directed against the structure in a manner most likely to disclose leaks. The opposite side of the structure must be inspected to detect and locate leaks. Allowable leakage: None.

3.5.8.11 Accomplish preservation of entire surface of exterior sides of each new installed access enclosure in accordance with the requirements of 3.5.1.

3.5.8.11.1 Accomplish preservation to identified alarm bell, siren on Forward and AFT enclosure, at designated locations by the SUPERVISOR.

3.5.8.11.2 Apply a topcoat coat in accordance with the requirements of MIL-PRF-24635, Type V or VI Grade B and manufacturer’s product data sheets. Alarm bell identification lettering shall be One inch in height and color shall Federal Standard 595 Color Number 17925 White.

(I) "BOARDING RAMP INSPECTION"

3.5.9 Weather Deck Boarding Ramp Inspect: Accomplish inspection of QTY (One EA) Weather Deck Boarding Ramp, FWD, Frame 35; QTY (One EA) Weather Deck Boarding Ramp, AFT, Frame 35; using drawing B-3739 as guidance.

3.5.9.1 Search for, mark, identify, and record defective (actual and suspect) areas that exhibit signs of severe pitting, excessive erosion, deterioration, damage, or which may be detrimental to the structural and watertight integrity of the caisson.

(I) "OPERATIONAL TEST"

3.5.9.2 Accomplish three complete open, close, dogging, and un-dogging cycles. Allowable sticking and binding: None.

3.5.10 Weather Deck Bolted Access Hatch Repair: Accomplish repairs to QTY (One EA) Bolted Access Hatch; using drawing PW 15959 as guidance.

3.5.10.1 Remove existing QTY (68 EA) Nuts, QTY (68 EA) Washers from item listed in 3.5.10.

3.5.10.1.1 Match mark each side of the Access Hatch to indicate orientation of the hatch for relocation on the Caisson.

3.5.10.2 Remove existing QTY (68 EA) Welded bolts from item listed in 3.5.10.

3.5.10.2.1 Grind surfaces flush and smooth in way of removals.

3.5.10.3 Modify Weather Deck Bolted Access Hatch listed in 3.5.10. Install QTY (68 EA) clearance holes in the location the removed existing nuts of 3.5.10.2.

3.5.10.4 Weld install new QTY (68 EA) ½-inch -13 Nuts, Steel, Grade 5 in the Operations Compartment Overhead in accordance with 3.5.10.8.

3.5.10.5 Install new QTY (68 EA) ½-inch -13 Bolts, CRES, MIL-DTL-1222, Type I, Grade 316, length to suit, in the Operations Compartment Overhead.

3.5.10.5.1 Install Anti-seize meeting the performance characteristics of CID-A-A-59004 on 100 percent of threaded surfaces of new equipment.

3.5.10.6 Install new QTY (68 EA) CRES/EPDM Neoprene Rubber bonded Sealing Washers, in the Operations Compartment Overhead.

3.5.10.7 Install new QTY (One EA) Gasket, 3-inch wide, ¼-inch thick, A-A-59588A, Class 3B, Grade 30, ZZ-R-765, in the Operations Compartment Overhead.

3.5.10.8 Weld attach each new nut of 3.5.10.4 with a 100% 3/16-inch fillet weld to surrounding structure of hatch opening, for each new nut. Assure each nut aligns with the new clearance holes previously installed in the operations deck compartment overhead.

(I) "FIT UP INSPECTION"

3.5.10.8.1 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.5.10.8.2 Accomplish visual inspection (VT).

3.5.10.9 Tap 100 percent each fully welded new nut installed in 3.5.10.4 prior to installation of new bolts listed in 3.5.10.5.

3.5.10.10 Repair each gasket-sealing surface of 3.5.10 by removing high spots, burrs, abrasions, and foreign matter by hand tools. Take precautions to maintain phonographic finish. Ensure no excessive material is removed causing loss of critical size or alignment.

(V) "VISUAL INSPECTION"

3.5.10.11 Accomplish visual inspection for each newly installed nut listed in 3.5.10.8. Acceptance criteria: maintain phonographic finish of sealing surface of 3.5.10. Ensure no excessive material is removed causing loss of critical size or alignment. (REQUIRED REPORT CDRL A070)

3.5.10.12 Remove existing and install new QTY (One EA) Gasket, 3 inch wide, ¼ inch thick, A-A-59588A, Class 3B, Grade 30, ZZ-R-765.

3.5.10.12.1 New gaskets shall remain free from preservation and coatings.

3.5.10.12.2 Install 3M 4000UV or equal temporary watertight sealing compound on 100 percent of each surface of the equipment listed in 3.5.10.12 immediately prior to installation the Bolted Access Hatch of 3.5.10.

3.5.10.13 Weld install new QTY (4 EA) swivel hoist ring foundations, into the Bolted Access Hatch of 3.5.10. Install A-36 steel, 1 inch thick. Diameter of each foundation shall be 2 1/2 inches. Template to existing shipboard conditions.

3.5.10.13.1 Locate each foundation of 3.5.10.13 six inches inboard of each hatch corner and adjacent to a structural reinforcement member.

3.5.10.13.2 Foundations QTY (4 EA) shall be spaced equidistantly from each other.

3.5.10.13.3 Foundations shall be located flush to the top surface of 3.5.10.

3.5.10.13.4 Weld install new up to QTY (4 LF) flat bar, A36 steel, ¼ inch ; as structural reinforcement on the underside of the Bolted Access Hatch of 3.5.10 as designated by the SUPERVISOR. Template to existing shipboard conditions.

3.5.10.13.5 Accomplish full penetration weld around 100 percent of the top surface of each foundation.

3.5.10.13.6 Accomplish 3/16 inch fillet weld around 100 percent of the remainder of each foundation and structural reinforcement.

(I) "FIT UP INSPECTION"

3.5.10.13.7 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.5.10.13.8 Accomplish nondestructive testing visual inspection (VT).

(I) "MAGNETIC PARTICLE TEST (MT)"

3.5.10.13.9 Accomplish nondestructive testing magnetic particle inspection (MT).

3.5.10.14 Drill and tap each foundation for each swivel hoist ring foundation of 3.5.10.13 to thread size of ½ inch – 13.

3.5.10.14.1 Chase and clean threads of each installed item prior to installation.

3.5.10.15 Install new QTY (4 EA) Lifting Hoist Ring, Heavy Duty, ½ inch – 13, 2250 lb. Working Load Limit, Equal to performance characteristic of ADB 33316, Grainger Part #39RJ41.

3.5.10.15.1 Equipment shall be tagged with engraved CRES metal tagged with OEM certified weight limit and test data of 3.5.10.16.1.

(I) "PULL TEST"

3.5.10.16 Accomplish a 10 minute pull test of each installed equipment listed in 3.5.10.15 at 2,500 pounds. (REQUIRED REPORT CDRL A071)

3.5.10.16.1 Create certification pull test data listing date of test, test duration, pull weight, equipment used, date of satisfactory test result, and person/company that performed each test.

3.5.10.16.2 Flush install new QTY (4 EA) Threaded Plugs, Brass, ½ inch – 13, Flush Fit within Swivel Hoist Foundation, Allen Head; using Teflon tape to accomplish watertight seal after completion of pull testing. Deliver Lifting Hoist Rings of 3.5.10.15 to the SUPERVISOR.

3.5.10.16.3 Grind surfaces flush and smooth in way of removals.

3.5.10.16.4 Install weather deck bolted access hatch of 3.5.10. Torque all fasteners using MIL-STD-777 for guidance.

(I) "WATER HOSE TEST"

3.5.10.17 After installation and bolt up of the weather deck bolted access hatch , accomplish a water hose test for 15 minutes, of the sealing surfaces of the weather deck bolted access hatch and each newly installed lifting threaded plug after all work is complete and plugs are installed in threaded holes.

3.5.10.17.1 Use a one and one-half inch hose with a minimum nozzle diameter of one-half inch. Pressure at the nozzle must be no less than 50 PSIG at a maximum distance of 10 feet from the surface being tested.

3.5.10.17.2 The stream of water must be directed against the structure in a manner most likely to disclose leaks. The opposite side of the structure must be inspected to detect and locate leaks. Allowable leakage: None.

3.5.11 Sounding Tube Repair: Remove existing and install Main Ballast and Trim Tank Weather Deck Socket, Deck Fitting and Striker Plate Repair; located at Weather Deck Frame FWD 26, Frame AFT 26, Frame FWD 29, and Frame 29 AFT; in accordance with MIL-STD-777.

3.5.11.1 Remove existing and install new QTY (4 EA) Deck Socket Fitting 1-1/2 inch NPS, Carbon Steel, Type B, equal in performance characteristics as to Tate Andale Part no. 58-50.

(I) "FIT UP INSPECTION"

3.5.11.1.1 Accomplish fit up inspection.

(I) "VISUAL INSPECTION"

3.5.11.1.2 Accomplish nondestructive testing visual inspection (VT).

3.5.11.1.3 Template exact size, configuration, and location from existing shipboard condition.

3.5.11.1.4 Install anti-seize sealant compound meeting the requirements of CID-A-A-59004-A to 100 percent of each threaded surface prior to installation of coupling.

3.5.11.2 Remove existing and install new QTY (4 EA) deck socket coupling, brass with ¾ inch square female key.

(I) "FIT UP INSPECTION"

3.5.11.2.1 Accomplish fit up inspection.

(I) "VISUAL INSPECTION"

3.5.11.2.2 Accomplish nondestructive testing visual inspection (VT).

3.5.11.2.3 Template exact size, configuration, and location from existing shipboard condition.

3.5.11.2.4 Chip and grind each surface flush in way of each repair.

3.5.11.2.5 Install anti-seize sealant compound meeting the requirements of CID-A-A-59004-A to 100 percent of each threaded surface prior to installation of coupling.

3.5.11.3 Provide new QTY (2 EA) deck socket fitting "T" Handle key, to the SUPERVISOR.

(I) "AIR HOSE TEST"

3.5.11.4 Accomplish local air test of newly deck socket fittings in 3.5.11.

3.5.11.4.1 Air hose nozzle must be placed at less than 3 inches, but greater than zero inches, from the area to be tested and pressure directed at the structure under test in a manner most likely to disclose leaks.

3.5.11.4.2 The minimum nozzle diameter must be 3/8 inch and the nozzle pressure must be 90 plus or minus 5 PSIG as monitored at the nozzle.

3.5.11.4.3 Apply a soapy solution to the opposite side of the structure and inspect for leaks.

3.5.11.4.4 Inspect joint or fitting for leakage by observing for formation of bubbles. Allowable leakage: None.

3.5.11.5 Accomplish piping system in accordance with the following:

(I) "HYDOSTATIC TEST"

3.5.11.5.1 Accomplish hydrostatic test at 2 PSI for 30 minutes of all piping, fittings, and components after installing in the caisson for evidence of external leakage and/or deformation. Allowable external leakage and/or deformation: None.

3.5.11.5.2 Provide a sketch of that portion of the system to be tested, showing the location of blanks, isolation valves, test connection, joints previously tested/inspected under 3.5.11 and the location of air vents to vent air. Sketch must be on the test site during the accomplishment of the test.

(I) "OPERATIONAL TEST"

3.5.11.5.3 Accomplish an operational test of each new and disturbed gravity drain piping for proper operation and unobstructed flow. Allowable obstruction of flow: None.

3.5.12 Weather Deck Handrail Stanchions Repair: Accomplish repairs to each weather deck handrail stanchion; using drawings PW 15957 and 804-5184155 as guidance.

3.5.12.1 Remove existing and QTY (69 EA) stanchions, locking clamps, removable and adjustable component(s), locking pins and safety chains.

3.5.12.1.1 Replace with new the safety chains, locking clamps, removable and adjustable component(s) and locking pins of 3.5.12.1.

3.5.12.2 Straighten QTY (30 EA) distorted existing stanchions.

(V) "VISUAL INSPECTION"

3.5.12.3 Accomplish a post blast visual inspection of each stanchion listed in 3.5.12. Search for, mark, identify, and record defective (actual and suspect) areas showing signs of cracks, pitting, corrosion, deterioration, damage, distortion, and unusual wear.

3.5.12.4 Install Anti-seize compound on 100 percent of each post blast faying surface of equipment listed 3.5.12, product meeting the requirements of CID-A-A-59004-A.

(I) "HAMMER TEST"

3.5.12.5 Hammer test for soundness and strength each deck socket for the stanchions listed in 3.5.12, in accordance with note 22 of drawing 804-5184155. Allowable Deformation: None.

(I) "OPERATIONAL TEST STANCHION ASSEMBLIES"

3.5.12.6 Install and operate each portable stanchion and safety chain to verify ease of installation, fit, and smooth operation of each stanchion’s removable and adjustable component(s).

(I) "STANCHION TEST PULL"

3.5.12.7 Accomplish a test pull for each installed component of paragraph 3.5.12 including all deck edge lifelines, turnbuckles, sockets, socket hooks and fittings when assembled as a line with its end fittings. Test shall be in accordance with the “Life Assemblies” paragraph of note 22 of drawing 804-

5184155. Allowable deformation: None. (REQUIRED REPORT CDRL A072)

(I) "INTERMEDIATE LIFELINE STANCHION TEST PULL"

3.5.12.8 Accomplish a test pull for each installed intermediate lifeline stanchion of paragraph 3.5.12. Test shall be in accordance with the "Intermediate Lifeline Stanchions" paragraph of note 22 of drawing 804-5184155. Allowable deformation: None. (REQUIRED REPORT CDRL A073)

(I) "OPERATIONAL TEST STANCHION ASSEMBLIES"

3.5.12.9 Install and operate each portable stanchion and safety chain of paragraph 3.5.12. to verify ease of installation, fit, and smooth operation of each stanchion's removable and adjustable component(s).

3.5.13 Shore Power Receptacles Replace: Remove existing and install new QTY (2 EA), Cable Connector; located at Weather Deck Forward, Frame 28; and Weather Deck AFT, Frame 28; using drawing MIL-STD-MIL-STD-2003-3B(SH) and MIL-STD-1310 REV H as guidance.

3.5.13.1 New cable connectors of 3.5.8 shall be equal in performance with Appleton Cat. No.: AR40144, 4W, 4P, Style 1.

3.5.13.2 Template exact size, configuration, and location from existing shipboard condition.

(I) "PRE CONTAINMENT AND STAGING REMOVAL INSPECTION"

3.5.14 Main Deck Pre-Staging Removal Inspection: Schedule and accomplish a Weather Deck closeout inspection of each location listed in 3.5.1.

3.5.14.1 Inspect cleanliness, cured paint, and completion of work prior to staging or containment removals.

3.5.14.1.1 If containment and scaffolding are removed in separate evolutions. Conduct an inspection at each evolution.

3.5.14.2 Verify that protections have been applied to the surfaces of the deck in the location listed in 3.5.1 to prevent the contamination of the deck by fugitive blast grit and dust generated during the removal of staging or containment.

(I) "POST CONTAINMENT AND STAGING REMOVAL INSPECTION"

3.5.15 Main Deck Post Staging Removal Inspection; Accomplish a Post Containment and Staging Removal Inspection no later than 24 hours after containment or staging is removed.

3.5.15.1 Entire weather deck shall be broom clean, dry, oxide films, oils, grease, dust, dirt, blast grit, rust and rust stains or foreign material.

3.5.15.2 Each deficiency shall be recorded on the checkpoint form and corrected within 24 hours.

3.5.15.3 All corrected deficiencies shall be verified by the SUPERVISOR as satisfactorily.

3.5.15.4 Notify the SUPERVISOR via email, 2 working days prior to accomplishing the weather deck closeout inspection(s).

3.6 **OPERATIONS DECK:** Accomplish repairs of operations deck as follows:

3.6.1 Identify Distinguishing Figures and Markings: Identify and record existing color schemes, markings, and figures in sketch form listing colors, locations, sizes, and distinguishing marking(s), and figure(s) information, using drawings PW 15969, PW 15970 and PW 15958 for guidance.

3.6.1.1 Record all name plate locations, material and data on each name plate.

3.6.1.2 Protect each name plate from blast and preservation damage. Alternatively, nameplates can be removed prior to blast and re-installed after operations deck preservation is complete.

(I) "ULTRASONIC TEST (UT)"

3.6.2 Operations Deck Ultrasonic Test (UT) Inspection: Accomplish UT measurement inspection of operations deck surfaces, including structural members in the location, in accordance with ANSI/AWS D1.1 in the locations of the operations deck, between forward frame 28 through aft frame 28, port to starboard at (G-4); operations compartment overhead, between forward frame 28 through aft frame 28, port to starboard at (G-5). Accomplish in accordance with drawing Spare Caisson 4&5 UT Grid, using PW 15958 PW 15969, PW 15970 and T9074-AS-GIB-010/271 and TABLE ONE for guidance. Record and submit readings on the drawing Spare Caisson 4&5 UT Grid. (REQUIRED REPORT CDRL A074)

3.6.2.1 List in one column the design thickness.

3.6.2.2 Record in a separate column if reading is taken over an existing doubler or insert plate.

3.6.2.3 List in another column, next to the design thickness, the actual readings taken.

3.6.2.4 List and identify corresponding specific measurements, locations, and/or areas in relationship to structural members or with measurements from suitable hull reference points.

3.6.2.5 A sketch of the tank plating and structural surfaces with appropriate data is an acceptable alternative.

3.6.2.6 Calculate, identify, and highlight measurements indicating a difference (wastage) greater than or equal to 25 percent of the designed plate thickness, using TABLE ONE for guidance.

3.6.3 Operations Compartment Preservation Inspection: Accomplish touch up preservation in Operations Compartment; using drawings PW 15958, PW 15959, PW 15961, PW 15962, PW 15963 and PW 15970 as guidance.

(V) "VISUAL PRESERVATION INSPECTION (VT)"

3.6.3.1 Accomplish a visual inspection of the interior surfaces including each deck, bulkhead, overhead, external and internal keyway surface, sounding tube, tank vent, piping, pipe support, structural member, foundation, motor valve operator floor stand, dewatering pump motor stand, hatch, appendage, attachment, combing, foundation, access, exterior side of access cover, ladder, ladder cage, and handrail and ventilation duct of the operations deck for structural deformation, deterioration, and/or defects in coatings, using PW 15958, PW 15959, PW 15961, PW 15962, PW 15963 and PW 15970 for guidance.

3.6.3.1.1 Search preservation coatings for and, mark, identify, and record defective coating (actual and suspect) areas showing signs of cracks, pitting, corrosion, deterioration, damage, distortion, and unusual wear.

(V) "VISUAL INSPECTION (VT)"

3.6.4 Operations Deck Visual Inspection: Accomplish a Visual Inspection (VT) of the entire surfaces listed in 3.6.3.1. Use drawings PW 15958, PW 15959, PW 15961, PW 15962, PW 15963 and PW 15970 for guidance.

3.6.4.1 Search for, mark, identify, and record defective (actual and suspect) areas that exhibit signs of severe pitting, excessive erosion, deterioration, damage, or which may be detrimental to the structural and watertight integrity of the caisson.

3.6.4.1.1 Unidentified damaged or deformed structure shall be reported within 48 hours of discovery to the SUPERVISOR.

3.6.4.1.2 Identify abandoned and unused appendages to be removed and ground flush.

3.6.4.1.3 A two dimensional, or AutoCAD 2016, sketch or drawing showing hull/structural configurations, arrangements, relationships, and a listing of the inspection location's data is an acceptable alternative.

3.6.5 The locations and dimensions of structural repairs identified by the Ultrasonic Inspection and Visual Inspection Reports of 3.6.2 and 3.6.4 will be listed on the Structural Metal Repair Plan of 3.3.1.1.

3.6.5.1.1 Unidentified damaged operations deck surfaces shall be reported within 48 hours of discovery to the SUPERVISOR.

(I) "STRUCTURAL REPAIR COMPLETION: OPERATIONS DECK"

3.6.5.2 Structural inspections and repairs of 3.6.2 and 3.6.4 shall be completed prior to accomplishing the requirements of 3.6.6. Verify all repairs are accomplished by the Contractor in the locations and dimensions as approved by the SUPERVISOR in the Repair Plan as identified on the Declining Log prior to SP-11 preservation of 3.6.6 and 3.6.7.

3.6.6 Operations Deck Preservation: Solvent clean the entire existing surfaces listed in 3.6.3.1. Accomplish the requirements of Surface Preparation Specification SSPC-SP-1.

3.6.6.1 Accomplish the requirements of Surface Preparation Specification SSPC-SP-11 of Systems and Specifications, SSPC Painting

Manual, Volume 2. Surface preparation areas will be identified by the SUPERVISOR based on the visual inspection report of 3.6.3.1. QTY not to exceed 1000 square feet. Any amount less than 1000 square feet that is not required will be descoped. Achieve a surface profile in accordance with SSPC Painting Manual, Volume 2 and the Manufacturers Product Data Sheet.

3.6.6.2 Apply preservation coatings in accordance with the requirements of MIL-PRF-23236, TYPE VII, CLASS 15B; Apply 2 coats. Install in accordance with the manufacturer's product data sheets to the entire surfaces listed in each location listed in 3.6.6.2.

3.6.6.3 Final topcoat color of disturbed existing surfaces that were colored white shall be Federal Standard 595 Color Number 17925 White.

3.6.6.4 Apply a topcoat coating to disturbed distinguishing figures and markings, in accordance with the requirements of MIL-PRF-24635, Type V or VI Grade B and manufacturer's product data sheets. Final topcoat color to match surrounding coating. Use the layout information recorded in 3.6.1 for guidance.

3.6.6.4.1 Final topcoat to disturbed handrails, handrail stanchions and safety rails shall be Federal Standard 595 color number 13591 OSHA Safety yellow.

3.6.6.4.2 Install new 6 inch high painted safety stripe to the entire inboard side of full length of horizontal structural member of (G-5).

3.6.6.4.3 Final topcoat color of the new safety stripes of 3.6.5.8.2 shall be Federal Standard 595 Color Number 14120 OSHA Safety Green for STBD, and Color Number 11140 OSHA Safety Red for PORT.

3.6.7 <u>Install Cosmetic Polymeric Deck Covering System:</u> Accomplish installation of a new deck covering system including underlayment, bond coat, basecoat, color coat, color flake and clear sealer conforming to the requirements of MIL-PRF-32584 type I or II, in accordance with the manufacturers product data sheet. Finish to be similar in appearance to Dex-O-Tex Elastaflake 431 Bright Gray. Deck system shall be applied with white chips. Installation to be located at Main Deck (G-4), from Forward frame 28 to AFT frame 28 and from starboard hull side shell to port hull side shell; Deck (G-5), from Forward frame 28 bulkhead to 4 inches forward of Forward frame 22 and from starboard hull side shell to port hull side shell; Deck (G-5), from AFT frame 28 bulkhead to 4 inches Aft of AFT frame 22 and from starboard hull side shell to port hull side shell; using drawings PW 15958 and PW 15959 as guidance.

3.6.7.1 Accomplish the requirements of Surface Preparation Specification SSPC-SP-11 of Systems and Specifications, SSPC Painting Manual, Volume 2 for the surfaces of 3.6.7 and 6 inches up each adjacent vertical surface. Achieve a surface profile in accordance with SSPC Painting Manual, Volume 2 and the Manufacturers Product Data Sheet.

3.6.7.2 Apply preservation coatings in accordance with the requirements of ONE COAT MIL-PRF-23236, TYPE V, VI, OR VII, CLASS 5. Apply two coats. Install in accordance with the manufacturer's product data sheets to the entire surfaces listed in each location listed in 3.6.7 and 6 inches up each adjacent vertical surface. Final topcoat color configuration shall be Federal Standard 595 color number 17925 OSHA Safety White.

3.6.7.3 Install new collars at each end of the foundations for the existing motor control panels, and at each end of the foundations for the new control console, for the equipment located on the operations deck. Collars shall span the distance between the foundations.

3.6.7.4 New collars shall be CRES Grade 316, 3/8-inch high by 3/8-inch thick and installed 1/4-inch peripherally to the areas listed in 3.6.7.3. Seal each collar to the deck, using epoxy compound conforming to the requirements of MIL-PRF-24176.

3.6.7.5 Remove unused remnants, clips, brackets, and weldments from decks and vertical surfaces receiving new deck coverings.

(I) "SURFACE CLEANLINESS"

3.6.7.6 Prior to installation of underlayment. Accomplish cleaning for each deck surface, including up the adjacent vertical surface intersecting the deck up to six inches above the complete deck covering system level. Cleaning shall be in accordance with SSPC SP-1. For components less than 1 inch from deck covering, the required surface preparation and primer application must be completed up vertical surfaces at the same height that the decking system will be installed. Final cleaned surface shall meet the surface preparation requirements of Surface Preparation Specification SSPC-SP-1. Document results on appendices of 1.16.1.1.

3.6.7.6.1 If solvent is used to clean the deck at any point in the installation process, the deck must be allowed to dry before application of any coating. No visible solvent must be present on deck surfaces prior to proceeding with the next process step.

3.6.7.7 Install underlayment conforming in way of low spots, dish pans, and high points that cannot be ground flush, to provide a smooth and fair surface. Slope and fair as required where installed. Underlayment shall be installed in accordance with manufacturer's product data sheet.

(V) "VISUAL INSPECTION"

3.6.7.8 Accomplish a visual inspection of the completely installed and cured base coat. Base coat must have a smooth, continuous surface that is free of air bubbles that penetrates any layer of the deck covering system. No embedded contaminants such as dust or fibers must be visible on the deck at 45 degrees to the surface when viewed using the as-installed ambient lighting source from a standing position. (REQUIRED REPORT CDRL A075)

3.6.7.9 Apply base coat shall be in accordance with manufacturer’s product data sheet.

3.6.7.10 Apply color coat in accordance with manufacturer’s product data sheet.

3.6.7.11 Apply color chips to attain approximately 20 percent chip coverage of the color coat. Apply in accordance with manufacturer’s product data sheet.

3.6.7.12 Apply one clear sealer coats. Apply in accordance with manufacturer’s product data sheet.

(V) "OPERATIONAL TEST”

3.6.7.12.1 Verify all operations deck fittings are free, removable, and operational. Fittings shall not be sealed over during the installation of sealer coats.

(V) "VISUAL INSPECTION"

3.6.8 <u>Operations Compartment Handrails and Stanchions Modify:</u> Accomplish alterations to Operations Compartment handrails and stanchions;

located at Operations Compartment, (G-5), Forward Mezzanine, Outboard of egress ladder; Operations Compartment, (G-5), AFT Mezzanine, Outboard of egress ladder; Operations Compartment, (G-5), Forward Mezzanine, Outboard of (G-4) to (G-5) ladder; Operations Compartment, (G-5), AFT Mezzanine, Outboard of (G-4) to (G-5) ladder; using drawings PW 15958 and 804-5184155 as guidance.

3.6.8.1 Remove existing QTY (4 EA), temporary safety chain, located at Outboard of Forward Egress Ladder; temporary safety chain, located at Outboard of Forward (G-4) to (G-5) Ladder; temporary safety chain, located at Outboard of AFT Egress Ladder; temporary safety chain, located at Outboard of AFT (G-4) to (G-5) Ladder. Remove associated handrails and stanchions of 3.6.8 including 100 percent of attachments.

3.6.8.2 Template exact sizes, locations, and configurations from existing shipboard conditions.

3.6.8.3 Install QTY (One EA) Handrail, Yellow, 24 inches long, equal to performance characteristics of Grainer 32X797 and QTY (2 EA) Deck Sockets, equal to performance characteristics of Grainer 2HER7.

3.6.8.3.1 Location for one set of Handrail and Sockets shall be Operations Compartment, (G-5), Forward Mezzanine, Outboard of egress ladder.

3.6.8.3.2 Location for one set of Handrail and Sockets shall be Operations Compartment, (G-5), AFT Mezzanine, Outboard of egress ladder.

3.6.8.3.3 Match template and drill 7/16-inch clearance holes through G-5 deck, to install Deck Sockets, equal to performance characteristics of Grainer 2HER7 to deck.

3.6.8.4 Install QTY (One EA) Handrail, Yellow, 48 inches long, equal to performance characteristics of Grainer 2HER9 and QTY (2 EA) Deck Sockets, equal to performance characteristics of Grainer 2HER7.

3.6.8.4.1 Location for one set of Handrail and Sockets shall be Operations Compartment, (G-5), Forward Mezzanine, Outboard of (G-4) to (G-5) ladder.

3.6.8.4.2 Location for one set of Handrail and Sockets shall be Operations Compartment, (G-5), AFT Mezzanine, Outboard of (G-4) to (G-5) ladder.

3.6.8.4.3 Match template drill 7/16-inch clearance holes through G-5 deck, to install Deck Sockets, equal to performance characteristics of Grainer 2HER7 to deck.

3.6.8.4.4 Fasten handrail and socket assemblies to (G-5) deck. Install new CRES fasteners including 3/8” x 16 threads per inch bolt (32 EA), 3/8-inch flat washers (32 EA), 3/8 x 16 threads per inch hex nut (32 EA), QTY (64 EA) Flat Washer, 3/8 inch, CRES, QTY (One EA) top and bottom.

3.6.8.4.5 Install anti-seize sealant compound meeting the requirements of CID-A-A-59004-A to each fastener thread prior to installation.

(I) “FASTENER TORQUE ”

3.6.8.4.6 Torque each bolt installed to 210-inch pounds in a diametrically opposed pattern at three separate and progressively incremental settings in accordance with manufacturer's instructions and the requirements of NSTM Chapter 075.

(I) “STRENGTH TEST”

3.6.8.4.7 Accomplish a strength test in accordance with drawing 804-5184155 note 22. (REQUIRED REPORT CDRL A076)

3.6.8.4.8 Near white blast entire surfaces of each deck socket of 3.6.8.3.

3.6.8.4.9 Faying surfaces shall not be blasted or painted.
3.6.8.4.8.2 Apply preservation coatings in accordance with the requirements of MIL-PRF-23236, TYPE VII, CLASS 15B; Apply 2 coats. Install in accordance with the manufacturer’s product data sheets to the entire surfaces listed in each new deck socket. Install in accordance with the manufacturer’s product data sheets. Final topcoat shall be Federal Standard 595 color number 13591 OSHA Safety Yellow.

3.6.8.5 Fabricate and install new 4 each Fall Protection Anchorage Points in using drawing M-1646 TRF Caisson 2 Fall Protection Anchorage as guidance. Template to existing shipboard conditions.

3.6.8.6 In addition to the installation of pieces #3, #4 and #5, install new 1 each (4 EA total), ½ inch grade 5 lock-washer with new each bolt.

(I) “FASTENER TORQUE"

3.6.8.6.1 Torque each bolt installed to 57-foot pounds in a diametrically opposed pattern at three separate and progressively incremental settings in accordance with manufacturer's instructions and the requirements of NSTM Chapter 075. (REQUIRED REPORT CDRL A077)

3.6.8.6.2 Weld attach 1 each D-Ring Assembly (piece #6) to each Fall Protection Anchorage Point Assembly (4 EA).

(I) “FIT UP INSPECTION”

3.6.8.6.3 Accomplish fit up inspection.

(V) “VISUAL INSPECTION (VT)”

3.6.8.6.4 Accomplish nondestructive testing visual inspection (VT).

(I) “MAGNETIC PARTICLE INSPECTION (MT)”

3.6.8.6.5 Accomplish nondestructive testing magnetic particle inspection (MT).

(I) “STATIC LOAD TEST”

3.6.8.6.6 Accomplish a static load test of each installed Fall Protection Anchorage Point. A test load of 3600 pounds shall be applied for 10 minutes. Remove the test load and inspect Fall Protection Anchorage Point and surrounding structure for evidence of damage or permanent deformation. Allowable damage: None. (REQUIRED REPORT CDRL A078)

3.6.8.6.7 After a satisfactory static load test has been accomplished, wire attach a metal tag to each installed I Beam (piece #2) with the following information:

Caisson 4-5 Spare
Static Load Tested to 3600 Pounds
Date of satisfactory test
Name of repair facility

3.6.8.6.8 Do not preserve new fasteners or D-Rings.

3.6.8.6.9 Faying surfaces shall not be blasted or painted.

3.6.8.6.10 Near white blast entire surfaces of each deck socket fall protection anchorage points of 3.6.8.5 prior to installation: Accomplish the requirements of Surface Preparation Specification SSPC-SP-10 of Systems and Specifications, SSPC Painting Manual, Volume 2.

3.6.8.6.11 Apply two coats MIL-PRF-23236, TYPE VII, CLASS 15B preservation coating to each fall protection anchorage point. Install in accordance with the manufacturer’s product data sheets to the entire surfaces listed in 3.6.8.5.

(V) "OPERATIONS DECK INSPECTION PRIOR TO STAGING REMOVALS"

3.6.9 Operations Deck Pre-Staging Removal Inspection: Accomplish a pre-closeout inspection in the locations listed in 3.6 after completion of preservation, and prior to staging removals.

3.6.9.1 Inspect cleanliness, cured paint, and completion of work prior to staging removals.

3.6.9.2 Each deficiency shall be recorded on the checkpoint form and corrected prior to staging removals.

3.6.9.3 Each corrected deficiency shall be verified by the SUPERVISOR as satisfactorily corrected prior to staging removals in the location listed in 3.6.

(V) "POST STAGING REMOVAL INSPECTION"

3.6.10 Operations Deck Post Staging Removal Inspection: Accomplish a Post Staging Removal Inspection no later than 24 hours after staging removals.

3.6.10.1 Demonstrate that the entire location listed in 3.6 is in a broom clean condition. Locations shall be free of visual contaminants such as oxide films, oils, grease, dust, dirt, blast grit, rust, rust stains or foreign material.

3.6.10.2 Each deficiency shall be recorded on the checkpoint form and corrected within 24 hours.

3.6.10.3 All corrected deficiencies shall be verified by the SUPERVISOR as satisfactorily corrected prior to the resumption of any other production in the location listed in 3.6.

3.7 **MAIN PROPULSION REPAIRS:** Not Used

3.8 **STEERING REPAIRS:** Not Used

3.9 **MACHINERY SYSTEMS:** Not Used

3.10 **ELECTRICAL SYSTEMS:** Accomplish repairs of Electrical Systems as follows:

3.10.1 Control Panel Install: Install new Control Panel; located in the Operations Deck; in accordance with drawing 980CAS-CNTRPNL-21, using PWD 67514, MIL-STD-2003-3B(SH), MIL-STD-1310, PW 15971, PW 48053, PW 51109 as guidance.

3.10.1.1 Control Panel shall be in equal performance characteristic as the material listed in 980CAS-CNTRPNL-21.

3.10.1.2 Template exact sizes, locations, and configurations from existing shipboard conditions.

3.10.1.3 Remove existing and install new each wire/cable for each equipment listed in 980CAS-CNTRPNL-21 in accordance with 980CAS-CNTRPNL-21 from load to source.

TABLE 3.10.1

Electrical Insulation	
Up to 5 Amp Load	2 Meg ohm
Up to 10 Amp Load	1 Meg Ohm
Up to 25 Amp Load	400,000 Ohm
Up to 50 Amp Load	250,000 Ohm
Over 50 Amp Load	100,00 Ohm

3.10.1.4 Weld install Total QTY (100 EA) electrical wireway hangers in locations as required to install new each wire/cable. Utilize existing wireways and hangers wherever possible. Installation of new hangers shall be for the new Control Panel of 3.10.1, each new Heater of 3.10.3 and each new Main Safety Switch, each new Main Breaker Cabinet and each new Power Distribution Panel of 3.10.4 using 980CAS-CNTRPNL-21 as guidance. Installation shall be in accordance with 302-5302690 Standard Cable Installation Requirements and MIL-STD-2003-3B(SH). Any amount less than Total QTY (100 EA) that is not required will be descoped.

3.10.1.5 Remove existing and accomplish weld installation of new foundations to include the new Control Panel of 3.10.1, each new Heater of 3.10.3 and each new Main Safety Switch, each new Main Breaker Cabinet and each new Power Distribution Panel of 3.10.4 and in accordance with 980CAS-CNTRPNL-21.

(I) "FIT UP INSPECTION"

3.10.1.5.1 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.10.1.6.2 Accomplish nondestructive testing visual inspection (VT).

3.10.1.6 Drill foundations and install new hardware to match foundation penetrations of the new equipment being mounted on the newly installed foundation to include the new Control Panel of 3.10.1, each new Heater of 3.10.3 and each new Main Safety Switch, each new Main Breaker Cabinet and each new Power Distribution Panel of 3.10.4 and in accordance with 980CAS-CNTRPNL-21.

3.10.1.6.1 Install grade 5 bolt (QTY One Each), lock-washer (QTY One Each), and self-locking nut (QTY One Each), at each mounting penetration in accordance with NSTM CH 075 Fasteners.

3.10.1.7 Bond and ground equipment in accordance with the requirements of 980CAS-CNTRPNL-21, MIL-STD-2003-3B(SH), and MIL-STD-1310. Grounding straps shall be CRES 316L.

(I) "OPERATIONAL TEST"

3.10.1.8 Energize accomplish operational testing each new installed equipment listed in 3.10.1. Equipment shall function to designed sequence of operation in conjunction with 3.13.1.

3.10.2 Interior Lighting System Replace: Remove existing and install new QTY (20 EA) Lighting Fixtures, new QTY (5 EA) Emergency Lighting Fixtures; located in Operations Compartment between (G-4) and (G-5); using drawing/manuals MIL-STD-2003-3B(SH), MIL-STD-1310, PW 15958 and PW 15971 as guidance.

3.10.2.1 Template exact sizes, locations, and configurations from existing shipboard conditions.

3.10.2.2 New QTY (20 EA) Lighting Fixtures shall be equal in performance characteristics of McMaster Carr 8295K62.

3.10.2.3 New QTY (5 EA) Emergency Lighting Fixtures shall be equal in performance characteristics of The Lighting Source SLEL-6-81-2-4WLED.

3.10.2.4 Create an electrical one line diagram sketch of emergency light system, including wiring and locations of each light fixture. (REQUIRED REPORT CDRL A079)

3.10.3 Heaters Replace: Remove existing and install new QTY (1 EA) heater; located at Operations Compartment, Forward End, Frame 20 to 22; Operations Compartment, (QTY 1 EA) AFT End, Frame 20 to 22; using MIL-STD-2003-3B(SH), MIL-STD-1310, PW 15971 and One Line Diagram Caiss 5 as guidance.

3.10.3.1 Template exact sizes, locations, and configurations from existing shipboard conditions.

3.10.3.2 Remove 100 percent of existing QTY (One EA), Chromalox Electric Heater; QTY (One EA), Heater Thermostat; QTY (200 LF), Electrical Cable, including mounting foundations and associated hardware; from each location in 3.10.3

3.10.3.3 Install new QTY (One EA), Heater, equal in performance characteristics of Modine model VE-200-3301, 480V, 3 Phase, 20kw with louver; new QTY (One EA) Thermostat, equal in performance characteristics of Siemens Line Volt Mechanical thermostat, SPDT, 40 -90 Deg F, 120 to 240VAC, Model 134-1084; new QTY (200 LF) Electrical Cable, NAVY, 3 conductor, 14 AWG, low smoke, watertight, non-flex, unarmored, LSTSGU-4, MIL-CL24643A; including mounting foundations and associated hardware; from each location in 3.10.3.

3.10.3.4 Equipment shall be installed in accordance with manufacturer’s instructions.

3.10.3.4.1 Utilize existing wire ways for new and existing wires/cables.

(I) "OPERATIONAL TEST"

3.10.3.5 Accomplish operational test of each newly installed equipment listed in 3.10.3.

3.10.3.6 Accomplish work in conjunction with 3.10.4, 3.13.1 and 3.13.2.

3.10.4 <u>Main Safety Switches & Power Distribution Panels Modification:</u> Remove existing and install new Main Circuit Panel; located Mezzanine Deck Compartment Forward bulkhead and Mezzanine Deck Compartment AFT bulkhead; in accordance with 980CAS-CNTRPNL-21 and using manuals/drawings MIL-STD-2003-3B(SH), MIL-STD-1310, PW15971, and One Line Diagram Caiss 5 as guidance.

3.10.4.1 Template exact size, configuration, material, and location from existing Asset conditions.

3.10.4.2 Electrically disconnect QTY (One EA), Existing Main Breaker Cabinet, each circuit breaker, and remove from foundation located at the Mezzanine Deck Compartment Forward bulkhead.

3.10.4.3 Remove existing and install new QTY (One EA) 600 AMP, 3-Pole, Heavy Duty Safety Switch, Fusible with Neutral, 600 VAC, 250 VDC, NEMA 3R, Uses IDSR Series Fuses, Enclosure Dims: H: 52.70", W: 24.00", D: 8.95". Safety switch is located at the Mezzanine Deck Compartment Forward bulkhead.

3.10.4.3.1 Remove existing and install new (One EA) Control Panel.

(I) "FIT UP INSPECTION"

3.10.4.3.4 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.10.4.3.5 Accomplish nondestructive testing visual inspection.

3.10.4.4 Electrically disconnect QTY (One EA), Existing Main Breaker Cabinet, each circuit breaker, and remove foundation located in Mezzanine Deck Compartment AFT bulkhead.

3.10.4.5 Remove existing and install new QTY (One EA) 600 AMP, 3-Pole, Heavy Duty Safety Switch, Fusible with Neutral, 600 VAC, 250 VDC, NEMA 3R, Uses IDSR Series Fuses, Enclosure Dims: H: 52.70", W: 24.00", D: 8.95". Safety switch is located at the Mezzanine Deck Compartment Aft bulkhead.

3.10.4.6 Power Distribution Panels Modification: Remove existing and install new Power Distribution Panels; located at Operations Deck (G-4); in accordance with 980CAS-CNTRPNL-21, using drawings/manuals, MIL-STD-2003-3B(SH), MIL-STD-1310, PW 15971, PW 48053 and PW 51109 as guidance.

3.10.4.6.1 Remove existing and install new each wire/cable for each equipment listed in the valve operator label in accordance with 980CAS-CNTRPNL-21 from load to source, in accordance with the requirements of 3.10.1.

3.10. 4.6.2 Electrically and mechanically disconnect and remove the existing power distribution panel circuit breakers and remove the existing foundation for QTY (One EA) existing power distribution panel P-5.

3.10.4.6.3 Install new QTY (One EA), New Power Distribution Panel P-5, 12 Circuit, 3 phase, 480V, Nema 12, Lockable Breakers, Locking Hasp.

3.10.4.6.4 Install new QTY (6 EA), 20 Amp, 3 Pole Power Breaker for electric-mechanical operator.

3.10.4.6.5 Install new QTY (One EA), 20 Amp, 3 Pole Power Breaker for Phase Meter.

3.10.4.6.6 Install new QTY (2 EA), 20 Amp, 3 Pole Power Breaker for Light Circuit.

3.10.4.6.7 Install new QTY (One EA), 50 Amp, 3 Pole Power Breaker for Forward Heater.

3.10.4.6.8 Install new QTY (One EA), 50 Amp, 3 Pole Power Breaker for Aft Heater.

3.10.4.7 Accomplish cable and wire connection using electrical hook-up data retained from 1.7.8.

3.10.4.8 Install new QTY (One EA), New Power Distribution Panel P-6, 8 Circuit, 3 phase, 480V, Nema 12, Lockable Breakers, Locking Hasp; <u>adjacent</u> to P-5 distribution panel.

3.10.4.8.1 Install new QTY (One EA), 20 Amp, 3 Pole Power Breaker for Equalizing Valve Control No. One.

3.10.4.8.2 Install new QTY (One EA), 20 Amp, 3 Pole Power Breaker for Equalizing Valve Control No. 2.

3.10.4.8.3 Install new QTY (One EA), 20 Amp, 3 Pole Power Breaker for Sea Valve Control No. One.

3.10.4.8.4 Install new QTY (One EA), 20 Amp, 3 Pole Power Breaker for Sea Valve Control No. 2.

3.10.4.8.5 Install new QTY (One EA), 20 Amp, 3 Pole Power Breaker for Discharge Valve Control No. One.

3.10.4.8.6 Install new QTY (One EA), 20 Amp, 3 Pole Power Breaker for Discharge Valve Control No. 2.

3.10.4.8.7 Electrically and mechanically connect each equipment.

3.10.4.8.8 Utilize existing wire ways for new and existing wires/cables.

3.10.5 Tank Level Indicator (TLI) Panel Relocate: Relocate the QTY (One EA) Tank Level Indicator (TLI) Panel; located at Operations Deck (G-4) in accordance with the requirements of drawing 980CAS-TLIRLC-21, using manual MIL-STD-2003-3B(SH) for guidance.

3.10.5.1 Accomplish weld installation of relocated TLI panel. Template exact locations, and configurations from existing shipboard conditions.

(I) "FIT UP INSPECTION"

3.10.5.2 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.10.5.3 Accomplish nondestructive testing visual inspection (VT).

3.10.6 Operations Compartment Junction Box Covers Repair: Remove existing and install new QTY (20 EA) Junction Box Covers and associated fasteners located at Operations Deck Overhead; using manual MIL-STD-2003-3B(SH) as guidance.

3.10.6.1 Locate and open all J-boxes that contain abandoned cables.

3.10.6.2 Remove existing abandoned cable and associated connectors and/or hardware.

3.10.6.3 Accomplish the installation of seals/plugs to the through penetrations of each J-Box.

3.10.6.4 All open penetrations through the equipment shall be made watertight.

3.10.6.5 Template exact size, configuration, and location from existing vessel conditions.

3.10.7 <u>Weather Deck Lighting Fixtures and Guards Replace:</u> Remove existing and install new QTY (12 EA), LED Light Fixture, equal in performance characteristics of Hubbell Lighting 54EN86, Gray Color, 120/277V, integrated LED, outdoor rated, watertight; located at Weather Deck (G-6). In accordance with WDL-SKTCH-980, using drawings 47978, 47983, PW 48054, and PW 70176 as guidance.

3.10.7.1 Template exact location, configuration, and materials from existing condition.

3.10.7.2 Remove existing and install new lighting guard foundation for each item listed in 3.10.7.

3.10.8 <u>Weather Deck Stuffing Tubes and Cables Replace:</u> Remove existing and install new Weather Deck, Light Mounts, Stuffing Tubes and Cables; located at Weather Deck (G-6) and Operations Compartment; using drawings/manuals MIL-STD-2003-3B(SH), PW 70178, 47983, PW 47978, PW 48054, and WDL-SKTCH-980 as guidance and in accordance with the Drawing PW 70176.

3.10.8.1 Remove existing and install new light mounts.

3. 10.8.1.1 Install new One-inch schedule 80 light mounts for the weather deck lights.

3. 10.8.1.2 Light mounts shall not penetrate the deck.

3. 10.8.1.3 Each weather deck light shall be installed on the light mount per the Drawing 70176.

3.10.8.1.4 Template exact location, configuration, and materials from existing Asset conditions.

3.10.8.2 Remove existing QTY (24 EA), Existing Lighting electrical cable stuffing tubes and packing; existing QTY (One EA) Red Light Alarm electrical cable stuffing tube and packing; existing QTY (2 EA) Alarm Bell electrical cable stuffing tubes and packing. Welding and blanks shall be in accordance with ANSI/AWS D1.1.

3.10.8.3 Weld install light mounts of 3.10.8.1 and blanks QTY (27 EA), in accordance with ANSI/AWS D1.1 one each for each location of 3.10.8.2.

(I) "FIT UP INSPECTION"

3.10.8.3.1 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.10.8.3.2 Accomplish nondestructive testing visual inspection (VT).

(I) "MAGNETIC PARTICLE TEST (MT)"

3.10.8.3.3 Accomplish nondestructive testing magnetic particle inspection (MT).

(I) "AIR TEST"

3.10.8.4 Accomplish local air test of newly installed blank.

3.10.8.4.1 Air hose nozzle must be placed at less than 3 inches, but greater than zero inches, from the area to be tested and pressure directed at the structure under test in a manner most likely to disclose leaks.

3.10.8.4.2 The minimum nozzle diameter must be 3/8 inch and the nozzle pressure must be 90 plus or minus 5 PSIG as monitored at the nozzle.

3.10.8.4.3 Apply a soapy solution to the opposite side of the structure and inspect for leaks.

3.10.8.4.4 Inspect joint or fitting for leakage by observing for formation of bubbles. Allowable leakage: None.

3.10.8.5 Install new QTY (12 EA), (One at each light location), new lighting electrical cable stuffing tube and packing in accordance with MIL-STD-2003-3B(SH).

(I) "FIT UP INSPECTION"

3.10.8.5.1 Accomplish fit up inspection in accordance with ANSI/AWS D1.1.

(V)"VISUAL INSPECTION"

3.10.8.5.2 Accomplish nondestructive testing visual inspection (VT).

(I) "AIR TEST"

3.10.8.6 Accomplish local air test of newly installed equipment.

3.10.8.6.1 Air hose nozzle must be placed at less than 3 inches, but greater than zero inches, from the area to be tested and pressure directed at the structure under test in a manner most likely to disclose leaks.

3.10.8.6.2 The minimum nozzle diameter must be 3/8 inch and the nozzle pressure must be 90 plus or minus 5 PSIG as monitored at the nozzle.

3.10.8.6.3 Apply a soapy solution to the opposite side of the structure and inspect for leaks.

3.10.8.6.4 Inspect joint or fitting for leakage by observing for formation of bubbles. Allowable leakage: None.

3.10.8.7 Install new QTY (150 Feet), Electrical Cable Equal in performance characteristics of TRICAB LSZH single conductor 444 MCM P/N BVAPVXD/1A444BK.

3.10.8.7.1 Route forward cable from the forward shore connection box through B1 and B2 of the forward main circuit breaker using drawing PW 48053 for guidance.

3.10.8.7.2 Route aft cable from the aft shore connection box through B8 and B7 of the aft main circuit breaker using drawing PW 48053 for guidance.

(I) "AIR TEST"

3.10.8.8 Accomplish local air test of newly installed equipment.

3.10.8.8.1 Air hose nozzle must be placed at less than 3 inches, but greater than zero inches, from the area to be tested and pressure directed at the structure under test in a manner most likely to disclose leaks.

3.10.8.8.2 The minimum nozzle diameter must be 3/8 inch and the nozzle pressure must be 90 plus or minus 5 PSIG as monitored at the nozzle.

3.10.8.8.3 Apply a soapy solution to the opposite side of the structure and inspect for leaks.

3.10.8.8.4 Inspect joint or fitting for leakage by observing for formation of bubbles. Allowable leakage: None.

3.10.9 Forward and AFT Access Enclosure Electrical Components Repair: Install new electrical components in new Forward Access Enclosure; located at Weather Deck and Operations Deck; using drawings/manuals 48055, PW 47978, PW 70178, PW 70147, PW 70187, PW 15971, Alarm Light Bracket, and MIL-STD-2003-3B(SH) as guidance.

3.10.9.1 Weld install QTY (One EA), Electrical cable stuffing tube and packing in the outboard interior deck of the new Forward Access Enclosure.

3.10.9.2 Each new stuffing tube shall conform to structural requirements and must be of correct size, material type, and design to accommodate cabling in accordance with MIL-STD-2003-3B(SH).

(I) "FIT UP INSPECTION"

3.10.9.2.1 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.10.9.2.2 Accomplish nondestructive testing visual inspection (VT).

(I) "AIR TEST"

3.10.9.2.3 Accomplish local air test of newly installed stuffing tubes.

3.10.9.2.4 Air hose nozzle must be placed at less than 3 inches, but greater than zero inches, from the area to be tested and pressure directed at the structure under test in a manner most likely to disclose leaks.

3.10.9.2.5 The minimum nozzle diameter must be 3/8 inch and the nozzle pressure must be 90 plus or minus 5 PSIG as monitored at the nozzle.

3.10.9.2.6 Apply a soapy solution to the opposite side of the structure and inspect for leaks.

3.10.9.2.7 Inspect joint or fitting for leakage by observing for formation of bubbles. Allowable leakage: None.

3.10.9.3 Disconnect and remove existing QTY (2 EA) Existing Light Switch & 2-gang box.

3.10.9.4 Weld install new QTY (One EA), Outlet box, with toggle switch and outlet box cover, minimum NEMA-3R; on the interior inboard surface of Forward Access Enclosure, 48 inches above the deck between the access enclosure door and top step of ladder.

(I) "FIT UP INSPECTION"

3.10.9.4.1 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.10.9.4.2 Accomplish nondestructive testing visual inspection (VT).

3.10.9.4.3 Install new QTY (100 LF), Cable equal in performance characteristics of cable LSTSGU-4, 14 AWG, Low Smoke, water-tight, non-flex from load to source.

3.10.9.4.4 Install new QTY (100 LF), Cable equal in performance characteristics of cable LSTSGU-4, 14 AWG, Low Smoke, water-tight, non-flex from load to source.

3.10.9.4.5 Equipment shall be wired to simultaneously operate access enclosure light and operations compartment lighting.

3.10.9.5 Disconnect and remove existing QTY (One EA) Existing Light; located at Operations Compartment.

3.10.9.5.1 Equipment is located on the forward vertical surface/face of the weather deck opening, above the ladder, overhead.

3.10.9.6 Weld install new QTY (One EA) Jelly Jar Light, equal in performance characteristics as McMaster Item 1616K111, Wash down Ceiling Light, Built-in LED, neutral 3,500k; at the center of the interior diagonal surface of the new Forward Access Enclosure.

(I) "FIT UP INSPECTION"

3.10.9.6.1 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.10.9.6.2 Accomplish nondestructive testing visual inspection (VT).

3.10.9.6.3 Install wiring between each newly installed outlet box and each associated newly installed jelly jar light.

3.10.9.6.4 Install new QTY (100 LF), Cable equal in performance characteristics of cable LSTSGU-4, 14 AWG, Low Smoke, water-tight, non-flex from load to source.

3.10.9.6.5 Install new QTY (100 LF), Cable equal in performance characteristics of cable LSTSGU-4, 14 AWG, Low Smoke, water-tight, non-flex from load to source.

3.10.9.6.6 Apply a 2 inch contrasting lettering on outside surface of items in AFT Access Enclosure to identify each alarm at location and color designated by the COR.

3.10.9.7 Apply coating to markings and distinguishing figures identified in 3.2.1. Coatings shall meet the requirements of MIL-PRF-23236, TYPE VII, CLASS 15B; and manufacturer's product data sheets using drawing S2804-921819 Rev C for guidance.

3.10.9.8 AFT Access Enclosure Electrical Components Repair: Install new electrical components in new AFT Access Enclosure; located at Weather Deck and Operations Deck; using drawings/manuals 48055, PW 47978, PW 70178, PW 70147, PW 70187, PW 15971, Alarm Light Bracket, and MIL-STD-2003-3B(SH) as guidance.

3.10.9.9 Each new stuffing tube shall conform to structural requirements and must be of correct size, material type, and design to accommodate cabling in accordance with MIL-STD-2003-3B(SH).

3.10.9.10 Weld install (One EA) Electrical cable stuffing tube and packing in the outboard interior deck of AFT Access Enclosure.

3.10.9.10.1 Construct and weld install new QTY (One EA), Alarm Light Bracket install the new bracket for mounting the new flooding and high-level beacon lights.

3.10.9.10.2 New Bracket shall be constructed from 3/8-inch, A-36 steel.

3.10.9.10.3 Mounting surface of the new bracket shall be 6 inches above the top surface of the new access enclosure.

(I) "FIT UP INSPECTION"

3.10.9.10.4 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.10.9.10.5 Accomplish nondestructive testing visual inspection (VT).

(I) "AIR TEST"

3.10.9.11 Accomplish local air test of newly installed stuffing tube. (REQUIRED REPORT CDRL A080)

3.10.9.11.1 Air hose nozzle must be placed at less than 3 inches, but greater than zero inches, from the area to be tested and pressure directed at the structure under test in a manner most likely to disclose leaks.

3.10.9.11.2 The minimum nozzle diameter must be 3/8 inch and the nozzle pressure must be 90 plus or minus 5 PSIG as monitored at the nozzle.

3.10.9.11.3 Apply a soapy solution to the opposite side of the structure and inspect for leaks.

3.10.9.11.4 Inspect joint or fitting for leakage by observing for formation of bubbles. Allowable leakage: None.

3.10.9.12 Disconnect and remove existing QTY (One EA) Existing Jelly Jar Light.

3.10.9.12.1 Equipment is located on the Aft vertical surface/face of the weather deck opening, above first ladder flight, overhead.

3.10.9.12.2 Mount and weld install new QTY (One EA) Light, equal in performance characteristics of McMaster Item 1616K111, Wash down Ceiling Light, Built-in LED, neutral 3,500k, at the center of the interior diagonal of the new AFT Access Enclosure.

3.10.9.12.3 Install wiring to the new QTY (One EA), Outlet box, with toggle switch and outlet box cover, minimum NEMA-3R, using previously recorded hook-up data in accordance with 1.7.8.

3.10.9.13 Disconnect electrically and mechanically existing and install new QTY (One EA) Flooding Alarm Beacon Light (Blue), equal in performance characteristics of McMaster Item 5846T23, 304 Stainless Steel Revolving Light, 120VAC, NEMA 4X.

3.10.9.13.1 Removals and blanking are accomplished in 3.10.9.

3.10.9.13.2 Install new QTY (One EA) Flooding Alarm Beacon Light (Blue), equal in performance characteristics of McMaster Item 5846T23, 304 Stainless Steel Revolving Light, 120VAC, NEMA 4X on the new Alarm Light Bracket.

3.10.9.14 Disconnect electrically and mechanically existing and install new QTY (One EA) Flooding Alarm Bell, equal in performance characteristics of McMaster Item 3038T39, Wash down Buzzer, Surface mount, 120VAC, 78-101dB @ 10 ft., NEMA 4X, Surface mount.

3.10.9.14.1 Removals and blanking are accomplished in 3.10.9.

3.10.9.14.2 Install new QTY (One EA) Flooding Alarm Bell, equal in performance characteristics of McMaster Item 3038T39, Wash down Buzzer, Surface mount, 120VAC, 78-101dB @ 10 ft., NEMA 4X, Surface mount.

3.10.9.15 Disconnect electrically and mechanically existing and install new QTY (One EA) Tank Level Beacon Light (Red), equal in performances characteristics of McMaster Item 5846T23, 304 Stainless Steel Revolving Light, 120VAC, NEMA 4X on the new alarm light bracket.

3.10.9.15.1 Removals and blanking are accomplished in 3.10.9.

3.10.9.15.2 Install new QTY (One EA) Tank Level Beacon Light (Red), equal in performances characteristics of McMaster Item 5846T23, 304 Stainless Steel Revolving Light, 120VAC, NEMA 4X, 6-10 inches from the front door edge, on the new alarm light bracket.

3.10.9.16 Disconnect electrically and mechanically existing and install new QTY (One EA) Tank Level Bell, equal in performance characteristics of McMaster Item 3038T39, and Wash down Buzzer, Surface mount, 120VAC, 78-101dB @ 10 ft., NEMA 4X, Surface mount.

3.10.9.16.1 Weld install new QTY (One EA) Tank Level Bell, equal in performance characteristics of McMaster Item 3038T39, Wash down Buzzer, Surface mount, 120VAC, 78-101dB @ 10 ft., NEMA 4X, Surface mount.

3.10.9.17 Install new QTY (20 FEET), Electrical Cable equal in performance characteristics of LSDSGU-4; from load to source, using hook-up data retained from 1.7.8, connecting /combining audible alarm circuit to the single horn.

3.10.9.18 Install new QTY (One EA), electrical cable stuffing tube and packing for the newly installed equipment. Install externally on the deck adjacent to the new AFT Access Enclosure.

(I) "FIT UP INSPECTION"

3.10.9.18.1 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.10.9.18.2 Accomplish nondestructive testing visual inspection (VT).

(I) "MAGNETIC PARTICLE TEST (MT)"

3.10.9.18.3 Accomplish nondestructive testing magnetic particle inspection (MT).

(I) "AIR TEST"

3.10.9.18.4 Accomplish local air test of newly installed equipment listed in 3.10.11.10.

3.10.9.18.5 Air hose nozzle must be placed at less than 3 inches, but greater than zero inches, from the area to be tested and pressure directed at the structure under test in a manner most likely to disclose leaks.

3.10.9.18.6 The minimum nozzle diameter must be 3/8 inch and the nozzle pressure must be 90 plus or minus 5 PSIG as monitored at the nozzle.

3.10.9.18.7 Apply a soapy solution to the opposite side of the structure and inspect for leaks.

3.10.9.18.8 Inspect joint or fitting for leakage by observing for formation of bubbles. Allowable leakage: None.

3.10.9.19 Install new QTY (30 LF), Wiring, for flooding circuit, equal to DSGA-3; from load to source, using hook-up data retained from 1.7.8, connecting /combining audible alarm circuit to the single horn.

3.10.9.20 Disconnect electrically and mechanically existing QTY (2 EA) Existing Light Switch & 2-gang box.

3.10.9.21 Weld install new QTY (One EA), Outlet box, with toggle switch and outlet box cover, minimum NEMA-3R., on the interior inboard surface of equipment of the new Aft Access Enclosure, 48 inches above the deck, between the access enclosure door and top step of ladder.

(I) "FIT UP INSPECTION"

3.10.9.21.1 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.10.9.21.2 Accomplish nondestructive testing visual inspection (VT).

3.10.9.22 Install QTY (One EA), Electrical cable stuffing tube and packing; in interior inboard deck surface of equipment of the new Aft Access Enclosure.

(I) "FIT UP INSPECTION"

3.10.9.22.1 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.10.9.22.2 Accomplish nondestructive testing visual inspection (VT).

(I) "MAGNETIC PARTICLE TEST (MT)"

3.10.9.22.3 Accomplish nondestructive testing magnetic particle inspection (MT).

(I) "AIR TEST"

3.10.9.23 Accomplish local air test of newly installed equipment listed in 3.10.11.10.

3.10.9.23.1 Air hose nozzle must be placed at less than 3 inches, but greater than zero inches, from the area to be tested and pressure directed at the structure under test in a manner most likely to disclose leaks.

3.10.9.23.2 The minimum nozzle diameter must be 3/8-inch, and the nozzle pressure must be 90 plus or minus 5 PSIG as monitored at the nozzle.

3.10.9.23.3 Apply a soapy solution to the opposite side of the structure and inspect for leaks.

3.10.9.23.4 Inspect joint or fitting for leakage by observing for formation of bubbles. Allowable leakage: None.

3.10.9.24 Install QTY (One EA), outlet box, with toggle switch and outlet box cover, minimum NEMA-3R. Equipment shall be wired to simultaneously operate access enclosure light and operations compartment lighting.

3.10.9.24.1 Install wiring and cables using the hook-up data in accordance with 1.7.8.

3.10.9.24.2 Install new QTY (120 FEET), Electrical Cable equal in performance of LSDSGU-4 load to source.

3.10.9.24.3 QTY (30 LF), Wiring, for flooding circuit, equal to DSGA-3; load to source.

3.10.9.24.4 QTY (30 LF), Wiring, for tank level circuit, equal to LSFSGU-3; load to source.

3.10.9.24.5 QTY (60 LF), Cable, for access enclosure and operations compartment lighting circuit, LSTSGU-4, 14 AWG, Low Smoke, Watertight, non-flex; load to source.

3.10.9.25 Apply a 2-inch contrasting lettering on outside surface of items in AFT Access Enclosure to identify each alarm at location and color designated by the SUPERVISOR.

3.10.9.25.1 Apply coating to markings and distinguishing figures identified in 3.2.1. Coatings shall be in accordance with the requirements of MIL-PRF-23236, TYPE VII, CLASS 15B; and manufacturer's product data sheets using drawing S2804-921819 Rev C for guidance.

(I) "OPERATIONAL TEST"

3.10.9.26 Accomplish an Operational Test for each newly installed equipment listed in 3.10.9. Acceptance criteria: All circuits, lights, alarms, horns and receptacles operate when energized. (REQUIRED REPORT CDRL A081

3.11 **PIPING REPAIRS:** Accomplish piping repairs as follows:

3.11.1 Piping, Drainage and Ballasting System (UT) Inspection: Accomplish inspections of piping, drainage and ballast system in Main Ballast Tank, Forward Trim Tank, AFT Trim Tank including items listed in TABLE 3; using drawings /manuals Spare Caisson 4&5 Piping UT Grid, T9074-AS-GIB-010/271, PW 50746 and PW C-1120.

TABLE 3
QTY (2EA) spool pieces, flanges, elbows and piping associated with 10-Inch Equalizer Valves No.1 and No.2
QTY (2EA) spool pieces, flanges, elbows and piping associated with 8-Inch Sea Valves No.1 and No.2
QTY (2EA) spool pieces, flanges, elbows and piping associated with 18-Inch Main Tank Dewatering Discharge Valves No.1 and No.2
QTY (2EA) spool pieces, flanges, elbows and piping associated with 18-Inch Main Tank Dewatering Discharge Check Valves No.1 and No.2
QTY (2EA) spool pieces, flanges, elbows and piping associated with 24-Inch Sinking Valves No.1 and No.2
QTY (4EA) spool pieces, flanges, elbows and piping associated with 36-Inch Flood through Valves No. One, No.2, No.3 and No.4

(I) "ULTRASONIC TEST (UT)"

3.11.1.1 Accomplish (UT) measurement inspection of each piece of equipment listed in 3.11. Accomplish in accordance with drawing Spare Caisson 4&5 Piping UT Grid, using T9074-AS-GIB-010/271, PW 50746 and PW C-1120 for guidance. Record and submit readings on the drawing Spare Caisson 4&5 Piping UT Grid. (REQUIRED REPORT CDRL A082)

3.11.1.1.1 List in one column the design thickness.

3.11.1.1.2 Record in a separate column if reading is taken over an existing doubler or insert plate.

3.11.1.1.3 List in another column, next to the design thickness, the actual readings taken.

3.11.1.1.4 List and identify corresponding specific measurements, locations, and/or areas in relationship to structural members or with measurements from suitable hull reference points.

3.11.1.1.5 A sketch of the underwater body and freeboard plating surfaces with appropriate data is an acceptable alternative.

3.11.1.1.6 Calculate, identify, and highlight measurements indicating a difference (wastage) greater than or equal to 25 percent of the designed plate thickness, using TABLE ONE for guidance.

(V) "VISUAL INSPECTION (VT)"

3.11.1.2 Accomplish Visual Inspection (VT) of each spool piece, flange, elbows and piping listed in Table 3 after completion of SP-6 blast and cleanup of 3.4.2.

3.11.1.2.1 Search for, mark, identify, and record defective (actual and suspect) areas that exhibit signs of severe pitting, excessive erosion, deterioration, damage, or which may be detrimental to the structural and watertight integrity of the Asset.

3.11.1.2.2 Inspection shall include all hull openings, over boards, attachments, and appendages.

3.11.1.2.3 A two dimensional, or Autocad 2016, sketch or drawing showing hull/structural configurations, arrangements, relationships, and a listing of the inspection location's data is an acceptable alternative.

3.11.1.3 Each location and dimension of structural repairs required, as identified by the reports of 3.11.1.1 and 3.11.1.2 will be listed on the Structural Metal Repair Plan of 3.3.1.1.

3.11.1.3.1 Unidentified damaged piping surfaces shall be reported within 48 hours of discovery to the SUPERVISOR.

(I) "STRUCTURAL REPAIR COMPLETION: PIPING"

3.11.1.4 Structural inspections and repairs of 3.4, and 3.11.1.1 and 3.11.1.2 shall be completed prior to accomplishing the requirements of 3.4.7. Verify all repairs are accomplished in the locations and dimensions as approved by the SUPERVISOR in the structural metal repair plan as identified on the declining log prior to final SP-10 blast.

Long Lead: The following materials are considered Long Lead Time Material and shall be ordered by the Contractor within 5 working days of contract award.

3.11.2 <u>Ballast and Trim Valves Replace:</u> Remove existing Ballast and Trim Valves and install new valves listed in TABLE 4, fasteners and gaskets; located in Main Ballast Tank; using drawings/ manuals PW- C-1120, PW 73742, PW 73743, PW 73744, PW 15969, PW 15970, S9086-RK-STM-010, NSTM Chapter 075 Fasteners, MIL-STD-777 as guidance.

TABLE 4
QTY (One EA) Sea Valve No. One, 8-Inch Gate, Resilient Wedge, Flanged, 125#, Face to face 11.5 inches, 8 bolts, epoxy painted interior, epoxy painted exterior to match tank color, painted in accordance with AWWA C500, one manufacturer, equal in performance characteristics of Mueller Co. Reliable Connections; A-2361-6, OS&Y Rising Stem, Open Left (Counter-clockwise), 2” square wrench nut, stainless stem, type 316, EPDM Disc and O-rings.

TABLE 4
QTY (One EA) Sea Valve No. 2, 8-Inch Gate, Resilient Wedge, Flanged, 125#, Face to face 11.5 inches, 8 bolts, epoxy painted interior, epoxy painted exterior to match tank color, painted in accordance with AWWA C500, one manufacturer, equal in performance characteristics of Mueller Co. Reliable Connections; A-2361-6, OS&Y Rising Stem, Open Left (Counter-clockwise), 2” square wrench nut, stainless stem, type 316, EPDM Disc and O-rings.
QTY (One EA) Equalizing Valve No. One, 10-Inch Gate, Resilient Wedge, Flanged, 125#, Face to face 13 inches, 12 bolts, epoxy painted interior, epoxy painted exterior to match tank color, painted in accordance with AWWA C500, one manufacturer, equal in performance characteristics of Mueller Co. Reliable Connections; A-2361-6, OS&Y Rising Stem, Open Left (Counter-clockwise), 2” square wrench nut, stainless stem, type 316, EPDM Disc and O-rings.
QTY (One EA) Equalizing Valve No. 2, 10-Inch Gate, Resilient Wedge, Flanged, 125#, Face to face 13 inches, 12 bolts, epoxy painted interior, epoxy painted exterior to match tank color, painted in accordance with AWWA C500, one manufacturer, equal in performance characteristics of Mueller Co. Reliable Connections; A-2361-6, OS&Y Rising Stem, Open Left (Counter-clockwise), 2” square wrench nut, stainless stem, type 316, EPDM Disc and O-rings.
QTY (One EA) Main Tank Dewatering Discharge Valve No. One, 18-Inch Gate, Resilient Wedge, Flanged, 125#, Face to face 17 inches, 16 bolts, epoxy painted interior, epoxy painted exterior to match tank color, painted in accordance with AWWA C500, one manufacturer, equal in performance characteristics of Mueller Co. Reliable Connections; A-2361-6, OS&Y Rising Stem, Open Left (Counter-clockwise), 2” square wrench nut, stainless stem, type 316, EPDM Disc and O-rings.
QTY (One EA) Main Tank Dewatering Discharge Valve No. 2, 18-Inch Gate, Resilient Wedge, Flanged, 125#, Face to face 17 inches, 16 bolts, epoxy painted interior, epoxy painted exterior to match tank color, painted in accordance with AWWA C500, one manufacturer, equal in performance characteristics of Mueller Co. Reliable Connections; A-2361-6, OS&Y Rising Stem, Open Left (Counterclockwise), 2” square wrench nut, stainless stem, type 316, EPDM Disc and O-rings.

TABLE 4
QTY (One EA) 18-Inch Main Tank Dewatering Discharge Check Valve No. One, Flanged 125#, Face to face 33 inches, 16 bolts, epoxy painted interior, epoxy painted exterior to match tank color, painted in accordance with AWWA C500, one manufacturer, equal in performance characteristics of DeZurik Check Valves; CSD, 18, 800, F1, D1, S2-S2*DTR-SB16*.
QTY (One EA) 18-Inch Main Tank Dewatering Discharge Check Valve No. 2, Flanged, 125#, Face to face 33 inches, 16 bolts, epoxy painted interior, epoxy painted exterior to match tank color, painted in accordance with AWWA C500, one manufacturer, equal in performance characteristics of DeZurik Check Valves; CSD, 18, 800, F1, D1, S2-S2*DTR-SB16*.
QTY (One EA), Sinking Valve No. One, 24-Inch Gate, Resilient Wedge, Flanged, 125#, Face to face 20 inches, 20 bolts, epoxy painted interior, epoxy painted exterior to match tank color, painted in accordance with AWWA C500, one manufacturer, equal in performance characteristics of Mueller Co. Reliable Connections; A-2361-6, OS&Y Rising Stem, Open Left (Counter-clockwise), 2” square wrench nut, stainless stem, type 316, EPDM Disc and O-rings.
QTY (One EA), Sinking Valve No. 2, 24-Inch Gate, Resilient Wedge, Flanged, 125#, Face to face 20 inches, 20 bolts, epoxy painted interior, epoxy painted exterior to match tank color, painted in accordance with AWWA C500, one manufacturer, equal in performance characteristics of Mueller Co. Reliable Connections; A-2361-6, OS&Y Rising Stem, Open Left (Counter-clockwise), 2” square wrench nut, stainless stem, type 316, EPDM Disc and O-rings.
QTY (One EA), Flood Through Valve No. One, 36 Inch Gate, Solid Disc, OS&Y Rising Stem, Ductile Iron body and bonnet, bronze body and disc seat rings, Bronze body and wedge guides for the entire length of travel, no bypass ports cast in body, 316 stainless Steel Stem, Flanged 125#, Face to Face 23.25 inches, 32 bolts, epoxy coated interior, epoxy coated exterior to match tank color, one manufacturer, painted in accordance with AWWA C500, equal in performance characteristics of American R/D (part of Kennedy Valves).

TABLE 4
QTY (One EA),Flood Through Valve No. 2, 36 Inch Gate, Solid Disc, OS&Y Rising Stem, Ductile Iron body and bonnet, bronze body and disc seat rings, Bronze body and wedge guides for the entire length of travel, no bypass ports cast in body, 316 stainless Steel Stem, Flanged 125#, Face to Face 23.25 inches, 32 bolts, epoxy coated interior, epoxy coated exterior to match tank color, one manufacturer, painted in accordance with AWWA C500, equal in performance characteristics of American R/D (part of Kennedy Valves).
QTY (One EA),Flood Through Valve No. 3, 36 Inch Gate, Solid Disc, OS&Y Rising Stem, Ductile Iron body and bonnet, bronze body and disc seat rings, Bronze body and wedge guides for the entire length of travel, no bypass ports cast in body, 316 stainless Steel Stem, Flanged 125#, Face to Face 23.25 inches, 32 bolts, epoxy coated interior, epoxy coated exterior to match tank color, one manufacturer, painted in accordance with AWWA C500, equal in performance characteristics of American R/D (part of Kennedy Valves).
QTY (One EA),Flood Through Valve No. 4, 36 Inch Gate, Solid Disc, OS&Y Rising Stem, Ductile Iron body and bonnet, bronze body and disc seat rings, Bronze body and wedge guides for the entire length of travel, no bypass ports cast in body, 316 stainless Steel Stem, Flanged 125#, Face to Face 23.25 inches, 32 bolts, epoxy coated interior, epoxy coated exterior to match tank color, one manufacturer, painted in accordance with AWWA C500, equal in performance characteristics of American R/D (part of Kennedy Valves).

(I) "RECEIPT INSPECT NEW VALVES"

3.11.2.1 Accomplish receipt inspection to verify each procured item meets the requirements listed in TABLE 4; with Certificates of Compliance. (REQUIRED REPORT CDRL A083)

3.11.2.1.1 At a minimum the Certificate of Compliance will list valve size, description, model number, PSI rating, hydro seat test pressure (PSI) and hydro shell test pressure (PSI).

(I) "HYDROSTATIC TEST NEW VALVES"

3.11.2.2 Accomplish shop hydrostatic test for each procured valve after successful receipt inspection. (REQUIRED REPORT CDRL A084)

3.11.2.3 Test for seat tightness alternately on each side of gate for double-seated valves, and on outboard side only on single-seated valves, with the opposite side open for inspection using clean fresh water and in accordance with manufacturer's instructions for each item listed in TABLE 4. Do not exceed maximum hand wheel closing force specified in table 505-11-2 of S9086-RK-STM-010.

3.11.2.3.1 Valve packing shall be loosened to allow the escape of air. Valve shall be filled with clean water. All efforts are to be made to assure the escape of air from the bonnet and body cavity.

3.11.2.3.2 Seat tightness test pressure shall be 50 (+2/-2) PSIG.

3.11.2.3.3 Test duration shall be 10 minutes.

3.11.2.3.4 Continue test in the event of visible leakage until accurate determination of leakage site can be made.

3.11.2.3.5 Maximum allowable leakage: 10 cubic centimeters (cc) per hour per inch of nominal pipe size.

3.11.2.4 Existing preservation system applied to interior and exterior of each new valve shall not be disturbed.

3.11.2.5 Install each procured item listed in TABLE 4 after successful completion of the requirements of 3.11.2.2.

3.11.2.5.1 Install new CRES fasters to include QTY (One EA) Bolt QTY (One EA) Nut QTY (2 EA) Flat Washers for each bolt hole of each valve listed in TABLE 4. Hardware shall comply with MIL-DTL-1222, Type I, Grade 316. Each Template new equipment to list in TABLE 4 and existing pipe flanges and spools. Tighten fasteners in a diametrically opposed pattern at three separate and progressively incremental settings in accordance with manufacturer's instructions and the requirements of NSTM Chapter 075 Fasteners.

3.11.2.5.2 Apply Anti-seize compound to 100 percent of threaded surfaces, product similar or equal to the performance characteristics of CID-AA- 59004.

3.11.2.5.3 Restore piping flange mating surfaces exposed by disassembly of piping system. Repair by removing high spots, burrs, abrasions, and foreign matter, where removal can be accomplished by hand tools. Take precautions to maintain phonographic finish on flanges that have it. Ensure no excessive material is removed causing loss of critical size or alignment.

(V) “VISUAL INSPECTION"

3.11.2.5.4 Visually inspect all piping, transition spools and operating linkage for damage, deterioration, or misalignment, which would preclude the smooth and proper operation of valves, actuators, and operating gear.

3.11.2.5.5 Modify each existing, pipe, piping flange, spool piece flange, valve operating shaft, valve operating shaft coupling, and valve stem yoke.

(I) "STRAIN FREE"

3.11.2.5.6 Accomplish piping to valve alignment by adjustment, modification on, or replacement of the hangers adjacent to each pipe.

3.11.2.5.7 Install new QTY (ONE EA PER VALVE) Gaskets, meeting the performance characteristics of Garlock 3200 1/8” thick.

(V) "VALVE FLANGE FASTENER TORQUE"

3.11.2.5.8 Accomplish torque of each newly installed Fastener in accordance with the requirements of NSTM Chapter 075 Fasteners.

(I) "OPERATIONAL DRY TEST"

3.11.2.6 Accomplish operational dry test installed item listed in TABLE 4. Operational test requirements shall be performed in conjunction with the requirements of 3.13.1.

3.11.2.6.1 Cycle each valve a minimum of 5 times, lubricating as necessary and adjust limits as necessary.

3.11.2.6.2 No abnormal noise, no binding, full open/close rate no less than 12 inches per minute, and no errors displayed on the actuator.

3.11.2.6.3 Take precautions when operating each motor driven valve actuator not to over drive reach rods or valves. Station a spotter a visual line of sight of the valve being tested and the reach rod. Spotter will maintain radio contact with the personnel operating the motor driven valve actuator.

(I) "VALVE SEAT TIGHTNESS TEST"

3.11.2.7 Accomplish a valve seat tightness test for 10 minutes on outboard side of each valve, where the inboard side of the gate is visible in accordance with manufacturer's instructions for each item listed in TABLE 4. Do not exceed maximum hand wheel closing force specified in table 505-11-2 of S9086-RK-STM-010.

3.11.2.7.1 Continue test in the event of visible leakage until accurate determination of leakage site can be made.

3.11.2.7.2 Maximum allowable leakage: 10 cubic centimeters (cc) per hour per inch of nominal pipe size.

(I) "OPERATIONAL WET TEST"

3.11.2.8 Accomplish operational wet test for each installed item listed in TABLE 4.

3.11.2.8.1 Test while caisson in conjunction with 3.13.1 and is in the dry dock in a partially submerged condition to assess the watertight integrity of piping and valves.

3.11.2.8.2 Cycle each valve a minimum of 5 times, lubricating as necessary and adjust limits as necessary.

3.11.2.8.3 Valves shall operate with no abnormal noise, binding, or errors displayed on the actuator/control panel/mimic board.

3.11.2.8.4 Full open/close rate shall be no less than 12 inches per minute.

3.11.2.8.5 Take precautions when operating each motor driven valve actuator not to over drive reach rods or valves. Station a spotter a visual line of sight of the valve being tested and the reach rod. Spotter will maintain radio contact with the personnel operating the motor driven valve actuator.

3.11.2.9 Accomplish a PDF and AutoCAD compatible Redline Drawing, for PW 15969 and PW 15970 showing modified structural configurations, arrangements, relationships, welds, and measurements. List data on the count, model number of valves, flange face to flange face measurements of valves, type, and dimensions of all materials installed, include cut sheets for each model of valve used to accomplish valve replacement. (REQUIRED REPORT CDRL A085)

3.11.2.9.1 Drawing modifications shall be marked in red. Each drawing shall be marked with the existing drawing number followed by the word "REDLINE" in the title box. Also include the contract number of the availability and the date the drawing was produced.

3.11.2.9.2 Provide a material list as part of the redline drawing of PW 15969 to include the manufacturer, model number and face to face flange dimension of each newly installed item listed in TABLE 4.

3.11.3 Discharge Check Valve No. One and No. 2 Grease Lines Replace: Remove existing and install new the equipment listed in QTY (One EA), 3/8 Inch Grease lines and Fittings for discharge check valve No. One and QTY (One EA), 3/8 Inch Grease lines and Fittings for discharge check valve No. 2; located in Main Ballast Tank and Operations Compartment Deck (G 4); using drawings 505-6814399, and PW 15970 as guidance.

3.11.3.1 Template exact size, configuration, and location from existing Asset conditions.

3.11.3.2 Fill each newly installed equipment listed in 3.11.3 with grease. Grease shall conform to the requirements of MIL-G-23549.

3.11.4 Valve Reach Rod Steady Bearing Grease Lines Removal: Remove existing Grease Lines, hangers, perpetrations and stuffing tubes listed in TABLE 5; located in Main Ballast Tank, Forward Trim Tank, AFT Trim Tank and Operations Compartment; using drawings C-1120 and PW 15970 as guidance.

TABLE 5
QTY (One EA), Grease lines, hangers, penetrations, and stuffing tubes for Dewatering Valve No. One reach rod bearings

TABLE 5
QTY (One EA), Grease lines, hangers, penetrations, and stuffing tubes for Dewatering Valve No. 2 reach rod bearings
QTY (One EA), Grease lines, hangers, penetrations, and stuffing tubes for Discharge Valve No. One reach rod bearings
QTY (One EA), Grease lines, hangers, penetrations, and stuffing tubes for Discharge Valve No. 2 reach rod bearings
QTY (One EA), Grease lines, hangers, penetrations, and stuffing tubes for Equalizing Valve No. One reach rod bearings
QTY (One EA), Grease lines, hangers, penetrations, and stuffing tubes for Equalizing Valve No. 2 reach rod bearings
QTY (One EA), Grease lines, hangers, penetrations, and stuffing tubes for Flood Through Valve No. One reach rod bearings
QTY (One EA), Grease lines, hangers, penetrations, and stuffing tubes for Flood Through Valve No. 2 reach rod bearings
QTY (One EA), Grease lines, hangers, penetrations, and stuffing tubes for Sea Valve, Valve No. One reach rod bearings
QTY (One EA), Grease lines, hangers, penetrations, and stuffing tubes for Sea Valve, Valve No. 2 reach rod bearings
QTY (One EA), Grease lines, hangers, penetrations, and stuffing tubes for Sinking Valve No. One reach rod bearings
QTY (One EA), Grease lines, hangers, penetrations, and stuffing tubes for Sinking Valve No. 2 reach rod bearings

3.11.4.1 Template exact sizes, locations, configurations and materials from existing Asset conditions.

3.11.4.2 Remove existing items listed TABLE 5.

3.11.4.3 Install new QTY (12 EA) CRES 316 threaded pipe plugs, in each removal location of TABLE 5 at each location listed in 3.11.4.

3.11.4.3.1 Weld install flush 3-inch circular blanks at each penetration and stuffing tube removal site.

(I) "FIT UP INSPECTION"

3.11.4.3.2 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.11.4.3.3 Accomplish nondestructive testing visual inspection (VT).

(I) "MAGNETIC PARTICLE TEST (MT)"

3.11.4.3.4 Accomplish nondestructive testing magnetic particle inspection (MT).

(I) "AIR TEST"

3.11.4.3.5 Accomplish local air test of newly installed equipment listed in 3.11.4.3.1.

3.11.4.3.5.1 Air hose nozzle must be placed at less than 3 inches, but greater than zero inches, from the area to be tested and pressure directed at the structure under test in a manner most likely to disclose leaks.

3.11.4.3.5.2 The minimum nozzle diameter must be 3/8 inch and the nozzle pressure must be 90 plus or minus 5 PSIG as monitored at the nozzle.

3.11.4.3.5.3 Apply a soapy solution to the opposite side of the structure and inspect for leaks.

3.11.4.3.5.4 Inspect joint or fitting for leakage by observing for formation of bubbles. Allowable leakage: None.

3.11.5 <u>Valve Reach Rod Assemblies Replace:</u> Remove existing and install new items listed in TABLE 6; located in Main Ballast Tank and Operations Deck; using drawings 980CAS-RRMOD-22, 980 CASPARE STAND 21, 2 Inch Reach Rod Packing Gland, PW 15969 and PW 15970 as guidance.

TABLE 6
QTY (One EA), 36-Inch Flood Through Valve No. One, Remote Valve Extension Shaft Assembly, From Top of Valve to Operating Deck Valve Operator

TABLE 6
QTY (One EA), 36-Inch Flood Through Valve No. 2, Remote Valve Extension Shaft Assembly, From Top of Valve to Operating Deck Valve Operator
QTY (One EA), 36-Inch Flood Through Valve No. 3, Remote Valve Extension Shaft Assembly, From Top of Valve to Operating Deck Valve Operator
QTY (One EA), 36-Inch Flood Through Valve No. 4, Remote Valve Extension Shaft Assembly, From Top of Valve to Operating Deck Valve Operator
QTY (One EA), 8-Inch Sea Valve No. One, Remote Valve Extension Shaft Assembly, From Top of Valve to Operating Deck Valve Operator
QTY (One EA), 8-Inch Sea Valve No. 2, Remote Valve Extension Shaft Assembly, From Top of Valve to Operating Deck Valve Operator
QTY (One EA), 10-Inch Equalizing Valve No. One, Remote Valve Extension Shaft Assembly, From Top of Valve to Operating Deck Valve Operator
QTY (One EA), 10-Inch Equalizing Valve No. 2, Remote Valve Extension Shaft Assembly, From Top of Valve to Operating Deck Valve Operator
QTY (One EA), 18-Inch Discharge Valve No. One, Remote Valve Extension Shaft Assembly, From Top of Valve to Operating Deck Valve Operator
QTY (One EA), 18-Inch Discharge Valve No. 2, Remote Valve Extension Shaft Assembly, From Top of Valve to Operating Deck Valve Operator
QTY (One EA), 24-Inch Sinking Valve No. One, Remote Valve Extension Shaft Assembly, From Top of Valve to Operating Deck Valve Operator
QTY (One EA), 24-Inch Sinking Valve No. 2, Remote Valve Extension Shaft Assembly, From Top of Valve to Operating Deck Valve Operator

3.11.5.1 Each valve reach rod diameter shall be 2 inches in diameter grade 316 stainless steel.

3.11.5.2 Prior to assembly remove high spots, burrs, abrasions, nicks, corrosion, and foreign matter from all mating surfaces.

3.11.5.3 Install anti-seize compound install on each threaded surface, product equal to the performance characteristics of CID-A-A-59004.

(V) "VALVE EXTENSION ALIGNMENT"

3.11.5.4 Installed alignment shall be within 1/32-inch from a relative linear axis from the valve coupling connection through each bearing to the actuator connection at each remote operating station.

(V) "OPERATIONAL TESTING"

3.11.5.5 Accomplish post assembly operational test of each valve reach rod shaft assembly to prove ease of operation and proper alignment of each valve stem extension. (REQUIRED REPORT CDRL A086)

3.11.5.5.1 Cycle each valve actuator from full closed to full open to full closed five times.

3.11.5.5.2 No abnormal noise, and no binding during open and close cycle operations.

3.11.6 Hull Exterior Openings Replace: Remove existing and install new QTY (8 EA) Flood Through Piping Cover and associated hardware for hinged doors; located at Flood Through Valve Piping; using drawings/ manual Hinged Cover Plate, CAS Hinged Cover Plate Hardware Replacement Detail, PW 15969, PW 15970, PW 50746 and MIL-STD-777 as guidance.

(I) "FIT UP INSPECTION"

3.11.6.1 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.11.6.1.1 Accomplish nondestructive testing visual inspection (VT).

(I) "MAGNETIC PARTICLE TEST (MT)"

3.11.6.1.2 Accomplish nondestructive testing magnetic particle inspection (MT).

3.11.6.2 Template exact sizes, locations, and configurations from existing Asset conditions.

3.11.6.3 Modify hinge plate holes to fit UHMW plastic bushings.

3.11.6.3.1 Install new bushing in accordance with drawing CAS Hinged Cover Plate Hardware Replacement Detail.

3.3.6.3.2 Accomplish preservation in conjunction and in accordance with 3.2.7 and 3.2.8.

3.11.6.4 Remove existing and install new each fastener and gasket for each equipment listed in TABLE 7.

TABLE 7
QTY (8 EA) Flood Through Piping Covers
QTY (2 EA) Sea Valve Piping Screens
QTY (2 EA) Dewatering Valve Piping Screens
QTY (2 EA) Sinking Valve Piping Screens

3.11.6.4.1 New fasteners shall conform to MIL-DTL-1222, Type I, Grade 316 CRES.

3.11.6.4.2 Install anti-seize sealant compound meeting the requirements of CID-A-A-59004-A to each fastener thread prior to installation.

3.11.6.4.3 Gaskets shall conform to HH-P-151, Class I, cloth inserted rubber, or MIL-PRF-1149, Type II, Class I, synthetic rubber.

Long Lead: The following materials are considered Long Lead Time Material and shall be ordered by the Contractor within 5 working days of contract award.

3.11.7 <u>Dewatering Pump and Motor Replace:</u> Remove existing and install new Dewatering Pumps and Motors; located in Main Ballast Tank and Operations Deck; using drawings/ manuals PW C-1120, PW 15969, PW 15970, PW 15959, PW 15960, PW 51109, S9086-KE-STM-010/CH-302 and S9086-RH-STM-010/CH. 503 as guidance.

3.11.7.1 Remove existing QTY (One EA), existing main ballast tank Dewatering Pump No. One and Motor and QTY (One EA), and existing main ballast tank Dewatering Pump No. 2 and Motor, including each motor, pump column, dewatering pump sump screen, support bracket and associated hardware.

3.11.7.2 Provide tech representative as listed in 1.26 for inspections, adjustment, pump foundation modification, pump screen modification, support bracket modification, and testing of the new pumps in accordance with manufacturer's instructions.

3.11.7.3 Weld fill and grind flush top surface of each unused dewatering pump foundation bolt hole.

(V) "VISUAL INSPECTION"

3.11.7.3.1 Accomplish nondestructive testing visual inspection (VT).

3.11.7.4 Modify welds and replace welds that are out of alignment and strained piping spool pieces to achieve correct strain free alignment of each piping spool piece.

(I) "FIT UP INSPECTION"

3.11.7.4.1 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.11.7.4.2 Accomplish nondestructive testing visual inspection.

(I) "MAGNETIC PARTICLE TEST (MT)"

3.11.7.4.3 Accomplish nondestructive testing magnetic particle inspection (MT).

3.11.7.5 Modify and weld each pump foundation sump screen to assure a snug fit to the dewatering pump column that will not allow trash or debris larger than the screen mesh to pass into the sump.

(I) "FIT UP INSPECTION"

3.11.7.5.1 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.11.7.5.2 Accomplish nondestructive testing visual inspection.

TABLE 8
QTY (One EA), Main Tank Dewatering Pump No. One, Equal in performance characteristics of Sulzer SJM Vertical Mixed Flow Pump, Vertical Turbine/Mixed flow Pump, GPM: 8000, Discharge Head: 28 Feet, Pump Length: 351 inches, Column OD: 18 inches, Column Flanges: 150# forged steel, Shaft & Coupling: 316 Stainless Steel, Line Bearings: Bronze Grade II, Pump Shaft: 316 Stainless Steel, Impeller: Aluminum Nickel Bronze C958, Pump Bowl: Cast Iron, Pump Bearings: Bronze Grade II, Pump Discharge Column: Flanged End 150# forged Steel, Pump/Column interior and exterior Preservation: Marine Environment Paint
QTY (One EA), Main Tank Dewatering Motor No. One, Equal In performance characteristics of Westinghouse Frame 607Y, HP:75, Voltage: 440V, Phase: 3, 60 Cycles, Insulation rating: 40C, Lube: Top/Bottom Bearing, Motor Heater, Seal: Packing, Amps per Terminal: 92.5, RPM at FL: 1,175, Class One, Locked KVA Code: D, Full Load Amps: 100, Hours: 24/ C Rise 40, Style: 2M334
QTY (One EA), Main Tank Dewatering Pump No. 2, Equal In performance characteristics of Sulzer SJM Vertical Mixed Flow Pump, Vertical Turbine/Mixed flow Pump, GPM: 8000, Discharge Head: 28 Feet, Pump Length: 351 inches, Column OD: 18 inches, Column Flanges: 150# forged steel, Shaft & Coupling: 316 Stainless Steel, Line Bearings: Bronze Grade II, Pump Shaft: 316 Stainless Steel, Impeller: Aluminum Nickel Bronze C958, Pump Bowl: Cast Iron, Pump Bearings: Bronze Grade II, Pump Discharge Column: Flanged End 150# forged Steel, Pump/Column interior and exterior Preservation: Marine Environment Paint

TABLE 8
QTY (One EA), Main Tank Dewatering Motor No. 2, Equal In performance characteristics of Westinghouse Frame 607Y, HP:75, Voltage: 440V, Phase: 3, 60 Cycles, Insulation rating: 40C, Lube: Top/Bottom Bearing, Motor Heater, Seal: Packing, Amps per Terminal: 92.5, RPM at FL: 1,175, Class One, Locked KVA Code: D, Full Load Amps: 100, Hours: 24/ C Rise 40, Style: 2M334

3.11.7.6 Install new each equipment listed in TABLE 8 in accordance with manufacturer's instructions.

3.11.7.6.1 Install new CRES fasteners conforming to MIL-DTL-1222, Type I, Grade 316, in way of the installation of each pump to its associated foundation, each dewatering pump sump screen and for each pump column support bracket.

3.11.7.6.2 Apply anti-seize compound that is similar or equal to the performance characteristics of CID-A-A-59004 for each fastener.

3.11.7.6.3 Modify each existing pump foundation for new pump motors.

(I) "FIT UP INSPECTION "

3.11.7.6.4 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.11.7.6.5 Accomplish nondestructive testing visual inspection (VT).

(I) "MAGNETIC PARTICLE TEST (MT)"

3.11.7.6.6 Accomplish nondestructive testing magnetic particle inspection (MT).

3.11.7.7 Install new gaskets conforming to HH-P-151, Class I, cloth inserted rubber, or MIL-PRF-1149, Type II, Class I, synthetic rubber.

3.11.7.8 Lubricate each bearing and coupling with new water resistant grease conforming to MIL-G-23549.

(V) "VERIFY ALIGNMENT"

3.11.7.9 Align the piping to each pump.

3.11.7.10 Piping shall be supported independently and shall not impose a strain.

(V) "OPERATIONAL TEST"

3.11.7.11 Accomplish an operational test of each new installed equipment listed in TABLE 8.

3.11.7.11.1 Inspect each pump for correct rotation.

3.11.7.11.2 Operate each pump for a minimum of One hour after temperatures stabilize (zero degree rise in temperature for 2 consecutive readings taken at 15 minute intervals).

3.11.7.11.3 Measure and record ambient temperatures and bearing temperatures at 15 minute intervals.

3.11.7.11.4 Bearing temperatures shall not exceed 180 degrees Fahrenheit at any time during the test.

3.11.7.11.5 Measure motor voltage and amperage during operation at 15 minute intervals.

3.11.7.11.6 Motor voltage and amperage shall be within the manufacturer's published tolerances for each newly installed motor.

3.11.7.11.7 Measure pump shaft revolutions per minute (RPM).

3.11.7.11.8 Allowable leakage at each pump seal: None.

3.11.7.11.9 Allowable leakage at new and disturbed joints: None.

3.11.7.11.10 Measure and record hot insulation resistance measurements between the motor windings to the Asset's grounding system using a 500-volt megger immediately upon completion of test.

3.11.7.11.11 Megger measurements shall be greater than One-meg ohm.

3.11.7.12 Final operational test requirements of this work item shall be performed in conjunction with the requirements of 3.13.1.

3.11.8 Actuator Stands Replace: Remove existing actuator stands including each existing actuator stand foundation and packing gland. Install new items listed in TABLE 9; located in Operations Deck, using drawings 980 CASPARE STAND 21, PW 51109, PW C-1120, PW 15969 and PW 15970 as guidance.

TABLE 9
QTY (One EA), Equalizing Valve Actuator Stand No. One
QTY (One EA), Equalizing Valve Actuator Stand No. 2
QTY (One EA), Sea Valve Actuator Stand No. One
QTY (One EA), Sea Valve Actuator Stand No. 2
QTY (One EA), Main Tank Dewatering Discharge Valve Actuator No. One
QTY (One EA), Main Tank Dewatering Discharge Valve Actuator No. 2
QTY (One EA), Main Tank Sinking Valve Actuator No. One
QTY (One EA), Main Tank Sinking Valve Actuator No. 2
QTY (One EA) Flood through Valve Actuator Stands No. One
QTY (One EA) Flood through Valve Actuator Stands No. 2
QTY (One EA) Flood through Valve Actuator Stands No. 3
QTY (One EA) Flood through Valve Actuator Stands No. 4

3.11.8.1 Template exact sizes, locations, and configurations from existing shipboard conditions.

3.11.8.2 Remove each existing stuffing box and foundation prior to installation of new equipment listed in TABLE 9.

3.11.8.2.1 Grind flush and smooth top surface of each weld removed in way of each foundation and stuffing box.

3.11.8.3 Install new insert in way of each existing stuffing box and foundation removal or weld fill each existing unused existing actuator stand hole in deck surface prior to installation of new equipment listed in TABLE 9.

3.11.8.3.1 Grind flush and smooth top surface of each weld filled hole.

(I) "FIT UP INSPECTION"

3.11.8.3.2 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.11.8.3.3 Accomplish nondestructive testing visual inspection (VT).

(I) "MAGNETIC PARTICLE TEST (MT)"

3.11.8.3.4 Accomplish nondestructive testing magnetic particle inspection (MT).

(I) "AIR TEST"

3.11.8.3.5 Accomplish local air test of newly installed equipment.

3.11.8.3.6 Air hose nozzle must be placed at less than 3 inches, but greater than zero inches, from the area to be tested and pressure directed at the structure under test in a manner most likely to disclose leaks.

3.11.8.3.7 The minimum nozzle diameter must be 3/8 inch and the nozzle pressure must be 90 plus or minus 5 PSIG as monitored at the nozzle.

3.11.8.3.8 Apply a soapy solution to the opposite side of the structure and inspect for leaks.

3.11.8.3.9 Inspect joint or fitting for leakage by observing for formation of bubbles. Allowable leakage: None.

3.11.8.4 Weld install to the deck Item #1 of drawing 980 CASPARE STAND 21. Install ½ -inch fillet weld around 100 percent of the interior and exterior surfaces of Item #1 in way of new equipment listed in TABLE 9.

(I) "FIT UP INSPECTION"

3.11.8.4.1 Accomplish fit up inspection.

(V) "VISUAL INSPECTION"

3.11.8.4.2 Accomplish nondestructive testing visual inspection (VT) .

(I) "MAGNETIC PARTICLE TEST"

3.11.8.4.3 Accomplish nondestructive testing magnetic particle inspection (MT).

3.11.8.5 Install Anti-seize compound on 100 percent of each threaded surface, product equal to the performance characteristics of CID-A-A-59004.

3.11.8.6 Accomplish the requirements of Surface Preparation Specification SSPC-SP-10 of Systems and Specifications, SSPC Painting Manual, Volume 2 for new equipment listed in TABLE 9. Achieve a surface profile in accordance with SSPC Painting Manual, Volume 2 and the Manufacturers Product Data Sheet.

3.11.8.6.1 Take care not to blast or damage faying or machined surfaces.

3.11.8.7 Apply two coats MIL-PRF-23236, TYPE VII, CLASS 15B preservation coating. Install in accordance with the manufacturer's product data sheets to the entire surfaces listed in each location listed in TABLE 9. Final topcoat color configuration shall be Federal Standard 595 color number 17925 OSHA Safety White.

3.11.8.7.1 Take care not to preserve faying or machined surfaces.

Long Lead: The following materials are considered Long Lead Time Material and shall be ordered by the Contractor within 5 working days of contract award.

3.11.9 Actuators Replace: Remove existing items listed in TABLE 10 including and install new items listed in TABLE 11; located Operations Compartment; using drawings/ manuals MIL-STD-2003-3B(SH), MIL-STD-1310, 980 CASPARE STAND 21, 980CAS-RRMOD-22, 980CAS-CNTRPNL-21, PW 48053, PW 51109, PW 15969, PW 15970 and PW 15971 as guidance.

TABLE 10
QTY (One EA), Existing Sinking Valve Actuator No. One
QTY (One EA), Existing Sinking Valve Actuator No. 2
QTY (One EA), Existing Flood Through Valve Actuator No. One
QTY (One EA), Existing Flood Through Valve Actuator No. 2
QTY (One EA), Existing Flood Through Valve Actuator No. 3
QTY (One EA), Existing Flood Through Valve Actuator No. 4

TABLE 11
QTY (One EA), Sinking Valve Actuator No. One, Equal in performance characteristics of Flowserve Limitorque MX-40 Series B
QTY (One EA), Sinking Valve Actuator No. 2, Equal in performance characteristics of Flowserve Limitorque MX-40 Series B
QTY (One EA), Flood Through Valve Actuator No. One, Equal in performance characteristics of Flowserve Limitorque MX-40 Series B
QTY (One EA), Flood Through Valve Actuator No. 2, Equal in performance characteristics of Flowserve Limitorque MX-40 Series B
QTY (One EA), Flood Through Valve Actuator No. 3, Equal in performance characteristics of Flowserve Limitorque MX-40 Series B
QTY (One EA), Flood Through Valve Actuator No. 4, Equal in performance characteristics of Flowserve Limitorque MX-40 Series B

3.11.9.1 Mechanically and electrically disconnect and remove the equipment listed in TABLE 10.

3.11.9.2 Provide a technical representative in accordance with paragraph 1.26 for disassembly, inspection, adjustment, and testing of the new valve actuator system in accordance with manufacturer's instructions.

3.11.9.2.1 Provide accurate sizing data for actuator to valve size and reach rods.

3.11.9.3 Mechanically and electrically connect new equipment listed in TABLE 11 to new actuator stands and associated electrical components.

3.11.9.3.1 Provide and install new actuator stems for equipment listed in TABLE 11.

3.11.9.3.2 Each actuator hand wheel orientation shall face inboard and position indicator shall be accessible.

3.11.9.3.3 Install new CRES fasteners conforming to MIL-DTL-1222, Type I, Grade 316.

3.11.9.3.4 Install anti-seize compound for each threaded surface a product that is similar or equal to the performance characteristics of CID-A-A-59004.

3.11.9.4 Install electrical cables from load to source using existing wire ways, for each equipment listed in TABLE 11, in accordance with the requirements of 3.10.1.

3.11.9.5 Manually cycle full open to full close each newly installed equipment listed in TABLE 11 one time, prior to electrical operation.

3.11.9.6 Each actuator shall Open/Close each valve at the maximum rate of the actuator/valve manufacturer's recommendations.

3.11.9.7 Accomplish Geared Position Limit Switch Settings and Torque Limit Switch Settings, in accordance with manufacturer’s instructions.

(I) “OPERATIONAL TEST”

3.11.9.8 Accomplish in conjunction with 3.13.1 an operational test the new valve actuator system for designed sequence of operation, in accordance with manufacturer’s instructions. (REQUIRED REPORT CDRL A087)

3.11.9.8.1 Cycle each valve actuator from full closed to full open to full closed five times and Record:

3.11.9.8.2 Ambient and surface temperature of actuator and actuator motor.

3.11.9.8.3 Three phase voltage during valve movement.

3.11.9.8.4 Three phase amperage during valve movement.

3.11.9.8.5 Time increments from full closed to full open and from full open to full close.

3.11.9.8.6 Travel rate of the actuator stem in inches.

3.11.9.8.7 Take precautions when operating each motor driven valve actuator not to over drive reach rods or valves. Station a spotter a visual line of sight of the valve being tested and the reach rod. Spotter will maintain radio contact with the personnel operating the motor driven valve actuator.

3.11.9.9 Install label plate with 1/2-inch letters for each new installed equipment listed in table 11 at locations designated by the SUPERVISOR.

3.11.10 Manual Actuators Remove and Replace: Remove existing items listed in TABLE 12 and install new items listed in TABLE 13; located Operation Compartment; using drawings/ manuals MIL-STD-2003-3B(SH), MIL-STD-1310, C-1120, 980 CASPARE STAND 21, 980CAS-RRMOD-22, 980CAS-CNTRPNL-21, PW 15970, PW 15971, PW 48053 and PW 51109 as guidance.

TABLE 12
QTY (One EA), Discharge Valve Manual Actuator No. One
QTY (One EA), Discharge Valve Manual Actuator No. 2
QTY (One EA), Equalizing Valve Manual Actuator No. One
QTY (One EA), Equalizing Valve Manual Actuator No. 2
QTY (One EA), Sea Valve Manual Actuator No. One
QTY (One EA), Sea Valve Manual Actuator No. 2

TABLE 13
QTY (One EA), Discharge Valve Actuator No. One, Equal in performance characteristics of Flowserve Limitorque MX-40 Series B

TABLE 13
QTY (One EA), Discharge Valve Actuator No. 2, Equal in performance characteristics of Flowserve Limitorque MX-40 Series B
QTY (One EA), Equalizing Valve Actuator No. One, Equal in performance characteristics of Flowserve Limitorque MX-20 Series B
QTY (One EA), Equalizing Valve Actuator No. 2, Equal in performance characteristics of Flowserve Limitorque MX-20 Series B
QTY (One EA), Sea Valve Actuator No. One, Equal in performance characteristics of Flowserve Limitorque MX-20 Series B
QTY (One EA), Sea Valve Actuator No. 2, Equal in performance characteristics of Flowserve Limitorque MX-20 Series B

3.11.10.1 Mechanically disconnect and remove the equipment listed in TABLE 12.

3.11.10.2 Provide a technical representative as in accordance with 1.26 for disassembly, inspection, adjustment, and testing of the new valve actuator system in accordance with manufacturer's instructions.

3.11.10.3 Mechanically and electrically connect new equipment listed in TABLE 13 to new actuator stands and associated electrical components.

3.11.10.3.1 Provide and install new actuator stems for equipment listed in TABLE 13.

3.11.10.3.2 Each actuator hand wheel orientation shall face inboard and position indicator shall be accessible.

3.11.10.3.3 Weld install each QTY (50 EA), Electrical Wire way Cable/Wire Hangers at locations designated by the SUPERVISOR.

3.11.10.3.4 Install new CRES fasteners conforming to MIL-DTL-1222, Type I, Grade 316.

3.11.10.3.5 Install anti-seize compound for each threaded surface a product that is similar or equal to the performance characteristics of CID-A-A-59004.

3.11.10.3.6 Install electrical cables from load to source using existing wire ways, for each equipment listed in TABLE 13 in accordance with the requirements of 3.10.1.

3.11.10.3.7 Manually cycle full open to full close each newly installed equipment listed in TABLE 13 one time, prior to electrical operation.

3.11.10.3.8 Each actuator shall Open/Close each valve at no less than 12 inches per minute or actuator/valve manufacturer's recommendation, whichever one is faster.

3.11.10.4 Accomplish Geared Position Limit Switch Settings and Torque Limit Switch Settings, in accordance with manufacturer’s instructions.

(I) “OPERATIONAL TEST”

3.11.10.5 Accomplish in conjunction with 3.13.1 an operational test of the new valve actuator system for designed sequence of operation, in accordance with manufacturer’s instructions. (REQUIRED REPORT CDRL A088)

3.11.10.5.1 Cycle each valve actuator from full closed to full open to full closed five times and Record:

3.11.10.5.2 Ambient and surface temperature of actuator and actuator motor.

3.11.10.5.3 Three phase voltage during valve movement.

3.11.10.5.4 Three phase amperage during valve movement.

3.11.10.5.5 Time increments from full closed to full open and from full open to full close.

3.11.10.5.6 Travel rate of the actuator stem in inches.

3.11.10.5.7 Take precautions when operating each motor driven valve actuator not to over drive reach rods or valves. Station a spotter a visual line of sight of the valve being tested and the reach rod. Spotter will maintain radio contact with the personnel operating the motor driven valve actuator.

3.11.10.6 Label plate install 1/2 inch letters each newly installed equipment listed in table 11 at locations designated by the SUPERVISOR.

3.11.11 Reach Rod Grease Lines Modify: Reach rod grease lines removed in paragraph 3.11.4; install new QTY (28 EA) UHMW flat surface mounted bearings; located in Main Ballast Tank, Forward Trim Tank and AFT Trim Tank; using drawings 980CAS_RR-UHMW_MOD 21 as guidance.

3.11.11.1 Template exact size, configuration, and location from existing Asset conditions.

3.12 **WEIGHT HANDLING SYSTEMS:** Not Used

3.13 **DOCK AND SEA TRIALS**:

3.13.1 Operational Test Support: Accomplish Sea-trial and dock trials of Caisson 4/5 equipment as follows:

3.13.1.1 Operational Responsibility: Navy Shipyard Personnel will operate all equipment during all tests and trials required by this paragraph. Equipment repaired or modified during this availability shall be operated in accordance with applicable technical manuals.

3.13.1.1.1 Whenever Government support is required for the operation of Asset equipment and/or systems, a minimum of 3 workday's prior notification of the operational/test support requirement shall be provided to the SUPERVISOR.

3.13.1.2 In Dry Dock Baseline Testing: In-Drydock baseline testing shall be conducted in local waters, during normal working hours, on a weekday. Trials shall not exceed 4 hours. The Contractor shall provide a minimum of one Contractor representative authorized to acknowledge and record sea trial discrepancies, one machinist, one pipefitter and one electrician authorized to witness the baseline sea trial. Coordinate test support requirements with PSNS & IMF via the SUPERVISOR.

3.13.1.2.1 Identify and record the following: leaks, vibrations, unusual noise, chattering, missing and loose fasteners, missing and damaged components, damaged wiring, gauge operation, excessive movement, component and linkage binding.

3.13.1.2.2 Accomplish and document results of paragraph 3.13.1.2 in accordance with the requirements of paragraph 3.13.2.1.

3.13.1.3 <u>In Dry Dock Final Trial</u>; In Dry dock final trial shall be scheduled for a 12-hour duration. Trials shall be conducted in local waters during normal working hours, on a weekday. Provide the services of one Contractor representative to acknowledge and direct corrections of test discrepancies and the standby services of One marine machinist, One pipefitter and One marine electrician for the duration of the dock trial test. Coordinate test support requirements with PSNS & IMF via the SUPERVISOR.

3.13.1.3.1 Trial shall not exceed a 12-hour duration. Testing will consist of cycling each valve, and the bumping of each dewatering pump to prove repairs are complete and operational to ensure the Caisson is operationally fit prior to crossing the dry dock sill.

3.13.1.3.2 Identify and record the following: leaks, vibrations, unusual noise, chattering, missing and loose fasteners, missing and damaged components, damaged wiring, gauge operation, excessive movement, component and linkage binding.

3.13.1.3.3 Accomplish and document results of paragraph 3.13.1.3 in accordance with the requirements of paragraph 3.13.2.12.

3.13.1.3.4 Contractor shall immediately correct any Contractor attributed operational test/equipment failure(s).

3.13.2 <u>Alarm Systems and Equipment Test:</u> Accomplish a Pre-Operational test of items listed in TABLE 14; using drawings PW 15959, PW 15960, 401-6896568, PW 48054, PW 15971, PW 51109, 73743, 73744 and C-1120 as guidance.

TABLE 14
Ships Loose Gear
Operations Deck Flooding Alarms
Tank Level Indicating Systems
Operations Deck, Weather Deck and Emergency Lighting Systems
Electrical Panels
Pump Controllers
Pump Motors
Valve Operators
Gauges and Metering Equipment
Heater System
Control Panels

(I) “OPERATIONAL TEST”

3.13.2.1 Accomplish a pre-operational test and inspection in conjunction with 3.13.1.2 prior to the start of for the equipment of TABLE 14 as follows: (REQUIRED REPORT CDRL A089)

3.13.2.1.1 Accomplish an inspection throughout the Asset for all ships loose gear, including mooring lines, shore power lines, lockers, furniture, and tools for the equipment identified in Ships Loose Gear.

3.13.2.1.2 Store ships loose gear identified aboard the asset in a secure and dry and secure location off the Asset.

3.13.2.1.3 Prepare a list of ships loose gear stored in 3.13.2.1.2. List to be submitted with the Required Report of 3.13.2.1.

3.13.2.1.4 Return ships loose gear stored in 3.13.1.2 to the Asset prior to accomplishing 3.13.2.12.

3.13.2.2 Test and inspect each high level alarm sensor for the equipment of Operations Deck Flooding Alarms.

3.13.2.2.1 Manually toggle each sensor. Sensors when toggled shall activate the audible alarms and lights of the flood alarm system.

3.13.2.2.2 Note any corrosion or deterioration of Flood Alarm Sensors.

3.13.2.3 Confirm that when energized each Digital Pressure Process Indicator of the equipment of Tank Level Indicating Systems displays the existing water level in each tank accurately for the forward trim tank, aft trim tank and main ballast tank.

3.13.2.4 Inspect each lighting circuit for the equipment of Operations Deck, Weather Deck and Emergency Lighting Systems.

3.13.2.4.1 Note any lights not functioning, any lights flickering and any transformers humming or making noise.

3.13.2.4.2 Test and confirm that each light switch energizes the associated lighting circuit in the on position. Test and confirm that each light switch de-energizes the associated lighting circuit in the off position.

3.13.2.4.3 Test and confirm each emergency lighting unit for proper operation.

3.13.2.4.4 Test each power receptacle to confirm power is present at the receptacle when the applicable circuit is energized.

3.13.2.4.5 Note any damage to lighting fixtures, wiring, junction boxes, switches and power receptacles.

3.13.2.5 Inspect each electrical panel for the equipment of Electrical Panels including the main power panel, and the 120 VAC power panel.

3.13.2.5.1 Inspect the cleanliness of the internal surfaces of each panel. Note any panel that has internal dirt, dust, blast grit or debris.

3.13.2.5.2 Note any circuit breaker or transformer humming or any visible damage to any components of the panel.

3.13.2.5.3 Test and confirm that each circuit breaker or switch energizes the associated circuit when placed in the on position. Test and confirm that each circuit breaker or switch de-energizes the associated circuit when placed in the off position.

3.13.2.5.4 Note any damage to wiring, wire ways and associated junction boxes emanating from each electrical panel to the point that each individual circuit terminates.

3.13.2.6 Inspect and test each pump controller for the equipment of Pump Controllers including the No. One dewatering controller and the No. 2 dewatering controller.

3.13.2.6.1 Inspect each controller for signs of physical damage.

3.13.2.6.2 Inspect the cleanliness of the internal surfaces of each controller. Note any panel that has internal dirt, dust, blast grit or debris.

3.13.2.6.3 Energize each controller. Note any unusual starting or stopping characteristics or noise.

3.13.2.7 Operate each pump motor for the equipment of Pump Motors including the No. One dewatering pump, and the No. 2 dewatering pump.

3.13.2.7.1 Note the operational status of each pump.

3.13.2.8 Operate each valve operator for the equipment of Valve Operators.

3.13.2.8.1 Note the operational status of each valve operator.

3.13.2.9 During the operation of equipment observe the operation of the equipment of gauges and metering equipment to include all gauges, metering equipment, indicator lights, alarm lights, audible alarms, control switches and buttons

3.13.2.9.1 Note any gauges, metering equipment, indicator lights, alarm lights, audible alarms, control switches and buttons that are damaged or not functioning correctly.

3.13.2.10 Operate the heater system for the equipment of Heater Systems.

3.13.2.10.1 Operate each heater and fan. Note any unusual noise or vibration. Note operation of thermostats.

3.13.2.10.2 Inspect each heater and fan for cleanliness. Note any heater of fan that has internal dirt, dust or debris.

3.13.2.11 Operate the equipment of Control Panels.

(I) "FINAL OPERATIONAL TEST"

3.13.2.12 Accomplish a final operational test and inspection in conjunction with 3.13.1.3 for the equipment listed in TABLE 14 as follows: (REQUIRED REPORT CDRL A090)

3.13.2.12.1 Verify all ships loose gear stored in 3.13.2.1.2 and listed in 3.13.2.1.3 have been returned to the Asset.

3.13.2.12.2 Test and inspect each high-level alarm sensor for the equipment of Operations Deck Flooding Alarms.

3.13.2.12.3 Manually toggle each sensor. Sensors when toggled shall activate the audible alarms and lights of the Flood Alarm System.

3.13.2.12.4 Note any corrosion or deterioration of flood alarm sensors.

3.13.2.12.5 Inspect and observe each Tank Level Indicating System.

3.13.2.12.6 Inspect the submersible depth level transmitter sensor pipe from the operations deck to the concrete level in the forward, aft and main trim tanks. Assure no blast grit, rust or debris is present.

3.13.2.12.7 During operational testing of 3.13.1 confirm that when energized each Digital Pressure Process Indicator of the equipment of Tank Level Indicating System displays the water level in each tank accurately for the forward trim tank, aft trim tank and main ballast tank.

3.13.2.12.8 Inspect each lighting circuit for the equipment of Operations Deck, Weather Deck and Emergency Lighting Systems.

3.13.2.12.9 Note any lights not functioning, any lights flickering and any transformers humming or making noise.

3.13.2.12.10 Test and confirm that each light switch energizes the associated lighting circuit in the on position. Test and confirm that each light switch de-energizes the associated lighting circuit in the off position.

3.13.2.12.11 Test and confirm each emergency lighting unit for proper operation.

3.13.2.12.13 Test each power receptacle to confirm power is present at the receptacle when the applicable circuit is energized.

3.13.2.12.14 Note any damage to lighting fixtures, wiring, junction boxes, switches and power receptacles.

3.13.2.12.15 Inspect each electrical panel for the equipment of Electrical Panel including the main power panel, the 120 VAC power panel, and each valve control panel.

3.13.2.12.16 Inspect the cleanliness of the internal surfaces of each panel. Note any panel that has internal dirt, dust, blast grit or debris.

3.13.2.12.17 Note any circuit breaker or transformer humming or any visible damage to any components of the panel.

3.13.2.12.18 Test and confirm that each circuit breaker or switch energizes the associated circuit when placed in the on position. Test and confirm that each circuit breaker or switch de-energizes the associated circuit when placed in the off position.

3.13.2.12.19 Note any damage to wiring, wire ways and associated junction boxes emanating from each electrical panel to the point that each individual circuit terminates.

3.13.2.12.20 Inspect and test each pump controller for the equipment of Pump Controllers including the No. One dewatering controller and the No. 2 dewatering controller.

3.13.2.12.21 Inspect each controller for signs of physical damage.

3.13.2.12.22 Inspect the cleanliness of the internal surfaces of each controller. Note any panel that has internal dirt, dust, blast grit or debris.

3.13.2.12.23 Energize each controller. Note any unusual starting or stopping characteristics or noise.

3.13.2.12.24 Operate each pump motor for the equipment of Pump Motors including the No. One dewatering pump, and the No. 2 dewatering pump.

3.13.2.12.25 Note the operational status of each pump.

3.13.2.12.26 Operate each valve operator for the equipment of Valve Operators.

3.13.2.12.27 Note the operational status of each valve operator.

3.13.2.12.28 During the operation of equipment observe the operation of the equipment of gauges and metering equipment to include all gauges, metering equipment, indicator lights, alarm lights, audible alarms, control switches and buttons.

3.13.2.12.29 Note any gauges, metering equipment, indicator lights, alarm lights, audible alarms, control switches and buttons that are damaged or not functioning correctly.

3.13.2.12.30 Operationally test newly installed control panel. Energize and accomplish operational testing in conjunction with each new installed equipment listed in 3.10.1 and 3.11.1. Equipment shall function to designed sequence of operation.

3.13.2.12.31 Operate the heater system for the equipment of Heater Systems.

3.13.2.12.32 Operate each heater and fan. Note any unusual noise or vibration. Note operation of thermostats.

3.13.2.12.33 Inspect each vent, heater and fan for cleanliness. Note any vent, heater of fan that has internal or external dirt, dust or debris.

4.0 **GOVERNMENT FURNISHED MATERIAL (GFM):**

	Quantity	**UI**	**Name of Part / Description**	**Pc No.**	**Ref No.**	**National Stock No.**	**Paragraph No.**
4.1	NA	NA	NA	NA	NA	NA	NA

5.0 **DATA SUBMISSION LIST:** The below listed reports shall be submitted to the SUPERVISOR.

DATA SUBMISSION LIST			
CA = Contract Award, DD = Drydocking, PCD – Project Completion Date, PM – Production Meeting, PSD = Production State Date			
CDRL No.	Paragraph No.	REPORT TYPE	Date Required
A001	1.5.1	Production Performance Schedule	Initial, PSD (- 5) Working Days; Weekly, PM (- 24) Hours

DATA SUBMISSION LIST			
CA = Contract Award, DD = Drydocking, PCD – Project Completion Date, PM – Production Meeting, PSD = Production State Date			
CDRL No.	Paragraph No.	REPORT TYPE	Date Required
A002	1.5.2	List of all subcontractors and extent of work being performed	Inspection +2 Working Days
A003	1.5.3	Technical Representatives	Contract Award + 5 Working Days
A004	1.5.4	Welder Qualification List	Inspection +2 Working Days
A005	1.5.5	Manufacturer's Product Data Sheet	Contract Award + 5 Working Days
A006	1.5.6	Safety Data Sheets for Materials Use	Contract Award + 5 Working Days
A007	1.5.7	Contractor Key Personnel, Position, and Contact Information	Inspection +2 Working Days
A008	1.5.8	Contractor's Inspection and Quality Control System	Inspection +2 Working Days
A009	1.5.9	Contractor's Environmental Protection and Hazardous Waste Management Plan	Inspection +2 Working Days
A010	1.7.3	Structural Inspection	Inspection +2 Working Days
A011	1.7.8	Cable Hookup Data	Inspection +2 Working Days
A012	1.7.8.3	Cable Insulation Resistance Measurements	Inspection +2 Working Days
A013	1.7.8.4	Electrical Continuity Test	Inspection +2 Working Days
A014	1.7.10	Photographs	Inspection +2 Working Days
A015	1.8.1.2	Cleanliness Inspection	Inspection +2 Working Days
A016	1.11	Contractor CFR Form	Initial, PSD (- 5) Working Days. Supporting Each Report, Inspection, Test
A017	1.13.2	Injuries to Contractor and Subcontractor Employees	Investigation Complete +24 Hrs.
A018	3.14.3.1	Visual Inspection VT Report	Inspection +2 Working Days
A019	3.14.3.2	Fit Up Inspection Report	Inspection +2 Working Days
A020	3.14.4.1	Magnetic Particle Inspection Report	Inspection +2 Working Days
A021	1.16.1	Preservation OQE, Appendices one Thru 8	SSPC SP1 + 1 Day; Surface Profiles + 1 Day; Surface Conductivity + 1 Day; Topcoat Cure + 1 Day
A022	1.16.2.1	Calibrated Electronic Data Logger record and maintain environmental conditions OQE	Inspection +2 Working Days

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CDRL No.	Paragraph No.	REPORT TYPE	Date Required
A023	1.17	Hazardous Waste Manifests and Waste Disposal Records	PCD + 30 Working Days
A024	1.18.2	CFM Status Report	Contract Award + 5 Working Days
A025	1.19	Weight Control	Undocking (-8) Working Days
A026	1.21	Integrated Logistics Support - O&M Information (operating instructions, manuals, drawings and catalog cuts)	Undocking (-3) Working Days
A027	1.22	NFPA Marine Chemist and Competent Person List	PSD + 8 Working Days; Revisions + 2 Working Days
A028	1.22.2	Outside Rescue Party Agreement	PSD + 8 Working Days; Revisions + 2 Working Days
A029	1.22.4	Marine Chemist Gas Free Certificate	Inspection +2 Working Days
A030	1.23.1	Test and Inspection Plan, Inspection and Quality Control System	Weekly, PM (- 24) Hours; Final +24 Hrs. after All Testing Complete
A031	1.23.6.2	G Point Documentation	Inspection + 24 Hours
A032	1.24	Process Control Procedure	Process Start Date + 15 Working Days
A033	1.25.2	Compartment Arrival / Closure Inspection Schedule	"PSD + 3 Working Days; PCD (-3) Working Days"
A034	1.25.3	Final Closeout Inspection	Inspection + 3 Working Days
A035	1.26.1	Technical Representative OEM Certification	Contract Award + 5 Working Days
A036	1.26.5	Technical Representative Reports, Recommendations and Acceptance Criteria	Inspection + 2 Working Days
A037	1.27	Heavy Weather Plan	CA + 5 Working Days
A038	1.28.3	FME Log	Inspection +2 Working Days
A039	3.1.2	Docking Procedure	Drydocking (-5) Working Days
A040	3.1.3	Blocking Plan	Drydocking (-5) Working Days
A041	3.1.12	Final Docking Report	Undocking +5 Working Days
A042	3.2.1	Hull Distinguishing Figures and Markings Report	Inspection +2 Working Days
A043	3.2.2.2	Mounting Studs Inspection	Inspection +2 Working Days
A044	3.2.3.3	Zinc Anode Locations Report	Drydocking +4 Working Days
A045	3.2.3.4	Anode Resistance Testing Report	Inspection +2 Working Days

DATA SUBMISSION LIST

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CDRL No.	Paragraph No.	REPORT TYPE	Date Required
A046	3.2.4	Hull and Freeboard UT Inspection Report	Inspection +2 Working Days
A047	3.2.6	Hull Visual Inspection	Inspection +2 Working Days
A048	3.2.6.3	Hull Structural Repair 100% Completion	Inspection +2 Working Days
A049	3.3.1.1	Steel Repair Plan	Inspection +2 Working Days
A050	3.3.1.5	Steel Repair Declining Log	Inspection +5 Working Days
A051	3.3.1.9.4	Air Hose Test Inspection Report	Inspection +2 Working Days
A052	3.3.2	Port Rubber Seal UT Inspection Report	Inspection +2 Working Days
A053	3.3.3	Port Rubber Seal VT Inspection Report	Inspection +2 Working Days
A054	3.3.9.1	Porthole Inspection Report	Inspection +2 Working Days
A055	3.3.9.3	Water Hose Test Report	Inspection +2 Working Days
A056	3.4.1.1	Tank/Void Residual Liquids Report	Inspection +2 Working Days
A057	3.4.2	Tank/Void Color Schemes and Markings Report	Inspection +2 Working Days
A058	3.4.3.5	Tank/Void Anode Locations Report	Inspection +2 Working Days
A059	3.4.3.6	Tank/Void Resistance Testing Report	Inspection +2 Working Days
A060	3.4.4	Tank/Void UT Inspection Report	Inspection +2 Working Days
A061	3.4.6	Tank Interior Tank Fastener Inspection	Inspection +2 Working Days
A062	3.4.11.8	Air Hose Test Inspection Report	Inspection +2 Working Days
A063	3.4.11.9	Chalk Test Inspection Report	Inspection +2 Working Days
A064	3.4.12.9.6	Pull Test Inspection Report	Inspection +2 Working Days
A065	3.4.13.7	Air Hose Test Inspection Report	Inspection + 2 Working Days
A066	3.4.15	Tank Ladders Inspection Report	Inspection +2 Working Days
A067	3.4.19	Tank Close Out Inspection Report	Inspection +3 Working Days
A068	3.5.3	Weather Deck UT Inspection Report	Inspection +2 Working Days
A069	3.5.5.3	Weather Deck Structural Repair 100% Completion	Inspection +2 Working Days
A070	3.5.10.11	Visual Inspection VT Report	Inspection +2 Working Days

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CDRL No.	Paragraph No.	REPORT TYPE	Date Required
A071	3.5.10.16	Pull Test Inspection Report	Inspection +2 Working Days
A072	3.5.12.7	Pull Test Inspection Report	Inspection +2 Working Days
A073	3.5.12.8	Pull Test Inspection Report	Inspection +2 Working Days
A074	3.6.2	Operations Deck UT Inspection Report	Inspection +2 Working Days
A075	3.6.7.8	Operations Deck Embedded Contaminates Inspection Report	Inspection +2 Working Days
A076	3.6.8.4.7	Strength Test Inspection Report	Inspection +2 Working Days
A077	3.6.8.6.1	Fastener Torque Inspection Report	Inspection +2 Working Days
A078	3.6.8.6.6	Static Load Test Inspection Report	Inspection +2 Working Days
A079	3.10.2.4	Electrical One Line Diagram	Inspection +2 Working Days
A080	3.10.9.11	Air Test Inspection Report	Inspection +2 Working Days
A081	3.10.9.26	Operational Test Inspection Report	Inspection +2 Working Days
A082	3.11.1.1	Pipe UT Inspection Report	Inspection +2 Working Days
A083	3.11.2.1	Receipt Inspection Report	Inspection +2 Working Days
A084	3.11.2.2	Hydrostatic Test Inspection Report	Inspection +2 Working Days
A085	3.11.2.9	Piping and Valve Redline Drawing Report	Inspection +2 Working Days
A086	3.11.5.5	Operational Test Inspection Report	Inspection +2 Working Days
A087	3.11.9.8	Valve Actuator Operational Test Report	Inspection +2 Working Days
A088	3.11.10.5	Valve Actuator Operational Test Report	Inspection +2 Working Days
A089	3.13.2.1	Pre-Operational Test Report	Inspection +2 Working Days
A090	3.13.2.12	Final Operational Test Report	Inspection +2 Working Days